S6_02_FAMILY
The period family, its identifications, and its degenerations

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Conventions and the notation dictionary

We use $B = \mathbb{P}^1_t$, $p_1 = 0$, $p_2 = 1$, $p_0 = \infty$, and $B^\circ = B \setminus \{0, 1, \infty\}$. Base meridians are counterclockwise in a local holomorphic coordinate. The analytic generators are clockwise root deck transformations, $s_j(g_j z) = e^{-2\pi i/m_j} s_j(z)$, with $g_1 g_2 g_0 = 1$. A positive base loop ends at the inverse deck translate; identifying its endpoint with the initial fibre applies g_j, giving homology monodromy A_j. The retained multiplicities are $m_1 = 3, m_2 = 4$; $f^* p_j = m_j S_j$ includes the nonreduced structure, whereas S_j is reduced. The cusp fibre $W = f^{-1}(\infty)$ is reduced.

$$\Lambda = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta} \rangle, \qquad \Pi_c = \begin{pmatrix} 6\mu & \tau & 1 & 0 \\ \beta^{\mathrm{part}} + c & \mu & 0 & 1 \end{pmatrix}, \qquad w = e^{-\pi i c}.$$

We fix $(\beta^{\mathrm{part}} + \tau)(p_0) = 0$. Write $D_{\mathrm{part}} = \Im\beta^{\mathrm{part}} - 6(\Im\mu)^2/\Im\tau$ and $M = \sup D_{\mathrm{part}}$. Engel's line bundle is $M_S = \mathcal{O}_S(P - O)$; his indices $0, 1, \infty$ correspond to our $1, 2, 0$. In his common integral basis, the columns of the following matrix are our ordered basis vectors:

$$P_{\mathrm{mark}} = \begin{pmatrix} 0 & -2 & -1 & -2 \\ 2 & -1 & 0 & -1 \\ 0 & 2 & 1 & 3 \\ -1 & 0 & 0 & 0 \end{pmatrix}, \quad T_0 P_{\mathrm{mark}} = P_{\mathrm{mark}} A_1, \quad T_1 P_{\mathrm{mark}} = P_{\mathrm{mark}} A_2, \quad \delta_{\mathrm{E}} = P_{\mathrm{mark}} \widehat{\delta}.$$

For our finite twist vectors v_1, v_2, his positive-root affine vectors are $a_0 = -P_{\mathrm{mark}} v_1/3$, $a_1 = -P_{\mathrm{mark}} v_2/4$. In additive fibre coordinates (ξ, η), put $v = e^{2\pi i \eta}$. The first period acts by $(\xi, v) \mapsto (\xi + 6\mu, e^{2\pi i \beta_c} v)$; Engel's contracting generator is its inverse, over $-6P$ with multiplier $e^{-2\pi i \beta_c}$. We choose his reference linearization to be this transported inverse period lift at $c = 0$. Then $\lambda_0 = 1$ and $u = w^2$. The comparison maps and the effect of changing this reference are proved in B.1 and B.2. Homology has integral coefficients unless stated otherwise; tori and the threefold have their complex orientations.

Each statement below has its own proof or an explicit unresolved target with the work obtained in attacking it. The report records every prompt item. No priority is claimed.

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1 A. The retained family and its intrinsic classification

Throughout this section $\mathcal{P} = \{c : \operatorname{Im} c < -M\}$, and X_c means the actual glued space with $v_1 = \varepsilon = (1, 2, -4, 0)^t$, $v_2 = -\varepsilon' = (-1, -3, 3, 0)^t$ and the untwisted cusp collar. We write $d = \widehat{\delta}$, $e = (0, 1)^t$, $E = d\gamma$, and use column vectors. Our peripheral generators are clockwise: $s_j(g_j z) = e^{-2\pi i/m_j} s_j(z)$ and $g_1 g_2 g_0 = 1$. Thus $s(g_0 z) = s(z) - 1$ at the cusp. All period comparisons use

$$A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 0 & 1 & 0 \\ -6 & -1 & -1 & 0 \\ -2 & 1 & 0 & 1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -6 & 1 & 0 & 0 \\ 3 & 0 & 1 & 1 \end{pmatrix}, \qquad R_g(z)\Pi_c(z) = \Pi_c(gz)A_g.$$

Here $R_1 = \left(\begin{smallmatrix} -1/\tau & 0 \\ (1-\mu)/\tau & 1 \end{smallmatrix}\right)$, $R_2 = \left(\begin{smallmatrix} 1/\tau & 0 \\ -\mu/\tau & 1 \end{smallmatrix}\right)$, and $R_0 = I$. The distinction between source and target markings in each equation below is retained by its displayed period equation.

1.1 A.1. Canonical sections and the finite covers

Proposition (A.1.1; Proved)**.** *For the constructed holomorphic map $f_c : X_c \to \mathbb{P}^1$, and also for every admissible choice of its affine translations and cusp collar, we have*

$$\omega_{X_c} \simeq \mathcal{O}_{X_c}(-2S_2), \qquad f_{c*}\mathcal{O}_{X_c}(kS_2) = \mathcal{O}_B(\lfloor k/4 \rfloor p_2) \quad (k \in \mathbb{Z}).$$

The latter equality is an equality of subsheaves of meromorphic functions under pullback. The anticanonical section ring is $\mathbb{C}[U, V]$, with $\deg U = 1$, $\deg V = 2$. The complete degree-two map is f_c followed by a projective coordinate isomorphism of its base. Every biholomorphism between two such threefolds is over the identity of B.

Proof. We compute the line and its evaluation map, including the multiple fibres. A holomorphic function restricts to a constant on every connected compact reduced fibre. To descend functions on a neighbourhood we first use regular fibres; equality then extends from their dense union by the identity theorem on the smooth total space. Away from the multiple fibres, evaluation on local sections descends it to a holomorphic base function. On a finite root cover this follows by evaluation on the zero section; invariance under $s \mapsto \zeta s$ leaves precisely the convergent series in s^m. At the cusp, a smooth point of the reduced fibre admits a local section, and constancy on each nearby compact fibre gives the same descent there by the identity theorem. Consequently $f_*\mathcal{O}_X = \mathcal{O}_B$.

Put $e_\wedge = d\zeta_1 \wedge d\zeta_2$ in relative fibre coordinates. At the finite centre j let $a_1 = 2, a_2 = 1$ and $\zeta_j = e^{-2\pi i/m_j}$. The canonical frame $ds_j \wedge e_\wedge$ has character $\chi_j(s) = \zeta_j \det R_j(s)$, with $\chi_j(0) = \zeta_j^{1+a_j}$. Take its holomorphic logarithm $\chi_j = \zeta_j^{1+a_j} \exp \psi_j$, where $\psi_j(0) = 0$. The finite group identity says $\sum_{r=0}^{m_j-1} \psi_j(\zeta_j^r s) = 0$. Writing $\psi_j = \sum c_n s^n$, we obtain $c_n = 0$ for $m_j \mid n$. Therefore

$$\varphi_j(s) = \sum_{m_j \nmid n} \frac{c_n}{1 - \zeta_j^n} s^n$$

is convergent and satisfies $\varphi_j(s) - \varphi_j(\zeta_j s) = \psi_j(s)$. The invariant coefficient module of the relative canonical frame $e^{\varphi_j}(ds_j \wedge e_\wedge) \otimes (dt_j)^{-1}$ is exactly $s_j^{m_j-1-a_j}\mathbb{C}\{s_j^{m_j}\}$. Notice that $(dt_j)^{-1}$ denotes a pulled-back line-bundle frame; no division by a vanishing differential is made. Since the total-space cover is etale, the evaluation divisor downstairs is $(m_j - 1 - a_j)S_j$, namely 0 and $2S_2$.

At the cusp the ordinary form

$$\Omega = \frac{dx_1}{x_1} \wedge \frac{dx_2}{x_2} \wedge dt_c$$

is nowhere zero. Indeed in a maximal unimodular cone chart, with coordinates z_0, z_1, z_2, the height-one condition gives $t_c = z_0 z_1 z_2$ and unimodularity gives $\Omega = \pm dz_0 \wedge dz_1 \wedge dz_2$. The same equality after inverting coordinates treats the other charts. Integral shears and holomorphic torus multipliers depending on t_c preserve this form: all their extra differential terms contain dt_c and vanish in the wedge. Thus the form descends to the cusp quotient and evaluation has no zero there. This calculation also shows that changing any collar by a fibre translation does not alter the canonical transition factors.

We next determine $\mathfrak{K} = f_*\omega_{X/B}$. On the simply connected period cover let $h^2 = E_6(\tau)$, and put $\Theta = E_4(\tau)^2 h/\Delta(\tau)$. The relation $j(\tau) = 1728t$ gives local degrees $\tau - \rho \sim s_1$, $\tau - i \sim s_2^2$ and $q = e^{2\pi i\tau} \sim t_c/1728$. The identities $E_4^3 = 1728t\Delta$ and $E_6^2 = 1728(t-1)\Delta$, together with the nonvanishing product $\Delta = q\prod_{n\geq 1}(1-q^n)^{24}$ on the upper half-plane, give orders $2, 1, -1$ for Θ. The same identities exclude every other zero or pole. The square root exists because the zeros of $E_6(\tau)$ have even order and the cover is simply connected. Its signs are determined locally: at ρ, $\rho^3 = -1$ and $h \neq 0$, whereas at i, h has order one in s_2. The weight transformation laws consequently give

$$\Theta(g_1 z) = -\Theta(z)/\tau(z), \qquad \Theta(g_2 z) = \Theta(z)/\tau(z), \qquad \Theta(g_0 z) = \Theta(z).$$

The invariant relative form $\Theta^{-1}e_\wedge$ is a unit in the finite local generator just calculated and has a simple zero in the cusp generator $e_\wedge = (2\pi i)^{-2}\Omega \otimes (dt_c)^{-1}$. It defines an isomorphism $\mathfrak{K} \simeq \mathcal{O}_B(p_0)$. The evaluation divisor now yields

$$\omega_X \simeq f^*\mathcal{O}_B(p_0) \otimes f^*\omega_B \otimes \mathcal{O}_X(2S_2).$$

In the fixed coordinate t, the function $(t-1)^{-1}$ has divisor $p_0 - p_2$, while $dt/(t-1)^2$ has divisor $-2p_2$. These explicit maps identify the last display with $\mathcal{O}_X(-4S_2 + 2S_2)$.

A meromorphic representative of a section of $\mathcal{O}_X(kS_2)$ is holomorphic off S_2, so the constancy argument makes it a pullback there. On the root cover its value along the zero section is meromorphic, with order at least $-k$, and is invariant under $s \mapsto -is$. Its Laurent series has only powers s^{4r}. The necessary and sufficient inequality is $4\,\mathrm{ord}_{p_2} b + k \geq 0$, which proves the direct-image formula, including negative k.

Choose the canonical divisor section u of $\mathcal{O}_X(2S_2)$ and identify this line with ω_X^{-1} using the preceding maps. Then $u^2 = f^*\sigma_{p_2}$. Choose $\ell \in H^0(B, \mathcal{O}_B(p_2))$ with $\ell(p_2) \neq 0$, and let $v = f^*\ell$ in degree two. In degree $m = 2r$ the monomials $u^{2r-2j}v^j$, $0 \leq j \leq r$, are the pullbacks of the usual basis $\sigma_{p_2}^{r-j}\ell^j$. In degree $2r+1$, multiplication by u gives the same basis because the allowed base pole order remains r. These are all sections by the direct-image formula. Thus $U \mapsto u, V \mapsto v$ is a graded algebra isomorphism, not only a Hilbert-series equality. The two degree-two sections have no common zero and give f in the projective coordinate $[\sigma_{p_2} : \ell]$.

A biholomorphism pulls back the canonical line, its sections and their evaluations; it therefore induces $\alpha \in \mathrm{PGL}_2(\mathbb{C})$ with $f'F = \alpha f$. The cusp fibre is reduced with nonnormal points. The two finite fibres have smooth reductions and nilpotence orders exactly three and four in their local equations $s^3 = 0, s^4 = 0$. These are intrinsic local-ring properties of the full fibres and distinguish all three. Every other fibre is a smooth reduced torus. Consequently α fixes $\infty, 0, 1$ individually, and the equation of a fractional linear transformation gives $\alpha = 1$. □

Proposition (A.1.2; Proved)**.** *The normalized base change $N_j \times_{D_j} \Delta_j \to \Delta_j$, $t_j = s_j^{m_j}$, is the original smooth torus family J_j'. The maps $E_1 \times F_1 \to S_1$ and $E_2 \times F_2 \to S_2$ described below are finite etale covers of degrees* 9 *and* 8*.*

Proof. On J_j', the deck generator is $(s, \zeta) \mapsto (\zeta_j s, R_j(s)\zeta + \Pi_c(g_j s)v_j/m_j)$. It has exact order m_j. Its rth power changes the invariant circle coordinate γ by $r\gamma(v_j)/m_j$; this is nonintegral

for $0 < r < m_j$, so the action is free. The quotient map is therefore a local biholomorphism. Its local base function is s^{m_j} and its fibre multiplicity is exactly m_j. The map

$$J_j' \longrightarrow N_j \times_{D_j} \Delta_j, \qquad y \longmapsto (q_j(y), s_j(y))$$

is finite, is an isomorphism over $s \neq 0$, and has normal source. Locally at the central fibre the base-change ring has equation $z^{m_j} - s^{m_j} = 0$. Its normalization is the disjoint union of branches $z = \zeta_j^r s$. The displayed map gives those branches in the covering charts. It is thus the normalization everywhere, with its specified root coordinate.

Put $\rho = e^{\pi i/3}$, $\mu_1 = (2-\rho)/3$, $\mu_2 = (1-i)/2$, $\kappa_1 = \beta_c(z_1) + 2\mu_1$, and $\kappa_2 = \beta_c(z_2) + 3\mu_2$. The fixed-point transformation laws for μ give the two displayed values directly. We have $\operatorname{Im}\kappa_j = D_c(z_j) < 0$. Define

$$E_j = \mathbb{C}/(\mathbb{Z} + \kappa_j\mathbb{Z}), \quad F_1 = \mathbb{C}/(\mathbb{Z}(\rho-2) - 3\mathbb{Z}), \quad F_2 = \mathbb{C}/(\mathbb{Z}(i-1) - 2\mathbb{Z}).$$

Let $e_1 = (0,1,-2,0)^t$, $f_1 = (0,0,-3,1)^t$, $e_2 = (0,1,-1,0)^t$, $f_2 = (0,0,-2,1)^t$. Direct solution of the fixed equations and inspection of the columns give

$$\Lambda^{A_1} = \mathbb{Z}\varepsilon + \mathbb{Z}d, \quad \Lambda^{A_2} = \mathbb{Z}\varepsilon' + \mathbb{Z}d, \quad (A_j - I)\Lambda = \mathbb{Z}e_j + \mathbb{Z}f_j.$$

The image lattices are saturated: they are the integer solutions of $\gamma = 0, 2u + w + 3\delta = 0$ and $\gamma = 0, u + w + 2\delta = 0$, respectively. The determinants of the fixed-plus-moving basis matrices are $-3, -2$.

The complex-linear maps $L_1(\xi,\eta) = (\eta, \xi - \eta/3)$ and $L_2(\xi,\eta) = (\eta, \xi - \eta/2)$ carry the product periods to those fixed and moving sublattices. They induce isogenies to the original central tori, with kernel generated respectively by $(2/3, \rho)$ and $(1/2, i)$, of exact orders three and two. For example $\widehat{u} = 2d/3 + e_1 - 2f_1/3 = d/2 + e_2 - f_2/2$ gives both kernel representatives.

The affine deck maps lift to

$$(\xi,\eta) \longmapsto (\xi + \kappa_1/3, (\rho-1)\eta), \qquad (\xi,\eta) \longmapsto (\xi - \kappa_2/4, -i\eta).$$

Substitution in L_j gives exactly $R_j L_j + \Pi_c v_j/m_j$. They commute with their kernel translations: the moving differences are periods, since $(\rho-2)\rho = -1-\rho = -(\rho-2) - 3$ can equally be checked in the lattice basis, and $(-i-1)i = 1 - i = -(i-1)$. More directly these assertions follow by the two-dimensional period matrices of the indicated rotations. A nonidentity element of either resulting product group translates the E_j coordinate by $2a/3 + b\kappa_1/3$ or $a/2 - b\kappa_2/4$, respectively. Real independence of $1, \kappa_j$ proves that this translation vanishes only for the identity element. Hence the groups $C_3 \times C_3$ and $C_2 \times C_4$ act freely and have quotient precisely S_j. □

1.2 A.2. The integral obstruction

Proposition (A.2.1; Proved)**.** *The rational centralizer of (A_1, A_2) is $\mathbb{Q}I + \mathbb{Q}E$. Every biholomorphism $X_c \to X_{c'}$ has $c' - c = n \in 2\mathbb{Z}$, complex-linear fibre part I_2, and source-to-target lattice map $I - nE$. In the full fixed lattices,*

$$\Lambda^{A_1}/N_1\Lambda \simeq \mathbb{Z}/3, \qquad \Lambda^{A_2}/N_2\Lambda \simeq \mathbb{Z}/4 \oplus \mathbb{Z}/2,$$

with d generating the second factor of the latter group and vanishing in the former.

Proof. The common invariant vector space is $\mathbb{Q}d$, and the common invariant covector space is $\mathbb{Q}\gamma$; this follows by intersecting the fixed bases in A.1.2 and the rows $\langle \gamma, 2u + w + 3\delta\rangle$, $\langle \gamma, u + w + 2\delta\rangle$. A commuting endomorphism preserves $\mathbb{Q}d \subset \ker\gamma$. Its induced matrix on $\ker\gamma/\mathbb{Q}d$ commutes with $Q_1 = \left(\begin{smallmatrix} 0 & 1 \\ -1 & -1 \end{smallmatrix}\right)$ and $Q_2 = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. Commutation with Q_2 makes it

$\left(\begin{smallmatrix} b & r \\ -r & b \end{smallmatrix}\right)$, and the Q_1 equation sets $r = 0$. Write the remaining columns as $H\widehat{\gamma} = (a, x, y, k)^t$, $H\widehat{u} = b\widehat{u} + u_0 d$, $H\widehat{w} = b\widehat{w} + w_0 d$, $Hd = d_0 d$. The equations on $\widehat{u}, \widehat{w}$ give $d_0 - w_0 = b + u_0$, $u_0 = 2w_0$ and $u_0 = w_0$, hence $u_0 = w_0 = 0, d_0 = b$. The first-column equations give $x = 2(a-b), y = -4(a-b), x = -y$. Thus $a = b, x = y = 0$, proving the assertion. Conversely $A_j E = E A_j = E$.

A fibre biholomorphism lifts to an affine map on $\mathbb{C}^2$: the derivative of a lifted map is lattice periodic, descends to a holomorphic function on a compact torus, and is constant. Its lattice map is locally constant and commutes with both monodromies. Write it as $H = aI + bE$; integrality and $\det H = a^4 = 1$ give $a = \pm 1, b \in \mathbb{Z}$. The equation $L\Pi_c = \Pi_{c'} H$, in columns three and four, gives $L = aI_2$; column one gives $a(c' - c) + b = 0$. Thus $n = c' - c \in \mathbb{Z}$.

The normalized base-change construction in A.1.2 is functorial for biholomorphisms over the base: the base-changed isomorphism transports integral elements of the total quotient ring to integral elements, and its inverse does the same. Its lift is over the identity of the root disc. It commutes with the prescribed deck generator because both compositions lift the same map $(x, s) \mapsto (F(x), \zeta_j s)$ on the unnormalized base change. This proves exact commutation, rather than commutation after an unchosen power of the generator.

Lift the translation of the resulting affine torus map holomorphically to a vector $B_j(s)$ on the simply connected root disc. Its deck equation, expressed in the source periods, is

$$B_j(g_j z) - R_j(z) B_j(z) = \Pi_c(g_j z) \left(\frac{(1-a) v_j + n\gamma(v_j) d}{m_j} + \ell_j \right), \qquad \ell_j \in \Lambda.$$

The integral vector is constant by continuity and real invertibility of the period matrix. Iterating the equation m_j times gives

$$N_j \ell_j = -(1-a) v_j - n\gamma(v_j) d.$$

Exact multiplication yields

$$N_1 = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 6 & 0 & 0 & 0 \\ -12 & 0 & 0 & 0 \\ 0 & 2 & 1 & 3 \end{pmatrix}, \quad N_2 = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 12 & 0 & 0 & 0 \\ -12 & 0 & 0 & 0 \\ 0 & 2 & 2 & 4 \end{pmatrix}.$$

In fixed bases their images are $3\mathbb{Z}\varepsilon + \mathbb{Z}d$ and $4\mathbb{Z}\varepsilon' + 2\mathbb{Z}d$. These explicit generators prove the asserted quotient groups. Applying γ to the norm equation excludes $a = -1$ already at $m_1 = 3$, since $3\gamma(\ell_1) = -2$. For $a = 1$, the second norm equation says exactly $2 \mid n$; the first imposes no condition. Program `code/ab_exact.py` verifies the matrices, Smith forms, centralizer and membership independently by Hermite-form comparisons. All its computations are over $\mathbb{Z}$ or $\mathbb{Q}$. □

1.3 A.3. Extension and coherent descent

Proposition (A.3.1; Proved)**.** *For each $n \in 2\mathbb{Z}$ there is a biholomorphism $F_{n,c} : X_c \to X_{c+n}$, holomorphic jointly in c, with $F_{2h,c+2k} F_{2k,c} = F_{2(h+k),c}$. The quotient family over $0 < |w| < e^{-\pi M}$, $w = e^{-\pi i c}$, is proper and holomorphic, with pairwise non-biholomorphic fibres.*

Proof. Integer shifts satisfy $\Pi_{c+n} = \Pi_c(I + nE)$, so the middle lattices agree. Set

$$r_1 = -\tfrac{n}{3}(\widehat{u} + \widehat{w}), \quad r_2 = \tfrac{n}{4}(\widehat{u} + \widehat{w}), \quad \ell_1 = -n\widehat{w}, \quad \ell_2 = \tfrac{n}{2}\widehat{u}, \quad b_j = \Pi_c r_j.$$

The four-entry equations are $(I - A_1) r_1 = (0, 0, -n, n/3)^t$ and $(I - A_2) r_2 = (0, n/2, 0, -n/4)^t$. Consequently $B_j = b_j$ solves the preceding deck equation for $a = 1$. The map $(z, \zeta) \mapsto$

$(z, \zeta + b_j(z))$ extends on the entire root family and descends, together with its inverse, to each finite filling.

Its middle translation on the punctured overlap is

$$a_j = b_j + n\gamma(v_j)\frac{\log s_j}{2\pi i}e.$$

This follows from the actual overlap $h_{j,c}(\zeta) = \zeta + \sigma_{j,c}$, where $\sigma_{j,c} = (\log s_j)/(2\pi i)\,\Pi_c v_j$: precisely $\sigma_{j,c+n} - \sigma_{j,c} = n\gamma(v_j)(\log s_j)/(2\pi i)e$. Changes of logarithm add periods. Since the chosen generator changes $(\log s_j)/(2\pi i)$ by $-1/m_j$, we have $a_j(g_j z) = R_j a_j(z) + \Pi_c(g_j z)\ell_j$.

On the universal cover of B°, extend ℓ_1, ℓ_2 by the integral cocycle law $\ell_{gh} = \ell_g + A_g\ell_h$. Freeness of $\pi_1(B^\circ)$ makes this well defined. At $g_0 = (g_1 g_2)^{-1}$ it gives

$$\ell_0 = -A_0(\ell_1 + A_1\ell_2) = \tfrac{3n}{2}\widehat{w} - \tfrac{n}{2}d.$$

Put $q_n = (-3n/2, n/2)^t$ and $a_0 = sq_n$. Then $a_0(g_0 z) - a_0(z) = \Pi_c\ell_0$. In cusp torus coordinates its map is $(x, t_c) \mapsto (t_c^{q_n}x, t_c)$. The integral linear map $(y, h) \mapsto (y + hq_n, h)$ of the fan lattice translates each height-one triangle by an integer vector. It therefore permutes every cone of the prescribed periodic triangulation. The map and its inverse extend across all strata, including both triple points and all three double curves after quotient. Moreover the cusp deck multipliers are identical at c and $c+n$, since $C_{c+n}\lambda - C_c\lambda = (0, n\lambda_1)^t$ has integral exponential. Thus the extended map descends with all its nonconstant units retained.

We supply the global analytic correction. Quotient the trivial affine vector bundle on the universal cover by $(z, y) \mapsto (gz, R_g(z)y + \Pi_c(gz)\ell_g)$. Its action law is exactly the cocycle law by period covariance. Specify at the finite points sections of the form $a_j + \eta_j$, with η_j holomorphic equivariant on the full root disc; at the cusp specify $a_0 + \eta_0(t_c)$ with η_0 holomorphic on the full disc. These prescriptions define a locally nonempty torsor under a rank-two locally free sheaf $\mathcal{V}$ on B.

Projection to the first coordinate gives

$$0 \longrightarrow \mathcal{O}_B e \longrightarrow \mathcal{V} \longrightarrow \mathcal{O}_B(-p_0) \longrightarrow 0.$$

Indeed its factors are $-1/\tau, 1/\tau, 1$. Multiplication by Θ identifies this first-coordinate sheaf with holomorphic functions at ordinary and finite points and functions vanishing at p_0: the finite vanishing orders are $2, 1$, and the cusp pole is one. At a finite stalk, any equivariant first component has an equivariant vector lift by averaging a preliminary lift $(u, 0)$ over the finite group. This proves stalkwise surjectivity and local freeness, including the off-diagonal entries of R_j.

Both $H^1(\mathbb{P}^1, \mathcal{O})$ and $H^1(\mathbb{P}^1, \mathcal{O}(-1))$ vanish. To make the analytic gluing explicit, split a smooth cocycle with a partition of unity, solve $\bar\partial$ locally by the Cauchy integral, and compare the two resulting solutions on an annulus. A Laurent series $\sum a_n t^n$ is the difference of its nonnegative and negative parts for $\mathcal{O}$; for $\mathcal{O}(-1)$ use the second frame t^{-1}, so the negative terms become $t^{-1}(-\sum_{n<0} a_n t^{n+1})$. Both series converge on their respective discs and leave no exponent unsolved. This kills the cocycle for any refinement. The exact sequence gives $H^1(B, \mathcal{V}) = 0$. Project the torsor cocycle to $\mathcal{O}(-1)$, solve it, lift the solutions, and solve the remaining scalar cocycle. We obtain one global section a, with $a = a_j + \eta_j$ at finite points and $a = sq_n + \eta_0$ at the cusp.

The three actual maps are now

$$\zeta \mapsto \zeta + a, \qquad \zeta \mapsto \zeta + b_j + \eta_j, \qquad (x, t_c) \mapsto (e^{2\pi i\eta_0(t_c)}t_c^{q_n}x, t_c).$$

The finite overlap identity is the equality of sums $\sigma_{j,c} + a = \sigma_{j,c+n} + b_j + \eta_j$. The cusp overlap identity is $e^{2\pi i a} = t_c^{q_n}e^{2\pi i\eta_0}$. The holomorphic-unit factors act on every toric chart. The local inverses satisfy the same identities; hence the maps glue to a biholomorphism.

For coherence, perform the additive solution only for $n = 2$ and multiply that solution by k for $n = 2k$. The cocycles have $\gamma(\ell_g) = 0$, so their affine terms $\Pi_c \ell_g$ are independent of c. The reference translations, $\mathcal{V}$, and the correction can all be chosen independent of c. Adding the translations and multiplying the toric factors proves the stated cocycle identity exactly.

Finally the original gluing constructs a holomorphic family over any parameter disc compactly contained in $\mathcal{P}$. In ordinary and finite charts, real invertibility of Π_c is uniform on compact parameter sets, so a fixed compact real parallelepiped proves properness and the lattice quotient is holomorphic. At the cusp write $Z = sB_0 + C(t_c, c)$ with $B_0 = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. On a compact parameter disc C is uniformly bounded; shrink the cusp radius until $\|(-2\pi \operatorname{Im} C)/\log|t_c|\| < 1/2$. The effective logarithmic lattice is then within $1/2$ of the unimodular B_0, uniformly. Reduce its logarithmic valuation modulo the effective logarithmic lattice to a bounded parallelogram. Choose a height-one triangle containing that reduced valuation. Its barycentric coordinates are nonnegative and sum to one; the corresponding three unimodular cone characters therefore have modulus at most one. This puts the point in the closed unit polydisc of that cone. Only finitely many triangles meet the fixed bounded parallelogram, uniformly on the parameter compactum. Boundary orbits have the same representatives by translating their height-one components to these triangles. Thus every orbit has a representative in this finite union of cone polydiscs. Those closed polydiscs times a compact parameter set form a compact surjective representative set. This proves properness at the cusp and proper discontinuity by the same uniform lower bound on nonzero lattice translations. The quotient charts are open subsets of a fixed smooth toric variety times the parameter disc, so projection to the parameter is a submersion. The explicit overlap maps are jointly holomorphic. These local families agree after shrinking cusp radii, and glue over $\mathcal{P}$. The coherent $2\mathbb{Z}$ action is properly discontinuous because its action on $\mathcal{P}$ is. Local inverse logarithms for $w = e^{-\pi i c}$ identify the quotient family with these proper local families. A.2.1 gives pairwise non-biholomorphic fibres. □

1.4 A.4. Albanese invariants and the derivative of the deformation

Proposition (A.4.1; Proved)**.** *With $\kappa_j(c) = \kappa_j^{\mathrm{part}} + c$, the Albanese elliptic curves of the reduced finite fibres are*

$$\operatorname{Alb}(S_1) = \mathbb{C}/(\mathbb{Z} + \kappa_1(c)\mathbb{Z}), \qquad \operatorname{Alb}(S_2) = \mathbb{C}/(\mathbb{Z} + \tfrac{1}{2}\kappa_2(c)\mathbb{Z}).$$

Writing $J(q) = q^{-1} + 744 + 196884q + \cdots$, their invariants are

$$j_1(w) = J(e^{-2\pi i \kappa_1^{\mathrm{part}}} w^2), \qquad j_2(w) = J(e^{-\pi i \kappa_2^{\mathrm{part}}} w).$$

These are convergent formulas because the displayed nome moduli are below one. Outside the explicitly defined countable exceptional set in the proof, equality of both Albanese invariants forces $c' - c \in 2\mathbb{Z}$. Thus the full generic Albanese cross-check of Lead 3 holds independently of the centralizer calculation.

Proof. In the product covers of A.1.2 the invariant holomorphic one-forms are exactly the multiples of $d\xi$: the moving factor is multiplied by a primitive third or fourth root. Integration thus identifies the Albanese with the quotient of the fixed line by the fixed-coordinate projections of every deck translation. These are, respectively,

$$\mathbb{Z} + \kappa_1\mathbb{Z} + \tfrac{2}{3}\mathbb{Z} + \tfrac{\kappa_1}{3}\mathbb{Z} = \tfrac{1}{3}\mathbb{Z} + \tfrac{\kappa_1}{3}\mathbb{Z}, \quad \mathbb{Z} + \kappa_2\mathbb{Z} + \tfrac{1}{2}\mathbb{Z} + \tfrac{\kappa_2}{4}\mathbb{Z} = \tfrac{1}{2}\mathbb{Z} + \tfrac{\kappa_2}{4}\mathbb{Z}.$$

Multiplication by three and two gives the stated maps of complex tori. In particular the Albanese computation includes the kernel of the product isogeny and the affine quotient, rather than just its fixed eigenspace. Using the positive-imaginary representatives $-\kappa_1$ and

$-\kappa_2/2$ in the standard modular j-series gives the formulas. At sufficiently small nome J is injective: $1/J(q) = q + O(q^2)$ has nonzero derivative at zero. Thus j_2 alone distinguishes parameters within a sufficiently small punctured subdisc when both parameters lie there. For the full generic assertion put $x = -\kappa_2(c)/2$ and $d_* = \kappa_1^{\mathrm{part}} - \kappa_2^{\mathrm{part}}$, so $-\kappa_1(c) = 2x - d_*$. The independently certified interval in `cd_special_intervals.py` gives $-0.2 < \operatorname{Im} d_* < -0.1$; in particular it is nonzero. Equality of the two elliptic invariants at c, c' gives matrices $M, N \in \mathrm{PSL}_2(\mathbb{Z})$ with $x' = M(x)$ and

$$N(2x - d_*) = 2M(x) - d_*.$$

Define the exceptional set to be the $c \in \mathcal{P}$ obtained from roots of these equations for all pairs for which the two rational functions are not identical. Each equation has at most two roots after clearing denominators, so this is a countable, explicitly specified set. For a pair giving an identity, let $C(x) = 2x - d_*$. Then $N = CMC^{-1}$. The real matrix M preserves both the real circle $\mathbb{R} \cup \{\infty\}$ and $C^{-1}(\mathbb{R} \cup \{\infty\})$, the parallel line with imaginary part $\operatorname{Im} d_*/2$ together with infinity. Their unique intersection is infinity, so M fixes infinity. An element of $\mathrm{PSL}_2(\mathbb{Z})$ fixing infinity is $x \mapsto x + b$, $b \in \mathbb{Z}$, and then $N(y) = y + 2b$. It follows that $c' - c = -2b$. This proves the generic necessity with its actual exceptional set; A.2.1 proves the necessity also on that set. □

Proposition (A.4.2; Proved)**.** *For every $c \in \mathcal{P}$, the Kodaira–Spencer class of $\partial/\partial c$ in $H^1(X_c, T_{X_c})$ is nonzero. On any fixed ordinary fibre, put $T = \operatorname{Im}\tau$, $m = \operatorname{Im}\mu$, $a = m/T$, $D = \operatorname{Im}\beta_c - 6m^2/T$. In the convention $\mathrm{KS}(\partial_c) = [\bar\partial h]$ for a smooth $(1,0)$ lift h, its restricted torus-deformation representative is*

$$-\frac{1}{2iD}\,\partial_{\zeta_2} \otimes (d\bar\zeta_2 - a\, d\bar\zeta_1).$$

Proof. Retain the same real marking $r \in \mathbb{R}^4$ and set $\zeta = \Pi_c r$. The two imaginary-part equations give $r_1 = (\operatorname{Im}\zeta_2 - a \operatorname{Im}\zeta_1)/D$. Thus at $c + \delta$ the marked real-linear identification is

$$w = \zeta + \frac{\delta}{2iD} P(\zeta - \bar\zeta), \qquad P = \begin{pmatrix} 0 & 0 \\ -a & 1 \end{pmatrix}, \qquad P^2 = P.$$

It is exactly $\Pi_{c+\delta}\Pi_c^{-1}$ as a real map, so it sends every marked lattice vector to the corresponding vector. Its determinant is $(D + \operatorname{Im}\delta)/D > 0$. The transported $(0,1)$ graph coefficient is $\nu_\delta = \delta P/(2iD + \delta)$. The derivative at zero is $P/(2iD)$. Equivalently the smooth lift of ∂_c in moving coordinates has vertical part $r_1\partial_{w_2}$ and $\bar\partial r_1 = -(d\bar\zeta_2 - a d\bar\zeta_1)/(2iD)$, giving the stated sign.

A nonzero constant tangent-valued $(0,1)$ form on a compact torus cannot be $\bar\partial$ of a global vector field. Write that field in the invariant holomorphic frame and integrate each derivative of its periodic coefficients against invariant probability volume. Every such derivative has integral zero, whereas the coefficient of $\partial_{\zeta_2} \otimes d\bar\zeta_2$ above has nonzero integral. This proves nonvanishing for the fibre deformation.

We explain its transfer to the threefold. Over the dual numbers $A = \mathbb{C}[\epsilon]/(\epsilon^2)$ at c, the chart proof of A.1.1 gives the relative line isomorphism $\omega_{X_A/A}^{-2} = f_A^*\mathcal{O}_B(p_2)$. Extra parameter differentials disappear in relative forms and all determinant characters are independent of c. The two pulled-back sections form an A-basis of $H^0(X_A, \omega^{-2})$: reduce a section modulo ϵ, subtract its linear combination of the two lifts, and use the exact sequence $0 \to L \xrightarrow{\epsilon} L_A \to L \to 0$ to write the remainder uniquely as ϵ times another such combination. The same argument proves independence.

A first-order trivialization would preserve this complete system and give a base automorphism reducing to the identity. It fixes all three special values even over A. On the

finite etale charts the relative critical ideal is $(s_j^{m_j-1})$, and $t_j = s_j^{m_j}$ is zero on that scheme; its constant parameter ring A injects. At a cusp triple chart, the critical ideal of $t_c = z_0z_1z_2$ is (z_0z_1, z_0z_2, z_1z_2) and has the same property. There are no other critical values. Therefore an infinitesimal displacement of a special value would remain nonzero in its critical chart, a contradiction. Writing a base infinitesimal automorphism as $t + \epsilon(b_0 + b_1t + b_2t^2)$, fixing $0, 1, \infty$ gives $b_0 = b_1 = b_2 = 0$. The trivialization would consequently trivialize the displayed nonzero torus deformation. Finally a zero Kodaira–Spencer class produces such a first-order trivialization by subtracting from local product charts the vector fields whose coboundary is the differentiated transition cocycle. This proves the claim. □

Proposition (A.4.3; Proved)**.** *The non-isotriviality argument of Viaclovsky, [Via26b, Proposition 4.7], follows for the present family from A.4.2 and A.1.1. His published countability argument proves local non-isotriviality and positive Kuranishi dimension; the computation in A.4.2 additionally proves nonzero derivative at every admissible parameter. The arithmetic method of Yang, [Yan26, Lemmas 6.6–6.9 and Proposition 6.15], specializes to the same centralizer, finite-norm and collar-gluing operations after replacing his displayed matrices by A_1, A_2; those replacements are evaluated in A.2.1 and A.3.1.*

Proof. On a fixed ordinary fibre, the three last periods remain fixed and span $\mathbb{C}^2$, whereas the first changes by δe. A marked complex-linear identification fixing those three periods must be the identity, so the marked period map is nonconstant. The set of integral markings of one torus is countable. A holomorphic nonconstant map from a disc has uncountable image; hence the unmarked fibre family is locally non-isotrivial. A.1.1 transfers this to total spaces. This is exactly the implication used in the cited Proposition 4.7, whose statement and proof we have read. A nonconstant classifying map gives positive dimension of its local target. It does not by itself determine the derivative at the base point, as the explicit holomorphic map $z \mapsto z^2$ has nonconstant image and derivative zero at zero. A.4.2 computes the derivative here.

Yang's cited lemmas use an integral commuting lattice map, an affine fibre lift, normalization of the root base change, and a norm-lattice defect. We have exhibited the respective maps and norm equations for the present matrices. His Proposition 6.15 obtains $6/r$ from the first-column scaling and its order-six norm image. Here the first-column scaling is one and the two computed norm images impose parity, yielding two. The common proof operations transfer through these explicitly stated matrix substitutions; no conclusion about one family is inferred merely from the label of the other. His nonzero absolute Kodaira–Spencer conclusion is obtained in the present setting by the explicit computation and intrinsic-pencil transfer above. □

1.5 A.5. The assembled theorem

Theorem (A.5.1; Proved)**.** *Fix the original cusp-regular period solution, the clockwise peripheral marking, the prescribed periodic triangular cusp fan, the affine vectors $(\varepsilon, -\varepsilon')$, and the untwisted cusp collar. For every $c, c' \in \mathcal{P}$, the constructed compact complex threefolds satisfy*

$$X_c \simeq X_{c'} \quad \Longleftrightarrow \quad c' - c \in 2\mathbb{Z}.$$

Every biholomorphism is over the identity of the original $\mathbb{P}^1$. The proper holomorphic family of their isomorphism classes has parameter $0 < |w| < e^{-\pi M}$ and nonzero Kodaira–Spencer class at every point.

Proof. A.1.1 makes the base intrinsic. A.2.1 gives the necessary parity and excludes the negative linear part. A.3.1 supplies the inverse maps, all collar identities, and descent of the jointly holomorphic family. A.4.2 proves its infinitesimal effectiveness. Finally $|e^{-\pi ic}| = e^{\pi \operatorname{Im} c}$

and equality of exponentials is exactly difference in $2\mathbb{Z}$, so the coordinate has precisely the stated image and fibres. □

Theorem (A.5.2; Proved)**.** *For every admissible c, the tangent cohomology of the actual compact member is*

$$(h^0(X_c,T_{X_c}),h^1(X_c,T_{X_c}),h^2(X_c,T_{X_c}),h^3(X_c,T_{X_c}))=(1,1,1,0).$$

Its Kuranishi germ is a smooth reduced disc. The family of A.5.1 is miniversal at every parameter: its Kodaira–Spencer map is an isomorphism, and every sufficiently small deformation is induced from this family.

Proof. We give the calculation, including the singular-fibre maps. The corresponding calculation appears in the historical first version [DGKX26a, Section 6, especially Proposition 6.10, Lemma 6.11, Theorem 6.12 and Proposition 6.16]; those deformation statements are omitted from version two. We use the verified maps and cohomology calculation below, rather than infer their validity from the earlier theorem label.

The base and its line bundles. Write $X=X_c$ and form the analytic root curve

$$\mathfrak{B}=\sqrt[3]{(B,p_1)}\times_B\sqrt[4]{(B,p_2)},\qquad \rho:\mathfrak{B}\longrightarrow B.$$

A root object is a line bundle with section whose indicated power is the pullback of the section cutting out p_j. The actual equalities $f^*p_1=3S_1$ and $f^*p_2=4S_2$ supply these objects as $(\mathcal{O}_X(S_j),1_{S_j})$, and hence supply a morphism $\bar f:X\to\mathfrak{B}$ with $\rho\bar f=f$. The explicit weighted quotient

$$\mathfrak{B}\simeq[(\mathbb{C}^2\setminus\{0\})/\mathbb{C}^*],\qquad \lambda(x,y)=(\lambda^3x,\lambda^4y),\qquad t=\frac{y^3}{y^3-x^4},$$

identifies the roots with $(\mathcal{O}(4),y)$ and $(\mathcal{O}(3),x)$ and the ordinary point p_0 with $y^3-x^4=0$. In particular $\rho^*\mathcal{O}_B(1)=\mathcal{O}(12)$. This isomorphism follows on each root chart by the displayed root sections; away from their zeros both sides have the same ordinary coarse coordinate, so the chart isomorphisms glue.

For completeness the analytic line-bundle computation can be made on the coarse curve. Removing the two stabilizer characters from any line bundle makes it descend to B: on each root chart, averaging a local frame whose central value is nonzero gives an invariant frame still nonzero at the centre, and its invariant coefficients are holomorphic in $s_j^{m_j}$. These frames give the descended line and its transition functions. Consequently its Picard group has generators H,R_1,R_2 and relations $3R_1=H=4R_2$. Thus it is freely generated by R_1-R_2, denoted $\mathcal{O}(1)$; $H=12\mathcal{O}(1)$, $R_1=4\mathcal{O}(1)$ and $R_2=3\mathcal{O}(1)$. The chart descent just used is exact because invariants of a finite group over $\mathbb{C}$ are an exact functor. With the clockwise root generators, a local invariant section of $\mathcal{O}(k)$ has vanishing exponents

$$r_3(k)\in\{0,1,2\},\quad r_3(k)\equiv k\pmod 3,\qquad r_4(k)\in\{0,1,2,3\},\quad r_4(k)\equiv -k\pmod 4.$$

Dividing by those local powers gives the precise coarse direct image

$$\rho_*\mathcal{O}(k)=\mathcal{O}_B(d(k)),\qquad d(k)=\frac{k}{12}-\frac{r_3(k)}{3}-\frac{r_4(k)}{4}. \tag{1}$$

Indeed $k-4r_3-3r_4=12d(k)$ identifies the line bundle with $\rho^*\mathcal{O}_B(d(k))\otimes R_1^{r_3}\otimes R_2^{r_4}$. The canonical root sections generate the invariant direct image of the last two factors; division by their local powers is the displayed line-bundle isomorphism on the coarse charts. Finite invariants also give $R^q\rho_*=0$ for $q>0$, as seen on each Stein quotient chart. For $k=0$ the degree is zero; for $k=1,-1,-2,-5,-6$ it is -1. It follows directly from the cohomology of

$\mathbb{P}^1$ that these five line bundles are acyclic, whereas $H^0(\mathcal{O}) = \mathbb{C}$ and $H^1(\mathcal{O}) = 0$. The local equation $dt_j = m_j s_j^{m_j-1} ds_j$ gives

$$K_{\mathfrak{B}} = \mathcal{O}(-24 + 2 \cdot 4 + 3 \cdot 3) = \mathcal{O}(-7), \qquad T_{\mathfrak{B}}(-p_0) = \mathcal{O}(-5).$$

The actual base changes and cusp cohomology. On a finite root disc Δ_{s_j} the local presentations are $[J_j'/\mu_{m_j}] \to [\Delta_{s_j}/\mu_{m_j}]$. The fibre-product identity

$$[J_j'/\mu_{m_j}] \times_{[\Delta_{s_j}/\mu_{m_j}]} \Delta_{s_j} \simeq J_j'$$

is induced by the quotient torsor and its equivariant map to the disc; its inverse sends a torsor with a chosen root-chart trivialization to the corresponding point of J_j'. The action on J_j' is free by A.1.2. Thus this is an etale atlas base change and is exactly the smooth torus family, with its given deck generator. Over the cusp the base change is the original cusp family. Locally it is $q = z_0 z_1 z_2$; the homomorphism $\mathbb{C}\{q\} \to \mathbb{C}\{z_0, z_1, z_2\}$ is flat because its target is torsion-free over the discrete valuation ring. These charts prove that $\bar{f}$ is representable, proper and flat, with connected Gorenstein fibres of dimension two. Properness on the cusp is the compact representative construction of A.3.1, and on the finite charts it is the compact torus quotient. Also $\bar{f}^* p_0 = W$ as reduced Cartier divisors.

Let $\nu : E \to W$ be the normalization, $E = \mathrm{dP}_6$, let C_h, C_v, C_d be the three double-curve normalizations, and let P_+, P_- be the two triple points. Each C_e is $\mathbb{P}^1$. Write $\rho_e^0, \rho_e^1 : C_e \to E$ for its two paired boundary maps and $j_e = \nu\rho_e^0 = \nu\rho_e^1$. The sheaf sequence is

$$0 \longrightarrow \mathcal{O}_W \longrightarrow \nu_* \mathcal{O}_E \xrightarrow{\delta^0} \bigoplus_{e=h,v,d} j_{e*} \mathcal{O}_{C_e} \xrightarrow{\delta^1} \mathbb{C}_{P_+} \oplus \mathbb{C}_{P_-} \longrightarrow 0, \tag{2}$$

where $\delta^0(s)_e = (\rho_e^1)^* s - (\rho_e^0)^* s$ and δ^1 evaluates $t_h - t_v + t_d$ at the two ordered triple points. Its exactness is local: for $xyz = 0$ it is the alternating restriction complex of the three coordinate planes, their pairwise lines and their common point. Decomposing convergent power series by the variables occurring in each monomial gives the augmented simplex complex, which is exact. The one- and two-branch cases are the same calculation with fewer planes. This verifies exactness for the actual boundary maps, including the torus multipliers in their gluing.

All positive-degree structure-sheaf cohomology of E, C_e vanishes. Consequently (2) computes cohomology by

$$\mathbb{C} \xrightarrow{0} \mathbb{C}^3 \xrightarrow{\begin{pmatrix} 1 & -1 & 1 \\ 1 & -1 & 1 \end{pmatrix}} \mathbb{C}^2.$$

It gives $(h^0, h^1, h^2)(\mathcal{O}_W) = (1, 2, 1)$. We also need its multiplication. The branch-intersection resolution is the augmented semisimplicial sheaf algebra with its restriction face maps; its local exactness makes the augmentation a multiplicative quasi-isomorphism. Its Alexander–Whitney product therefore computes the sheaf-cohomology cup product. In the ordered triangles $[0, e_1, e_2]$ and $[e_1, e_2, e_1 + e_2]$ it is

$$\alpha \smile \beta = (\alpha_h \beta_d, \alpha_d \beta_h), \qquad [\alpha] \smile [\beta] = \alpha_h \beta_d - \alpha_d \beta_h \quad \text{in } \mathbb{C}^2/\mathbb{C}(1,1).$$

The cocycles $(1, 1, 0)$ and $(0, 1, 1)$ pair to one. Thus $\bigwedge^2 H^1(\mathcal{O}_W) \to H^2(\mathcal{O}_W)$ is an isomorphism, including its product rather than just its dimensions.

Vertical fields and the direct images. Set $\mathcal{T} = T_X(-\log W)$ and $\mathcal{V} = \ker(d\bar{f}^{\log})$. The vertical transitions on the core are precisely $R_g(z)$. Finite collar translations have vertical differential the identity, and at the cusp the frame ∂_{ζ_i} corresponds to $2\pi i x_i \partial_{x_i}$. The cusp transformations $x \mapsto u(q) q^{B_0 \lambda} x$ fix the coefficients of these relative logarithmic fields. These exact identifications construct a bundle $\mathcal{E}$ on $\mathfrak{B}$ with $\mathcal{V} = \bar{f}^* \mathcal{E}$. The fixed vector $e = (0, 1)^t$ is nonzero in every logarithmic frame, so there is an exact sequence

$$0 \longrightarrow \mathcal{O} \xrightarrow{e} \mathcal{E} \longrightarrow L \longrightarrow 0.$$

The factors of L are $-1/\tau$ and $1/\tau$. Therefore the usual weight-twelve modular discriminant makes $1/\Delta(\tau)$ a meromorphic section of L^{12}. It has no finite zero or pole, and its cusp pole is simple: $\Delta(\tau) = q\,u(q)$ with $u(0) \neq 0$. Hence $L^{12} = \mathcal{O}(-p_0) = \mathcal{O}(-12)$. The Picard calculation gives $L = \mathcal{O}(-1)$, and the extension splits since $H^1(\mathcal{O}(1)) = 0$. Thus

$$\mathcal{E} \simeq \mathcal{O} \oplus \mathcal{O}(-1).$$

There is also the exact sequence

$$0 \longrightarrow \bar{f}^*\mathcal{E} \longrightarrow \mathcal{T} \longrightarrow \bar{f}^*\mathcal{O}(-5) \longrightarrow 0. \tag{3}$$

On the smooth root charts it is the relative tangent sequence. At the cusp its last map sends $\sum a_i z_i \partial_{z_i}$ to $(\sum a_i) q\partial_q$, so it is onto and its kernel is the two height-zero lattice directions already used above. Taking its determinant and using $W = \bar{f}^* p_0$ gives the actual relative dualizing line

$$\omega_{X/\mathfrak{B}} = \bar{f}^*(\det \mathcal{E})^{-1}. \tag{4}$$

In particular this computation gives $K_X = \bar{f}^*\mathcal{O}(-6) = \mathcal{O}_X(-2S_2)$, with the last identification supplied by $\bar{f}^* R_2 = \mathcal{O}_X(S_2)$; it agrees through this actual pullback with A.1.1.

The fibres of $\bar{f}$ all have structure-sheaf cohomology dimensions $(1, 2, 1)$: on ordinary and finite root charts they are tori, and at the cusp we have just computed them. Analytic proper flat base change on these complex-space charts gives locally free $R^q\bar{f}_*\mathcal{O}_X$ with these fibres. Its canonical comparison maps agree on the atlas double overlaps, so they descend to $\mathfrak{B}$. The unit gives $R^0\bar{f}_*\mathcal{O}_X = \mathcal{O}$. The source charts are smooth and the fibres are Cartier divisors, so relative analytic duality applies in its Gorenstein form. The relative trace pairing, compatible on the same overlaps, and (4) give

$$(R^2\bar{f}_*\mathcal{O}_X)^\vee = \bar{f}_*\omega_{X/\mathfrak{B}} = (\det \mathcal{E})^{-1}, \qquad R^2\bar{f}_*\mathcal{O}_X = \mathcal{O}(-1).$$

Here local freeness removes the higher sheaf-Ext terms in the duality spectral sequence. This is precisely the pointwise base-change and top-degree duality application spelled out in the cited Lemmas 6.8–6.9.

Put $G = R^1\bar{f}_*\mathcal{O}_X$. Cup product and the verified fibre comparisons show $\det G = R^2\bar{f}_*\mathcal{O}_X = \mathcal{O}(-1)$; at W this uses the nonzero pairing above. We construct the required subline of G. On an ordinary fibre, constant lattice cocycles identify

$$H^1(F_t, \mathcal{O}_{F_t}) = \Lambda_{\mathbb{C}}^* / \operatorname{rowspan}_{\mathbb{C}} \Pi_c.$$

Indeed a coboundary among constant cocycles is given by an entire function with constant period differences. Its derivatives are periodic holomorphic functions on the compact torus, hence constant; the function is affine linear. This gives exactly the displayed kernel, and both sides then have dimension two. The class of $\gamma = (1, 0, 0, 0)$ is nonzero, since evaluation on columns $\widehat{w}, \widehat{\delta}$ would force both coefficients of a putative period-row combination to vanish. The matrices satisfy $\gamma A_j = \gamma$. Translations act trivially on $H^1(\mathcal{O})$, so this holomorphic class extends on both finite root charts and respects their gluing.

At the cusp γ kills $\Lambda_{\mathrm{tor}} = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle$. For local sections of the free toric quotient with changes $s_i = \Psi_{\lambda_{ij}} s_j$, the constants $\gamma(\lambda_{ij})$ therefore form a holomorphic Cech cocycle on the entire neighbourhood. With $B_0 = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, the edges $e_1, e_2, e_2 - e_1$ have deck vectors $(0, 1), (-1, 0), (-1, -1)$. The central cocycle values are consequently $(0, -1, -1)$, which pair to one with $(1, 0, -1)$. Its value on W is nonzero. On the punctured overlap the exponential map kills precisely Λ_{tor}, so its full-lattice cocycle is exactly $\lambda \mapsto \gamma(\lambda)$, proving equality with the preceding class. Over the Stein cusp disc, the Leray edge map $H^1(\bar{f}^{-1}D_0, \mathcal{O}_X) \to H^0(D_0, G)$ is

an isomorphism, since $H^1(D_0,\mathcal{O}) = H^2(D_0,\mathcal{O}) = 0$. Thus this is a holomorphic, nonvanishing extension of $[\gamma]$. It supplies a global subline and exact sequence

$$0 \longrightarrow \mathcal{O} \xrightarrow{[\gamma]} G \longrightarrow \mathcal{O}(-1) \longrightarrow 0.$$

Again $H^1(\mathcal{O}(1)) = 0$ splits it. We have proved

$$R^q\bar{f}_*\mathcal{O}_X = \begin{cases} \mathcal{O}, & q = 0, \\ \mathcal{O} \oplus \mathcal{O}(-1), & q = 1, \\ \mathcal{O}(-1), & q = 2. \end{cases} \tag{5}$$

From logarithmic to ordinary tangent cohomology. The projection formula gives, for $q = 0, 1, 2$ respectively,

$$R^q\bar{f}_*\mathcal{V} : \quad \mathcal{O} \oplus \mathcal{O}(-1), \quad \mathcal{O} \oplus 2\mathcal{O}(-1) \oplus \mathcal{O}(-2), \quad \mathcal{O}(-1) \oplus \mathcal{O}(-2).$$

For $\bar{f}^*\mathcal{O}(-5)$ the three expressions are $\mathcal{O}(-5)$, $\mathcal{O}(-5) \oplus \mathcal{O}(-6)$ and $\mathcal{O}(-6)$. Formula (1) makes every nontrivial summand in this list acyclic. Leray, whose base has cohomological dimension one, therefore gives $h^\bullet(\mathcal{V}) = (1, 1, 0, 0)$ and zero cohomology for the horizontal term. Sequence (3) yields $h^\bullet(\mathcal{T}) = (1, 1, 0, 0)$.

The normalization map $\nu : E \to X$ is an immersion on each local branch. Let $N_\nu = \operatorname{coker}(T_E \to \nu^* T_X)$. The exact normal-derivative map is

$$0 \longrightarrow \mathcal{T} \longrightarrow T_X \longrightarrow \nu_* N_\nu \longrightarrow 0. \tag{6}$$

For a local equation $z_0 \cdots z_{r-1} = 0$, it sends $\sum b_i \partial_{z_i}$ to the r independent normal coefficients $b_i|_{z_i=0}$. These can be lifted independently, proving surjectivity; the kernel is exactly the logarithmic tangent module. This also specifies the global map under changes of branch charts. The cusp three-form

$$\Omega = d\log x_1 \wedge d\log x_2 \wedge dq$$

is invariant under $x \mapsto u(q)q^{B_0\lambda}x$ because the extra logarithmic differentials are multiples of dq. In any unimodular maximal cone the height-one equation is $q = z_0 z_1 z_2$, and the determinant-one character change gives $\Omega = \pm dz_0 \wedge dz_1 \wedge dz_2$. It therefore trivializes K_X on the entire cusp neighbourhood, including every branch intersection. Taking determinants of the immersion tangent sequence now gives the explicit line identification $N_\nu = K_E \otimes (\nu^* K_X)^{-1} \simeq K_E$. Since E is the rational projective surface dP_6, Serre duality gives $h^\bullet(E, K_E) = (0, 0, 1)$. The cohomology sequence of (6) proves $h^\bullet(X, T_X) = (1, 1, 1, 0)$, as asserted.

The reduced deformation germ. A.4.2 gives a nonzero Kodaira–Spencer vector at this same c; the just computed $h^1 = 1$ makes its map an isomorphism. Embed the actual Kuranishi germ in its one-dimensional tangent space and write its local analytic algebra as $\mathbb{C}\{t\}/I$. The existing proper family has a local classifying map, whose pullback sends t to $g(s)$ with $g'(0) \neq 0$. The holomorphic inverse-function theorem makes $\mathbb{C}\{t\} \to \mathbb{C}\{s\}$, $t \mapsto g(s)$, an isomorphism. Every element of I maps to zero, so $I = 0$. This proves smoothness and reducedness even though $h^2(T_X) = 1$. The classifying map is consequently a local isomorphism of germs; completeness of the Kuranishi deformation then gives the claimed miniversality and exhaustion of sufficiently small deformations. □

Scope and a precise changed-input test. The hypotheses of A.5.1–A.5.2 fix the original period system and multiplicities $(3, 4)$; B.2 treats its arbitrary affine and cusp data, and B.3 classifies its stipulated higher-multiplicity variants. A concrete extension beyond these hypotheses begins with $y^2 = x^3 + tx + 1$ in [Via26c, Sections 2–3 and Proposition 6.1]. Already

one rank-one filling over a specified root p_1 of $4t^3+27=0$ replaces the hexagonal normalization calculation by

$$E' = \mathbb{C}^*/\langle c_0\rangle, \qquad R = \mathbb{P}_{E'}(\mathcal{O}_{E'} \oplus L),$$
$$R = (\mathbb{P}^1 \times \mathbb{C}^*)/\langle (z,v) \mapsto (\gamma_0 z, c_0 v)\rangle.$$

with its boundary sections identified by $v \mapsto \beta_0 v$. Here c_0, γ_0, β_0 are exactly the nonzero central multipliers in that paper's equation (3.1). The invariant nodal volume form there trivializes the ambient canonical line, so the branch-normal map has quotient $\nu_* K_R$ by the same local coefficient map as (6). The ruling has $\pi_*\mathcal{O}_R = \mathcal{O}_{E'}$ and $R^1\pi_*\mathcal{O}_R = 0$; Leray and Serre duality therefore give $h^\bullet(K_R) = (0,1,1)$. The first unevaluated global datum for transferring the deformation argument is the connecting map from this single $H^1(R,K_R) = \mathbb{C}$ summand into $H^2(T_X(-\log D))$, where D is the union of the three reduced singular fibres of that model. Its direct attack is to compute the boundary-translation Cech cocycle using the full holomorphic unit functions in that equation (3.1), their central values (γ_0, c_0, β_0), and the corresponding vertical frames. The global identification argument also requires the integral centralizer and collar defects of that model's four monodromy matrices. These are specified substitutions to perform for the stated one-fibre test; no classification or deformation conclusion for that changed period system is included in A.5.1–A.5.2.

2 B. Discrete data and the geometric parameter

2.1 B.1. A typed fibre dictionary

Proposition (B.1.1; Proved)**.** *Let $E_\tau = \mathbb{C}/(\mathbb{Z}+\tau\mathbb{Z})$, $P=[\mu]$, and $L_\mu = \mathcal{O}_{E_\tau}(P-O)$. The holomorphic map*

$$[\xi,\eta] \longmapsto [\xi, e^{2\pi i\eta}]$$

identifies $\mathbb{C}^2/\Pi_c\Lambda$ with the quotient of $L_\mu^\times$ by the lift $A(\xi,W) = (\xi+6\mu, e^{2\pi i\beta_c}W)$. Engel's generator, in Definition 2.3 of the attached paper, is its inverse, with translation $-6P$ and multiplier $e^{-2\pi i\beta_c}$. The section $P=[\mu]$ is a generator of the Mordell–Weil group of the stated pointed Weierstrass surface.

Proof. First quotient by columns two, three and four. Exponentiation kills column four; columns three and two then give

$$(\xi,W) \sim (\xi+1,W) \sim (\xi+\tau, e^{2\pi i\mu}W).$$

For the odd theta function with periods $1,\tau$, the quotient $\theta(\xi-\mu)/\theta(\xi)$ is a meromorphic section of this line bundle: its factors are $1, e^{2\pi i\mu}$ and its divisor is $P-O$. Sending its divisor section to this quotient is the asserted line-bundle isomorphism. Quotienting the remaining first column gives exactly A. On sufficiently small simply connected charts the logarithm of W is a holomorphic inverse; changes of logarithm are column-four periods. Thus the displayed map is a biholomorphism of the two quotient complex spaces, with its inverse explicitly defined by local logarithms. Choosing A^{-1} changes the generator of the same infinite cyclic group and gives precisely Engel's declared inverse-translation convention.

For clarity the elliptic surface in this dictionary is the displayed cubic, rather than only a curve with the same j-value. Let f_{sc} be the branch satisfying

$$f_{\mathrm{sc}}^4 = \frac{12t^3(t-1)}{g_2(\tau)}, \qquad f_{\mathrm{sc}}^6 = -\frac{8t^4(t-1)^2}{g_3(\tau)}, \qquad f_{\mathrm{sc}}/t \longrightarrow i\sqrt{3}/\pi.$$

The identities $g_2^3/(g_2^3-27g_3^2) = t$ make these equations compatible. Their zero and pole orders give a holomorphic branch on the period cover; the indicated cusp sign fixes it. The map

$\xi \mapsto (f_{\mathrm{sc}}^2 \wp_\tau(\xi), \frac{1}{2} f_{\mathrm{sc}}^3 \wp'_\tau(\xi))$ is a pointed isomorphism to $y^2 = x^3 - 3t^3(t-1)x + 2t^4(t-1)^2$. Its factors are $f_{\mathrm{sc}}(g_1 z) = -f_{\mathrm{sc}}/\tau$, $f_{\mathrm{sc}}(g_2 z) = f_{\mathrm{sc}}/\tau$; therefore this isomorphism descends over B°.

We first verify that $P = [\mu]$ gives a rational section of the resolved elliptic surface. On B° its cubic coordinates descend by the displayed scale and affine laws. At the two finite centres, μ_j is not a lattice point, while f_{sc} has orders two and one in the root parameter. Its cubic coordinates are therefore holomorphic and invariant, hence holomorphic in the base coordinate. At the cusp μ is holomorphic and bounded, and the convergent multiplicative expansion of $\wp$ gives meromorphic cubic coordinates with finite pole orders. Even when its exponential meets one, the pole is controlled by its finite vanishing order; it cannot be identically one because the two finite values of μ differ. The coordinates are thus meromorphic on $\mathbb{P}^1$ and rational. The closure of its graph on the proper resolved surface is an irreducible curve; its normalization maps finitely with degree one to the smooth base, so is the base itself. This gives the unique section of that surface extending P.

We also verify that the specified section is a Mordell–Weil generator using its full logarithm. A rational section of this cubic pulls back to a holomorphic section of the good elliptic family at both finite covering points: multiply its projective coordinates by a common power to remove poles, and use the nonsingular limiting Weierstrass equation. At the cusp put $v = 1/t$, $X = v^2 x, Y = v^3 y$. Its equation is $Y^2 = X^3 - 3(1-v)X + 2(1-v)^2$. A section cannot pass through the node $(1, 0)$: substitution of $X = 1 + O(v)$, $Y = O(v)$ leaves the coefficient $+1$ of v on the left-hand side of the defining equation. At every other central point the multiplicative uniformization is a local isomorphism. Explicitly, at $v = 0$ it is

$$X = 1 + \frac{12U}{(U-1)^2}, \qquad Y = \frac{12\sqrt{3}U(U+1)}{(U-1)^3}, \qquad U = e^{2\pi i \xi},$$

with inverse $U = (Y/(X-1) + \sqrt{3})/(Y/(X-1) - \sqrt{3})$ away from O; a local parameter at O is X/Y. The convergent lattice expansion of $\wp$ gives the same holomorphic map for small v. Hence a section has a holomorphic unit $U(v)$ and a bounded logarithm at the cusp. Lifting the section on the simply connected cover gives a global logarithm ν, with this cusp branch. Its only remaining ambiguity is an integer, since a nonzero multiple of τ is unbounded.

Its two affine laws have integral defects:

$$\nu(g_1 z) = -\nu/\tau + \alpha_1 + \beta_1 \tau(g_1 z), \quad \nu(g_2 z) = \nu/\tau + \alpha_2 + \beta_2 \tau(g_2 z).$$

Composing them and using cusp invariance gives $\alpha_2 = -\beta_1$, $\beta_2 = \alpha_1 + \beta_1$. Put $n = -\alpha_1 - 2\beta_1$ and $a = \alpha_1 + \beta_1$. Substitution shows that $n\mu + a$ has exactly these defects. Their difference is a global section of the first-coordinate sheaf $\mathcal{O}_B(-p_0)$ from A.3.1, so is zero. Every rational section is therefore nP. The same cusp argument and the distinct fixed values of μ show that P has infinite order. This proves $\mathrm{MW}(S) = \mathbb{Z}P$. □

Proposition (B.1.2; Proved)**.** *The exact set of nonzero parameters for which every ordinary fibre and both normalized finite root fibres of Engel's period-reference quotient are compact complex two-tori is*

$$0 < |u| < e^{-2\pi M}.$$

This is the parameter domain for the compact family with both smooth finite root models. The condition on ordinary fibres alone at equality is determined by whether a maximizing point of D_{part} is ordinary; this boundary issue is treated in Part C. For an independent reference, the radius is $|\lambda_0| e^{-2\pi M}$.

Proof. The function $N(\xi, W) = |W| \exp(2\pi \operatorname{Im}\mu \operatorname{Im}\xi / \operatorname{Im}\tau)$ is invariant under both line-bundle period identifications. The inverse lift multiplies it by $e^{2\pi D_c}$. More directly the real determinant of the four period columns is a nonzero sign times $(\operatorname{Im}\tau) D_c$. Thus the four

columns generate a full lattice exactly when $D_c \neq 0$, and then the quotient is a compact torus by a closed fundamental parallelepiped. At $D_c = 0$ their real span has dimension three; a rank-four lattice in $\mathbb{R}^4$ cannot result, so the quotient cannot be a compact complex two-torus. The function D_{part} descends continuously to the connected base with the cusp removed and tends to $-\infty$ at the cusp. Its image is therefore $(-\infty, M]$: the maximum is attained, and the intermediate value theorem fills all smaller values. Consequently D_c is nonzero at every base point exactly when $\operatorname{Im} c < -M$. Since $|u| = e^{2\pi \operatorname{Im} c}$, the assertion follows, with the indicated reference scaling. □

Proposition (B.1.3; Proved)**.** *For the original affine logarithm μ, we have $\wp_\tau(\mu) = 0$ identically. Equivalently the generator $P = [\mu]$ of B.1.1 is one of the two sections $P_* = (0, \sqrt{2}t^2(t-1))$ and $-P_*$. This assertion does not use a numerical evaluation of μ or of β.*

Proof. We give the intersection calculation that identifies the generator. At $t = 1$, put $u = t - 1$ and blow up the origin. In the u-chart $x = uX, y = uY$, the strict equation is $Y^2 = uX^3 - 3(1+u)^3X + 2(1+u)^4$. Its exceptional curve is smooth, and P_* meets it at $(X, Y) = (0, \sqrt{2})$. In the other chart $u = xU, y = xY$, the equation is $Y^2 = x - 3(1+xU)^3U + 2(1+xU)^4U^2$. The implicit function theorem solves for x at the common point; the two fibre branches $U = 0, x = 0$ meet with intersection multiplicity two. These charts cover the tangent conic and resolve the singularity. The two components have multiplicity one; the one met by P_* is the nonidentity component D, with $D^2 = -2$ from intersection with the full fibre.

At $t = 0$, two blow-ups in the successive t-charts give

$$Y^2 = tX^3 + 3t^2(1-t)X + 2t^2(1-t)^2, \quad Y_2^2 = t^2X_2^3 + 3t(1-t)X_2 + 2(1-t)^2.$$

The second equation has two smooth multiplicity-one components $E_{2,\pm}$ at $Y_2 = \pm\sqrt{2}$, and P_* meets one of them. The remaining singular point, in the second X-chart, has

$$V^2 = H(X,T), \quad H = X^2T + (2+3X)T^2 - (4X+3X^2)T^3 + 2X^2T^4, \qquad t = XT.$$

Let $T_0(X)$ solve $H_T(X, T_0) = 0$. It has $T_0 = -X^2/4 + O(X^3)$. Taylor's formula gives $H - H(X, T_0) = (T - T_0)^2 h_0$, $h_0(0,0) = 2$, and $H(X, T_0) = X^4 e(X)$, $e(0) = -1/8$. These identities follow by substitution in the displayed polynomial, not by discarding its higher terms. With $Z = (T - T_0)\sqrt{h_0}$, $a = V - Z$, $b = (V + Z)/e(X)$, the equation becomes exactly $ab = X^4$ through a local analytic isomorphism. Its resolution has charts

$$a = U_i^{i+1}V_i^i, \quad b = U_i^{3-i}V_i^{4-i}, \quad X = U_iV_i \quad (0 \le i \le 3),$$

with changes $U_{i+1} = 1/V_i, V_{i+1} = U_iV_i^2$. They give a chain of three (-2)-curves. These are the charts of the two successive point blow-ups resolving $ab = X^4$, so the resolution is proper. Retaining $t = XT$ in the comparison gives multiplicities $2, 3, 2$ on that chain. The two new multiplicity-one components meet its ends; the first exceptional curve meets its middle component, and the original strict fibre meets the other end of that first exceptional curve. All these statements follow by substituting the displayed monomials and their orders; the only fibre components of multiplicity one are the strict component and $E_{2,\pm}$.

Order the six nonidentity components by the graph $\Theta_1 - \Theta_2 - \Theta_3 - \Theta_4$ and $\Theta_2 - \Theta_5 - \Theta_6$, with $E_{2,\pm} = \Theta_4, \Theta_6$. Their intersection matrix is $-C$, where

$$C = \begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & -1 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & -1 & 2 & 0 & 0 \\ 0 & -1 & 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}, \quad C^{-1} = \frac{1}{3}\begin{pmatrix} 6 & 9 & 6 & 3 & 6 & 3 \\ 9 & 18 & 12 & 6 & 12 & 6 \\ 6 & 12 & 10 & 5 & 8 & 4 \\ 3 & 6 & 5 & 4 & 4 & 2 \\ 6 & 12 & 8 & 4 & 10 & 5 \\ 3 & 6 & 4 & 2 & 5 & 4 \end{pmatrix}.$$

Multiplication verifies this inverse. Self-intersections follow as well from the complete multiplicity vector and $F \cdot \Theta_i = 0$.

The form $dx \wedge dt/y$ is holomorphic and nonvanishing on the finite resolved charts. In the last charts its expression is a unit times $dX \wedge da/a = -dU_i \wedge dV_i$; the earlier blow-up equations cancel the apparent denominators at $y = 0$. At infinity it is $-v^{-1}dX \wedge dv/Y$ and has one fibre pole. Thus $K_S = -F$. Adjunction gives $Q^2 = -1$ for every section Q, since $F \cdot Q = 1$. A section meets a unique multiplicity-one component transversely at a smooth fibre point: the pullback of the full base equation to the section has order one, excluding all other possibilities.

Let v_Q be the intersection vector with Θ_i and let $d_Q = Q \cdot D$. The possibilities are $v_Q = 0, e_4, e_6$ and $d_Q = 0, 1$. Define the rational divisor class

$$\varphi(Q) = Q - O - (Q \cdot O + 1)F + \sum \Theta_i (C^{-1} v_Q)_i + \tfrac{1}{2} d_Q D.$$

It is orthogonal to O, F, Θ_i, D by direct multiplication. The chord-and-tangent rational function on the generic cubic has divisor $(Q) + (R) - (Q + R) - (O)$; extending it to the surface adds only vertical components. Applying the displayed orthogonal projection therefore proves $\varphi(Q + R) = \varphi(Q) + \varphi(R)$. Expansion gives

$$h(Q) = -\varphi(Q)^2 = 2 + 2Q \cdot O - v_Q^t C^{-1} v_Q - \tfrac{1}{2} d_Q.$$

For $Q \neq O$, the nonnegative intersection $Q \cdot O$ gives $h(Q) \geq 2 - 4/3 - 1/2 = 1/6$. The visible P_* is disjoint from O everywhere, and the explicit blow-up charts place it on both nonidentity components. Hence $h(P_*) = 1/6$. The logarithm argument in B.1.1 already proved $P_* = nP$ for an integer $n \neq 0$, without identifying the visible section. Additivity gives $1/6 = n^2 h(P) \geq n^2/6$, so $n = \pm 1$. The pointed cubic map in B.1.1 has $x(P) = f_{\mathrm{sc}}^2 \wp_\tau(\mu)$ with $f_{\mathrm{sc}} \neq 0$ on the ordinary locus. This proves the asserted zero identity. □

Proposition (B.1.4; Proved)**.** *The line-bundle comparison of the period family with Engel's family over B° transports its inverse period lift to a global section of* $\mathrm{Hom}(t_{6P}^* M_S, M_S)$ *with divisor F_∞. Take this transported section for $c = 0$ as the reference linearization. Then $\lambda_0 = 1$ exactly and $u = e^{-2\pi i c} = w^2$. For another independently fixed reference the precise constant is its scalar ratio with this section. With the cusp branch fixed in Part C and $f_{\mathrm{sc}}/t \to i\sqrt{3}/\pi$, the section is $P = (0, -\sqrt{2}t^2(t-1))$.*

Proof. The zero identity $\wp_\tau(\mu) = 0$, proved without numerical input in B.1.3, identifies P with the signed visible section. With the cusp branch in Part C and the specified scale, substitution of $U = -(5 + 2\sqrt{6})$ in the nodal uniformization of B.1.1 gives $Y = -\sqrt{2}$, so $P = (0, -\sqrt{2}t^2(t-1))$.

To compare the line bundles as families, use their common divisor $P - O$ and rigidify at the zero section. Put $F_\theta(\xi) = \theta(\xi - \mu)/\theta(\xi)$, $r_\theta = \theta(-\mu)/\theta'(0)$, and let s_{P-O} be the canonical meromorphic divisor section. The map

$$[\xi, W] \longmapsto \frac{2r_\theta}{f_{\mathrm{sc}}} W \frac{s_{P-O}(\xi)}{F_\theta(\xi)}$$

is a holomorphic line-bundle isomorphism, including at P, O after cancellation of the equal divisors. At O it sends the chosen frame to dx/y, since $F_\theta \sim r_\theta/\xi$ and $x/y \sim -\xi/f_{\mathrm{sc}}$ with the corresponding canonical divisor frame. Two such maps that agree at O differ by a holomorphic scalar on each compact elliptic fibre and hence agree everywhere. This proves descent under the actual period identifications.

Engel's [Eng26, Lemma 2.1], which we have read together with its proof, identifies $\mathrm{Hom}(t_{6P}^* M_S, M_S)$ with $\pi_S^* \mathcal{O}_B(1)$. The lemma applies because the preceding pointed cubic map and Mordell–Weil calculation identify his (S, O, P) with ours, including the chosen sign.

To determine which section is transported, it suffices to compute its order at one generic smooth point of each special fibre, since a section of this pulled-back line is a base section. At a finite centre write $6\mu = p + r\tau + \delta$ with $(p, r) = (4, -2), (3, -3)$ and δ/f_{sc} holomorphic; this divisibility follows by linearizing the affine laws, whose homogeneous orders are two and one. Evaluate at $\xi = f_{\mathrm{sc}} z$ for generic fixed z. The factor $(2r_\theta/f_{\mathrm{sc}})/F_\theta(f_{\mathrm{sc}} z)$ tends to $2z$, a unit. Restoring the lattice shift $p + r\tau$ at the target changes the fibre multiplier to $e^{-2\pi i(\beta_c - r\mu)}$, whose finite limits are $e^{-2\pi i\kappa_1}$ and $e^{-2\pi i\kappa_2}$. The source and target theta ratios remain units. Therefore the base section has order zero at both finite points. At the cusp f_{sc} has a simple pole and r_θ has a nonzero limit; the two factors $2r_\theta/f_{\mathrm{sc}}$ cancel between source and target. The remaining multiplier is $e^{-2\pi i\beta_c} = qe^{-2\pi i(\beta_c+\tau)}$, of exact order one. Thus the transported section has divisor F_∞. Engel's [Eng26, Corollary 2.2] now identifies it with a nonzero scalar multiple of his reference, since two base sections with the same divisor have constant ratio. Changing β by c multiplies it exactly by $e^{-2\pi ic}$. With our stated period-reference normalization, this scalar is one. For any independently chosen reference ϕ_{ref}, the scalar is instead the precisely defined ratio $\lambda_0 = \phi_{\mathrm{part}}/\phi_{\mathrm{ref}}$, evaluated on any ordinary fibre. Scaling the reference by $a \in \mathbb{C}^*$ changes λ_0 to λ_0/a; no scalar value is invariant under that declared choice. □

2.2 B.2. All affine vectors and cusp sections

Proposition (B.2.1; Proved)**.** *Let*

$$w_1 = r_1\varepsilon + k_1 d, \qquad w_2 = r_2\varepsilon' + k_2 d, \qquad 3 \nmid r_1, \quad 2 \nmid r_2,$$

and use an arbitrary holomorphic cusp collar section σ in the construction of [CA26, Section 6.4]. Write $l_0 = \gamma(c(\sigma))$ in its positive cusp-winding convention, and set $p = 12l_0 - 4r_1 - 3r_2$. Let primes denote a second such datum with the same period solution and cusp fan. Then

$$X(c; w_1, w_2, \sigma) \simeq X(c'; w_1', w_2', \sigma')$$

holds exactly for the following arithmetic conditions:

$$n = c' - c \in \mathbb{Z}, \qquad p' = ap \text{ for some } a \in \{1, -1\}, \qquad n \equiv k_2 - k_2' \pmod 2.$$

All maps are over B. A map with the indicated a has fibre-linear part aI_2. In particular the complete invariants are

$$(|p|,\ e^{-\pi i(c+k_2)}), \qquad |p| \geq 1, \quad \gcd(|p|, 6) = 1.$$

Every value of this positive integer occurs. The full collar section enters the construction, but its additional analytic part and its second integral winding contribute no further invariant.

Proof. The determinant calculation in A.1.1 was independent of translations, so the base is still intrinsic. The centralizer proof gives $c' = c + n$ and aI_2 exactly as before. Put $S_n = I + nE$ and compare all target periods in source coordinates. On the finite root cover the exact deck equation is

$$B_j(g_j z) - R_j B_j(z) = \Pi_c(g_j z)\left(\frac{d_j}{m_j} + \ell_j\right), \quad d_j = S_n w_j' - a w_j, \quad N_j \ell_j = -d_j. \tag{B2.1}$$

Necessity follows by the normalized-base-change lift already proved. Conversely the norm equation has a rational solution r_j^* of $(I - A_j)r_j^* = d_j/m_j + \ell_j$, since $\ker N_j$ over $\mathbb{Q}$ equals the image of $I - A_j$. We can choose $\gamma(r_j^*) = 0$: the columns $\widehat{u}, \widehat{w}$ already span that image over $\mathbb{Q}$. Then $B_j = \Pi_c r_j^*$ is holomorphic on the whole root disc and solves the exact equation. Thus

the norm condition is sufficient for a local isomorphism, not only for conjugacy on the central real torus.

The computed norm images translate (B2.1) into

$$r_1' \equiv ar_1 \pmod 3, \quad r_2' \equiv ar_2 \pmod 4, \quad ak_2 - k_2' - nr_2' \equiv 0 \pmod 2. \tag{B2.2}$$

There is no condition on k_1. Since r_2' is odd, the last condition is the displayed parity. On the finite collar the middle translation is $A_j = B_j + (\log s_j)/(2\pi i)\,\Pi_c d_j$ and has integral defect ℓ_j. Its γ defect is $\alpha_j = (ar_j - r_j')/m_j$.

We prove the precise cusp necessity used next. A holomorphic isomorphism between cusp neighbourhoods with fibre-linear part aI_2 has, after lifting near a generic smooth central point to the toric covers, the form

$$(x, t_c) \longmapsto (t_c^q u(t_c) x^a, t_c), \qquad q \in \mathbb{Z}^2, \quad u(0) \in (\mathbb{C}^*)^2. \tag{B2.3}$$

Here the assertion concerns this local lift near a generic central point. To prove it, take a source chart near the generic point of its component and a target chart near its image. The etale toric quotient permits a local lift over a simply connected neighbourhood. The torus characters in a chart at the height-one ray v are $x_i = t_c^{v_i} X_i$, with X_i invertible. On $t_c \neq 0$ the affine fibre equation gives x_i'/x_i^a independent of the X coordinates. Its expression in the two charts is therefore $t_c^{v_i' - av_i}$ times a holomorphic unit of t_c, including zero. This proves (B2.3). Taking logarithms shows that its winding is in $\Lambda_{\mathrm{tor}} = \mathbb{Z}\widehat{w} + \mathbb{Z}d$. Changing a lift adds $(I - A_0)\Lambda = \Lambda_{\mathrm{tor}}$, so the vanishing of its class in $\bar{\Lambda} = \Lambda/\Lambda_{\mathrm{tor}}$ is intrinsic. This proof supplies the necessary extension test; no assertion about arbitrary punctured translations is used.

We retain the actual collar convention. The map from the cusp piece to the middle family is $T_{-\sigma} G_0^{-1}$. Consequently the middle translation induced by (B2.3) is

$$A_0 = sq + a\sigma - \sigma' + \eta_0,$$

with η_0 holomorphic on the cusp disc after a logarithm of the unit. For the clockwise generator $s \mapsto s - 1$, the class of its defect in $\bar{\Lambda}$ is $-ac(\sigma) + c(\sigma')$. On the other hand every middle integral cocycle satisfies $\ell_0 = -A_0(\ell_1 + A_1\ell_2)$, whose γ coordinate is $-\alpha_1 - \alpha_2$. Equality of these coordinates is exactly

$$l_0' - al_0 = -(ar_1 - r_1')/3 - (ar_2 - r_2')/4,$$

or $p' = ap$. This proves the remaining necessity. Notice that $p \equiv -r_1 \pmod 3$ and $p \equiv r_2 \pmod 2$, so p is nonzero and coprime to six.

For sufficiency, $p' = ap$ implies the first two congruences of (B2.2) by reduction modulo three and four. The parity supplies the remaining one. Choose integral solutions ℓ_1, ℓ_2 of (B2.1). Their γ coordinates already have the required values. Adding any element of $\ker N_1 = (I - A_1)\Lambda$ preserves (B2.1). In particular adding an arbitrary integer multiple of $e_1 = (0, 1, -2, 0)^t$ changes the $\widehat{u}$ coordinate of ℓ_0 by an arbitrary integer while leaving its γ coordinate fixed. We can therefore arrange $[\ell_0] = -ac(\sigma) + c(\sigma')$ in the entire rank-two group $\bar{\Lambda}$. Choose lifts of σ, σ' on the cusp half-plane. Their remaining defect differs from ℓ_0 by an actual toric lattice vector; an integral q makes $sq + a\sigma - \sigma'$ have exactly defect ℓ_0.

The finite translations and this cusp reference now define a locally nonempty affine torsor with linear difference sheaf precisely $\mathcal{V}$ of A.3.1. The terms $a\sigma - \sigma'$ are retained, even when their chosen lifts have essential singularities at the puncture; they belong to the specified collar reference, while differences of allowed extensions are holomorphic η_0. The proof of $H^1(B, \mathcal{V}) = 0$ gives a global middle translation. The finite maps descend by (B2.1), and the cusp map is $x \mapsto t_c^q u(t_c) x^a$. The fan is preserved by all integer translations and by $y \mapsto -y$;

exponentials of the integer period shift agree. Hence this cusp map and its inverse extend over every stratum. The finite overlap is the equality defining A_j, and the cusp overlap is the equality defining A_0. Thus the maps glue globally.

Finally parity is equivalent to equality of the two displayed exponentials. Passing from p to $-p$ is effected by $a = -1$ and the same proof; for fixed sign the congruences have just been shown sufficient. Every positive integer coprime to six is obtained by choosing $r_1 \in \{1, 2\}$ and odd r_2 in the appropriate residue classes modulo three and four, and then taking $l_0 = (p + 4r_1 + 3r_2)/12$. This also proves that the extra collar data have been accounted for, rather than omitted from the statement. □

Proposition (B.2.2; Proved)**.** *For multiplicities* $(3, 4)$ *and* $|p| = 1$*, every admissible discrete member is biholomorphic to one member of A.5.1, uniquely modulo* $2\mathbb{Z}$*. At fixed nonzero admissible Engel parameter* u *there are exactly two isomorphism classes, exchanged by a positive circuit of* u *about zero. In the convention that Engel's root generator is the inverse of our clockwise one and his circle class is* d*, the identity of the middle quotient sends*

$$X_{c+1} \longleftrightarrow X(u; a_0 - d/3,\ a_1 + d/4).$$

Reversing both declared root generators reverses these two displayed signs.

Proof. Take the canonical target data $(r_1', r_2', k_1', k_2', l_0') = (1, -1, 0, 0, 0)$, for which $p' = -1$. Choose a so that $p' = ap$, and an integer $n \equiv k_2 \pmod 2$. B.2.1 supplies the isomorphism to X_{c+n}. The uniqueness is A.5.1. At fixed $u = e^{-2\pi i c}$, its logarithms differ by integers; the invariant $e^{-\pi i(c+k_2)}$ takes exactly the two square roots of u. Both occur, already as X_c, X_{c+1}, and A.2.1 distinguishes them.

We spell out the integral comparison to Engel's common basis $B_0 = (e_1, e_2, e_3, e_4)$, retaining his $Cf_4 = e_1 + e_4$:

$$P_{\mathrm{E}} = \begin{pmatrix} 0 & -2 & -1 & -2 \\ 2 & -1 & 0 & -1 \\ 0 & 2 & 1 & 3 \\ -1 & 0 & 0 & 0 \end{pmatrix}, \qquad T_0 P_{\mathrm{E}} = P_{\mathrm{E}} A_1, \quad T_1 P_{\mathrm{E}} = P_{\mathrm{E}} A_2.$$

Its determinant is -1, $P_{\mathrm{E}} d = \delta_0$, $\psi P_{\mathrm{E}} = -\gamma$, $P_{\mathrm{E}} \varepsilon = -e_4$ and $P_{\mathrm{E}} \varepsilon' = \delta_0 - Cf_4$. These are exact lattice homomorphism identities, verified independently through Viaclovsky's intermediate basis in `ab_geometry.py`. They also agree with the complex periods: at z_1 the complex-linear map $(\zeta_1, \zeta_2) \mapsto (-\zeta_1/(\rho+1), \zeta_2 + \zeta_1/3)$ carries $\Pi_c P_{\mathrm{E}}^{-1}$ to Engel's columns $(1, 0), (\rho - 1, 0), ((1 + \rho)/3, 1/3), (0, -\kappa_1)$. At z_2 his Cf_4 has preimage $d - \varepsilon'$ and fixed period $1 - \kappa_2$. Thus the comparison fixes the positive circle class and the actual local complex structures, in addition to their monodromy matrices.

For the sign assertion, inverting the concrete clockwise affine generator changes its invariant translation $\Pi_c v_j / m_j$ to $-\Pi_c v_j / m_j$. Thus our dictionary is $a_{j-1} = -P_{\mathrm{E}} v_j / m_j$ in Engel's lattice marking (and its period image in the torus). Integer shift gives $\Pi_{c+1} v_j = \Pi_c (v_j + \gamma(v_j) d)$, with $\gamma(v_1) = 1, \gamma(v_2) = -1$, proving the two signs. A positive u circuit changes $\log u$ by $2\pi i$ and hence changes c by -1, exchanging the square roots. This gives the claimed permutation with its orientation. □

Proposition (B.2.3; Refuted, with corrected family)**.** *The assertion in [Eng26, Remark 8.1] of a global proper family over the original punctured* u*-disc for fixed sphere logarithmic data requires a double-cover qualification. The retained family with such data does not descend through* $u = w^2$*. It does give a proper holomorphic family on each simply connected* u*-open set, and a global proper family over the punctured* w*-disc as in A.3.1.*

Proof. Descent through $u = w^2$ would furnish an isomorphism between the two pulled-back fibres at w and $-w$, namely X_c and X_{c+1}. A.2.1 excludes that isomorphism by the nonzero class of d in $\Lambda^{A_2}/N_2\Lambda$. This is an exact order-two obstruction on the normalized finite cover. It applies to all sphere data by B.2.1 because r_2 is odd.

There is also a direct analytic witness near zero. The relative Albanese invariant of the multiplicity-four reduced fibre in a holomorphic descended family would be a single-valued holomorphic function of u. After pullback it would be even in w. But A.4.1 gives $j_2(w) = C^{-1}w^{-1} + 744 + O(w)$, where $C = e^{-\pi i\kappa_2^{\mathrm{part}}} \neq 0$; its Laurent coefficient of w^{-1} is nonzero. Thus it is not even on any punctured neighbourhood. This verifies the obstruction independently of a choice of isomorphism lift. Local inverse logarithms of u give the proper local families already constructed; coherent even shifts give their global double-cover family. The untwisted middle quotient itself has the integer shear identification, so this refutation concerns the fixed sphere logarithmic data in the stated remark. □

2.3 B.3. Higher multiplicities

Proposition (B.3.1; Proved)**.** *Let k, l be positive integers with $\gcd(3k, 4l) = 1$, and put $m_1 = 3k, m_2 = 4l$. Consider Viaclovsky's sphere members expressed in the same clockwise marking, with $w_j = r_j\varepsilon_j + s_j d$, $\gcd(r_j, m_j) = 1$, cusp winding l_0, and $p = m_1m_2l_0 - m_2r_1 - m_1r_2 = \pm 1$, where $\varepsilon_1 = \varepsilon, \varepsilon_2 = \varepsilon'$. Two such members are biholomorphic precisely when their multiplicity pair agrees and, for some $a \in \{1, -1\}$ and integer $n = c' - c$,*

$$p' = ap, \quad r'_j \equiv ar_j \pmod{m_j}, \quad nr'_1 \equiv as_1 - s'_1 \pmod{k}, \quad nr'_2 \equiv as_2 - s'_2 \pmod{2l}.$$

The congruences on r'_j already follow from $p' = ap$. For fixed discrete vectors and cusp collar the identification subgroup is $2kl\mathbb{Z}$. For fixed admissible u and this multiplicity pair, all the stated sphere translation data give exactly $2kl$ isomorphism classes, permuted in one cycle by a circuit of u about zero.

Proof. We first recover the base for these sphere members without using the degree-two canonical pencil. A biholomorphism $F : X \to X'$ gives $g = f'F$. Since $H^2(S^6, \mathbb{R}) = 0$, the closed form $g^*\omega_{\mathrm{FS}}$ is exact. Restrict to a regular torus T and wedge with a flat Kahler form on T. Stokes' theorem gives integral zero, while the integrand is nonnegative and strictly positive wherever $dg|_T \neq 0$. Thus g is constant on each regular fibre. The image of (f, g) is a compact analytic curve whose projection to B has degree one; its normalization gives a holomorphic $h : B \to B$ with $g = hf$. Apply the same construction to F^{-1} to see that h is an automorphism. The three homological local monodromy orders are three, four and infinity, independently of k, l; their intertwining maps under F therefore fix each special point. Hence $h = 1$. The nonreduced fibre multiplicities now force equal k, l. This is the full reduction in the read Proposition 4.7 of Viaclovsky's paper, written here with its exact maps.

The root model of degree m_1 is the pullback of the degree-three model by $s \mapsto s^k$, and likewise at the other point. Its linear action remains A_1 or A_2, while its root generator has order m_j. The primitive invariant coordinate r_j/m_j proves freeness of every nonidentity power. The normalization and deck-commutation arguments of A.1.2 and A.2.1 apply to these root covers without changing a generator. Their norms are

$$N_{3k} = kN_1, \qquad N_{4l} = lN_2,$$

so the fixed-coordinate norm images are $3k\mathbb{Z}\varepsilon + k\mathbb{Z}d$ and $4l\mathbb{Z}\varepsilon' + 2l\mathbb{Z}d$. The norm equation (B2.1), with m_j replaced by these actual covering degrees, gives exactly the displayed congruences. The centralizer and first-column comparison remain unchanged.

For completeness the integer p records the primitive sphere relation in this marking. The common torus coinvariants are detected by γ, since $(A_1 - I)\Lambda + (A_2 - I)\Lambda = \ker\gamma$ by their

explicit saturated bases. The finite meridian relations are $x^{m_1} = c^{r_1}$, $y^{m_2} = c^{r_2}$, and the cusp relation is $xy = c^{l_0}$ with c central. Eliminating y gives an abelian relation matrix with determinant $-p$ and primitive first elementary divisor because $\gcd(m_1, m_2) = 1$. Thus the sphere data have $p = \pm 1$. Reducing $p' = ap$ modulo m_1, m_2 gives the two congruences for r_j.

To prove sufficiency, use the local rational solutions of the norm equation and the identical cusp comparison in B.2.1. The first-coordinate extension on the new root discs has vanishing orders $2k$ and l, precisely the orders of the pulled-back Θ. Dividing by it again gives $\mathcal{O}_B(-p_0)$, and finite averaging again gives the same rank-two extension sheaf. Thus $H^1(B, \mathcal{V}) = 0$ still removes the analytic cocycle. The images of $I - A_j$ and their arbitrary $\widehat{u}$ adjustments are unchanged, so $p' = ap$ is exactly the remaining cusp winding condition. Every local map and overlap is the one already written in B.2.1, with the new denominator m_j; this proves global sufficiency.

For identical data, $p \neq 0$ excludes $a = -1$. The congruences reduce to $k \mid nr_1$ and $2l \mid nr_2$. The two primitivity assumptions imply $k \mid n$ and $2l \mid n$. Since $\gcd(k, 2l) = 1$, their least positive common multiple is $2kl$. For different sphere translation data, r_1' is invertible modulo k, r_2' modulo $2l$, and the Chinese remainder theorem gives exactly one integer shift class modulo $2kl$. The values of s_1 modulo k and s_2 modulo $2l$ all occur without affecting freeness or p. Thus there are exactly $2kl$ classes at fixed u. The change $c \mapsto c - 1$ on a positive u circuit acts by a generator of this cyclic residue group. Program `ab_exact.py` verifies the repeated norms and shift steps for five coprime pairs; the preceding symbolic identities prove all pairs. □

2.4 B.4. Automorphisms and the remaining real-structure calculation

Proposition (B.4.1; Proved)**.** *For each member of the retained* $(3, 4)$ *family,* $\mathrm{Aut}(X_c)$*, including all its components, is* $\mathbb{C}/\mathbb{Z} \simeq \mathbb{C}^*$*. Its action on the ordinary fibres is* $\zeta \mapsto \zeta + be$*, and the exponential group isomorphism is* $b \mapsto e^{2\pi i b}$*.*

Proof. A.1.1 and A.2.1 give base identity, $n = 0$ and fibre-linear part I. Let ℓ_1, ℓ_2 be the integral defects of a lift of its translating section to the universal cover of B°. Finite extension gives $N_j \ell_j = 0$. The saturated image calculations in A.1.2 give the integral equality $\ker N_j = (I - A_j)\Lambda$. The cusp valuation argument (B2.3) gives $\ell_0 \in \Lambda_{\mathrm{tor}}$. Write

$$\ell_1 = ae_1 + bf_1, \qquad \ell_2 = ce_2 + df_2.$$

The $\widehat{u}$ coordinate of $\ell_1 + A_1\ell_2$ is $a - c - 2d$, so the cusp condition is exactly $a = c + 2d$. Changing the global lift by a period λ changes this pair by the simultaneous coboundary $((I - A_1)\lambda, (I - A_2)\lambda)$. In the four displayed coefficient coordinates its matrix is

$$C = \begin{pmatrix} -6 & 1 & -1 & 0 \\ 2 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ -3 & 0 & -1 & 0 \end{pmatrix}.$$

The lattice of all allowed pairs has basis matrix $K = \left(\begin{smallmatrix} 0 & 1 & 2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$. The first three columns of C expressed in this basis have determinant one. Hence every allowed pair is an integral coboundary; the computation is exact, including its saturation, and is independently checked by Hermite form in `ab_exact.py`. After changing the lift, all its defects vanish.

The corrected lift extends as an equivariant holomorphic vector on the finite discs. A holomorphic local lift exists on the whole root disc by lifting the translating section through its covering vector bundle. Its defect is in $\ker N_j = (I - A_j)\Lambda$; subtracting an integral period kills that defect while preserving holomorphy. The corrected global lift and this local defect-zero

lift differ by an invariant period, which is holomorphic through the centre. At the cusp, the local toric logarithm in (B2.3) is $sq + \eta(t_c)$. The chosen global lift differs from it by $\Pi_c(s)\lambda = sB_0\bar{\lambda} + C(t_c)\bar{\lambda} + \lambda_{\mathrm{tor}}$. Defect zero gives exactly $q + B_0\bar{\lambda} = 0$, so the logarithmic terms cancel and the remaining vector is holomorphic. This proves the extension at the cusp as well. We therefore obtain a global section of $\mathcal{V}$. Its exact sequence in A.3.1 gives $H^0(B, \mathcal{V}) = \mathbb{C}e$. Conversely each such section defines the displayed translations on the finite covers and torus multiplication at the cusp, hence a global automorphism. Its kernel consists of global invariant periods. The common fixed lattice is $\mathbb{Z}d$, and $\Pi_c d = e$, giving kernel exactly $\mathbb{Z}$. This proves the full group assertion and its connectedness. □

Proposition (B.4.2; Proved)**.** *Use the normalization* $(\beta^{\mathrm{part}} + \tau)(p_0) = 0$. *There is an antiholomorphic diffeomorphism* $X_c \to X_{c'}$ *exactly when*

$$c' - (1 - \bar{c}) \in 2\mathbb{Z}, \qquad \textit{equivalently } w' = -\bar{w}.$$

Consequently $\overline{X_c} \simeq X_{1-\bar{c}}$. *The members admitting antiholomorphic self-maps are exactly those with* $\operatorname{Re} c \in \frac{1}{2} + \mathbb{Z}$, *equivalently* $w = -\bar{w}$. *On each such member the group of holomorphic and antiholomorphic diffeomorphisms is*

$$\mathrm{Aut}^{\pm}(X_c) \simeq \mathbb{C}^* \rtimes \mathbb{Z}/2, \qquad z \longmapsto 1/\bar{z}$$

for the action of the second factor. Its antiholomorphic maps form the nonidentity coset. For all other members the full group is the holomorphic group of B.4.1.

Proof. We first construct the comparison before returning its affine data to the canonical choice. Lift conjugation of the real orbifold base to the antiholomorphic involution ι of its universal cover whose cusp branch has $\tau(\iota z) = -\overline{\tau(z)}$. A lift exists by the universal covering property; it extends at the finite orbifold points, whose local equations are $s_j^{m_j} = t_j$. Choose the usual paths from a real point $t > 1$, so reflection fixes the second centre and sends the first chosen lift to $g_2^{-1} z_1$. The induced involution of the deck group is

$$\phi(g_1) = g_2^{-1} g_1^{-1} g_2, \qquad \phi(g_2) = g_2^{-1}, \qquad \phi(g_0) = g_0^{-1}.$$

These identities also follow by reflecting the lifted peripheral paths; their product and their squares agree with $g_1 g_2 g_0 = 1$. The cusp branch fixes the lift uniquely.

All reflection identities for the periods can be checked before invoking uniqueness. Denote their affine transformation formulas by G_1, G_2 on the triple (τ, μ, β) and put $K(\tau, \mu, \beta) = (-\bar{\tau}, 1 - \bar{\mu}, -\bar{\beta})$. Direct rational substitution gives

$$KG_1K = G_2^{-1} G_1^{-1} G_2, \qquad KG_2K = G_2^{-1}.$$

For example $G_2^{-1}(\tau, \mu, \beta) = (-1/\tau, (1-\mu)/\tau, \beta + 3 - 6(1-\mu)^2/\tau)$ and $G_1^{-1}(\tau, \mu, \beta) = (1/(1-\tau), 1 - \mu/(1-\tau), \beta - 2 + 6\mu^2/(1-\tau))$. These formulas verify every constant in the two equalities. The reflected μ is cusp bounded and has the same affine laws, so their difference is a global section of $\mathcal{O}_B(-p_0)$ and vanishes. After this equality, the difference between the reflected and original β^{part} is a holomorphic invariant base function, hence constant; its cusp value is zero under the stated normalization. Therefore

$$\mu(\iota z) = 1 - \bar{\mu}(z), \qquad \beta^{\mathrm{part}}(\iota z) = -\overline{\beta^{\mathrm{part}}(z)}.$$

The original τ identity follows from its j equation and its specified cusp branch. In particular $D_{-\bar{c}}(\iota z) = D_c(z)$, so this comparison stays in the exact admissible domain.

Let

$$H = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -6 & 0 & -1 & 0 \\ 0 & -1 & 0 & -1 \end{pmatrix}.$$

Then $H^2 = I$, $HA_2H = A_2^{-1}$, $HA_1H = A_2^{-1}A_1^{-1}A_2$, and

$$\Pi_{-\bar{c}}(\iota z)H = -\overline{\Pi_c(z)}.$$

Thus $(z, \zeta) \mapsto (\iota z, -\bar{\zeta})$ is a well-defined antiholomorphic torus map. It also descends under the middle deck action. Indeed covariance and the displayed lattice conjugacies imply $R_{\phi(g)}(\iota z) = \overline{R_g(z)}$ by multiplying the full real-invertible period matrices. This proves exact compatibility, not merely an equality of fibre lattices.

At the first finite centre compose this map with the target deck identification g_2 to return the centre $g_2^{-1}z_1$ to z_1. Its lattice map becomes A_2H and it conjugates the clockwise generator to the inverse generator. At the second centre its lattice map is H with the same reversal. Consequently the target affine vectors are exactly

$$w_1' = -A_2H\varepsilon = -\varepsilon + d, \qquad w_2' = -H(-\varepsilon') = \varepsilon' - 3d.$$

The minus signs arise because the inverse of a generator with invariant translation w_j'/m_j has translation $-w_j'/m_j$. This calculation verifies the full affine actions on the root covers. For the overlaps choose the root coordinates of the real local base coordinates t and $t-1$ compatible with these local reflections. This choice has an explicit comparison with any original root coordinate: their ratio is a holomorphic unit, so changing it changes the collar by $(2\pi i)^{-1}\log h\,\Pi w_j$, a translation extending on the entire root family. Its inverse gives a holomorphic isomorphism of the two glued descriptions. Thus the extension property and final classification transfer to the original coordinates. Then $\log s_j' = \overline{\log s_j}$ on paired branches. Under $\zeta \mapsto -\bar{\zeta}$, conjugation of $1/(2\pi i)$ supplies the compensating sign, and the identity of the two logarithmic collar translations follows exactly. Two roots of the same fixed real coordinate differ by an m_jth root of unity; this changes the comparison only by a constant period-coordinate translation, and the additive correction of A.3.1 glues those constants. The preceding holomorphic-unit comparison handles a different original local base coordinate. Its inverse has the same construction.

At the cusp choose $s = (2\pi i)^{-1}\log t_c$, so $s \mapsto -\bar{s}$. The full holomorphic unit matrix satisfies

$$C_{-\bar{c}}(\bar{t_c}) = -\overline{C_c(t_c)} + \begin{pmatrix} 6 & 0 \\ 0 & 1 \end{pmatrix}.$$

The integral extra matrix has trivial exponential. Hence $(x, t_c) \mapsto (\bar{x}, \bar{t_c})$ intertwines the full cusp deck groups, with the same integral index λ. The toric fan is defined over $\mathbb{Z}$, so this map extends antiholomorphically on every toric chart and stratum. It preserves the untwisted cusp collar. We have therefore constructed an antiholomorphic map to the actual target datum with parameter $-\bar{c}$, $r_1' = -1, k_1' = 1, r_2' = 1, k_2' = -3, l_0' = 0$, whose p' is $+1$.

Apply B.2.1 to normalize this target back to the canonical $p = -1$ datum. Take $a = -1$ and the odd shift $n = 1$. This gives a holomorphic map to $X_{1-\bar{c}}$. Every other antiholomorphic map, composed with the inverse of this one, is a holomorphic comparison between canonical members. A.5.1 gives exactly the stated congruence. Exponentiating it gives $w' = -\bar{w}$.

It remains to verify the group extension rather than infer its square from H^2. For a real member, choose any resulting antiholomorphic self-map J. Its derivative on ordinary fibre coordinates is +complex conjugation: the initial reflection has derivative −complex conjugation, while the normalization just used has linear part $-I_2$. Hence it

conjugates translation by $b \in \mathbb{C}/\mathbb{Z}$ to translation by $\bar{b}$. Its square is a holomorphic translation d. Commuting J with J^2 shows $d = \bar{d}$ in $\mathbb{C}/\mathbb{Z}$. This equality forces the imaginary part of a representative of d to vanish, so take $d \in \mathbb{R}$. Replacing J by the composition of translation by $-d/2$ with J changes its square to $d - d/2 - d/2 = 0$. We have an antiholomorphic involution. Under $b \mapsto e^{2\pi i b}$, its conjugation action is $z \mapsto 1/\bar{z}$. This proves the split group and specifies its antiholomorphic coset. □

3 C. The normalized periods, quantitative bounds, and the boundary

3.1 C.1. A theta expression for the missing period

We use the odd Jacobi function

$$\vartheta_1(v,\tau) = 2\sum_{n\geq 0}(-1)^n e^{\pi i\tau(n+1/2)^2}\sin((2n+1)\pi v)$$

and the Dedekind function $\eta_D(\tau) = e^{\pi i\tau/12}\prod_{n\geq 1}(1-e^{2\pi i n\tau})$. The subscript distinguishes this function from the invariant integral class. We choose the cusp lift reached from $\tau = iT$, $T > 1$, on which $\mu = 1/2 - iy(T)$, $0 < y(T) < T/2$. Put $a = 5 + 2\sqrt{6}$ and $\rho = e^{\pi i/3}$.

Proposition (C.1.1; Proved)**.** *For the period section identified in Statement B.1.3, the particular solution with $(\beta^{\mathrm{part}} + \tau)(p_0) = 0$ is*

$$\beta^{\mathrm{part}} = \frac{6i}{\pi}\operatorname{Log}\frac{\vartheta_1(\mu,\tau)}{2\sqrt{3}\,\eta_D(\tau)}. \tag{7}$$

Here the logarithm is continued on the simply connected uniformizing half-plane from its cusp expansion. Its special values are

$$\mu_0 = \frac{1}{2} - \frac{i\log a}{2\pi}, \qquad \mu(z_1) = \frac{2-\rho}{3}, \qquad \mu(z_2) = \frac{1-i}{2},$$

$$\beta^{\mathrm{part}}(z_2) = i\left(\frac{3}{2} - \frac{3}{\pi}\log 6\right), \qquad \operatorname{Re}\beta^{\mathrm{part}}(z_1) = -\frac{1}{2}.$$

In particular $e^{2\pi i\mu_0} = -a$.

Proof. Statement B.1.3 identifies the elliptic point represented by μ with one of the sections $(0, \pm\sqrt{2}t^2(t-1))$. The uniformization map $(v,\tau) \mapsto (f^2\wp_\tau(v), f^3\wp'_\tau(v)/2)$ therefore gives $\wp_\tau(\mu) = 0$. No zero of $\wp$ is a lattice point. Thus the numerator of (7) is nonzero; the denominator is nonzero as well. A holomorphic logarithm exists on the simply connected cover.

For completeness the cusp value follows directly from the normally convergent Fourier expansion, rather than from a quoted Eichler–Zagier constant:

$$\frac{\wp_\tau(v)}{\pi^2} = \csc^2(\pi v) - \frac{1}{3} + 8\sum_{n\geq 1}\frac{nq^n}{1-q^n}(1-\cos(2\pi n v)), \qquad q = e^{2\pi i\tau}. \tag{8}$$

This identity follows by differentiating the logarithm of the Jacobi product twice and using $E_2 = 1 - 24\sum nq^n/(1-q^n)$. Boundedness of the chosen elliptic logarithm at the cusp permits passage to the limit. We obtain $\sin^2(\pi\mu_0) = 3$, equivalently $u^2 + 10u + 1 = 0$, $u = e^{2\pi i\mu_0}$. On the chosen imaginary-axis branch $\operatorname{Im}\mu < 0$, so $u = -a$. One can see the branch without a numerical convention: on the rectangular elliptic curve with $\tau = iT$, $T > 1$, $\wp$ is real on the segment $1/2 - iy$, $0 \leq y \leq T/2$. Its endpoint values are the largest and middle real roots of its cubic, respectively positive and negative. Its derivative has no zero in the open segment, since

the zeros of $\wp'$ are the three nonzero half-periods. There is exactly one zero there; this gives the indicated continuation. Consequently $2\sin(\pi\mu_0) = 2\sqrt{3}$.

Put $R = \vartheta_1/\eta_D$. Poisson summation for the defining Gaussian series and its index shift give

$$\begin{aligned} R(v,\tau+1) &= e^{\pi i/6} R(v,\tau), \\ R(v/\tau,-1/\tau) &= -ie^{\pi i v^2/\tau} R(v,\tau), \\ R(1-v,\tau) &= R(v,\tau), \qquad R(v+1,\tau) = -R(v,\tau). \end{aligned}$$

Inserting the actual transformed arguments gives

$$\begin{aligned} R(\mu\circ g_1, \tau\circ g_1) &= e^{-\pi i/3} e^{\pi i(1-\mu)^2/\tau} R(\mu,\tau), \\ R(\mu\circ g_2, \tau\circ g_2) &= ie^{\pi i\mu^2/\tau} R(\mu,\tau). \end{aligned}$$

Multiplication of the logarithms by $6i/\pi$ gives exactly $2-6(1-\mu)^2/\tau$ and $-3-6\mu^2/\tau$, modulo constants in $12\mathbb{Z}$. These constants vanish: each defect is constant on the connected cover, and evaluation at the corresponding fixed point makes both its two sides zero. The fixed-point equations themselves give $\mu(z_1) = (2-\rho)/3$ and $\mu(z_2) = (1-i)/2$. The Jacobi product has cusp expansion $R(\mu,\tau) = 2q^{1/12}\sin(\pi\mu)(1+O(q))$. Equation (7) consequently gives $\beta^{\mathrm{part}} + \tau \to 0$. The difference from the specified particular solution is an invariant holomorphic function extending across all three points of B, hence a constant, and the cusp normalization makes it zero.

At $\tau = i$, the half-period translation formula gives $\vartheta_1((1-i)/2, i) = e^{\pi/4}\vartheta_3(0,i)$. The identities $2\eta_D^3 = \vartheta_2\vartheta_3\vartheta_4$, $\vartheta_2(0,i) = \vartheta_4(0,i) = 2^{-1/4}\vartheta_3(0,i)$ give $R = e^{\pi/4}\sqrt{2}$, proving the displayed value of β. At ρ, q is negative real, $\mu = 1/2 - i/(2\sqrt{3})$, and the product for R has phase $\pi/12$, with all its remaining factors positive real. This gives $\operatorname{Re}\beta = -1/2$. □

Proposition (C.1.2; Proved)**.** *The following decimal evaluations of the exact expressions in C.1.1 agree in two independent evaluations through the displayed twenty decimal places:*

μ_0	$0.5 - 0.36485183191106784792\,i$
$\mu(z_1)$	$0.5 - 0.28867513459481288225\,i$
$\mu(z_2)$	$0.5 - 0.5\,i$
$\beta^{\mathrm{part}}(z_1)$	$-0.5 - 1.27101206057537346287\,i$
$\beta^{\mathrm{part}}(z_2)$	$-0.21100425815613411418\,i$

These are evaluations of explicit convergent expressions, with working precision ninety decimal digits. The formula (7), together with $j(\tau) = 1728t$ and $\wp_\tau(\mu) = 0$, determines the periods throughout a fundamental domain.

Proof. The executable `cd_numerics.py` evaluates μ_0 first by its algebraic logarithm and then by solving $\pi^2\csc^2(\pi v) - \pi^2/3 = 0$. For β, its first calculation uses the Jacobi theta series and the eta product; its second uses

$$R(v,\tau) = 2e^{\pi i\tau/6}\sin(\pi v)\prod_{n\ge 1}(1 - 2q^n\cos(2\pi v) + q^{2n}).$$

The product is truncated after 120 factors. At the two specified points $|q| \le e^{-\pi\sqrt{3}}$ and $|\operatorname{Im} v| \le 1/2$, so the omitted logarithm is bounded by $\sum_{n>120}(2|q|^n\cosh\pi + |q|^{2n})/(1-2|q|^n\cosh\pi - |q|^{2n})$, which is less than 10^{-280}. The measured discrepancies of the ninety-digit calculations are below 10^{-88}. We do not promote floating-point rounding to an interval certificate. A further sixty-digit interval computation, `cd_special_intervals.py`, evaluates the real positive product at ρ, including its forty-factor tail, and the exact logarithmic expression at i. It encloses both imaginary parts within intervals narrower than $3\cdot 10^{-59}$ and therefore certifies the displayed

twenty decimal places. It also proves $-0.2 < \operatorname{Im}(\kappa_1^{\mathrm{part}} - \kappa_2^{\mathrm{part}}) < -0.1$. The exact formulas, product-tail bounds, independent algorithms, and full outputs accompany the decimals. The special values of μ are also checked directly in the fixed-point equations, an exact algebraic calculation. □

3.2 C.2. Bounds for the sharp parameter disc

Proposition (C.2.1; Proved)**.** *For the normalization of C.1.1, the exact admissibility maximum is finite and satisfies*

$$-\frac{3}{\pi}\log 6 \;\le M < 2.$$

More precisely the proof below gives $M < 1.635$. Consequently the sharp radius $R_w = e^{-\pi M}$ satisfies $e^{-2\pi} < R_w \le 216$, and for the reference linearization with $u = w^2$, $e^{-4\pi} < R_u \le 46656$.

Proof. Equation (7) gives

$$D_{\mathrm{part}} = \frac{6}{\pi}\log\frac{\mathcal{N}(\mu,\tau)}{2\sqrt{3}}, \qquad \mathcal{N}(v,\tau) = e^{-\pi(\operatorname{Im} v)^2/\operatorname{Im}\tau}|R(v,\tau)|. \tag{9}$$

The theta transformation formulas prove that $\mathcal{N}$ is unchanged by integral period translations of v, sign change of v, and the simultaneous modular change of v,τ. We may therefore use the comparison isomorphism $\mathbb{C}/(\mathbb{Z}+\tau\mathbb{Z}) \to \mathbb{C}/(\mathbb{Z}+\tau'\mathbb{Z})$, $[v] \mapsto [v/(c\tau+d)]$, to place τ in the standard modular domain and then choose $|\operatorname{Im} v| \le T/2$, $T = \operatorname{Im}\tau \ge \sqrt{3}/2$. Both operations preserve the equation $\wp(v) = 0$ and (9).

For $\sqrt{3}/2 \le T \le 2$, the theta series and $\prod(1-r^n) \ge \exp(-r/(1-r)^2)$ imply

$$\mathcal{N} \le 2e^{\pi/6}\left(1 + \frac{e^{-\pi\sqrt{3}/2}}{1-e^{-3\pi\sqrt{3}/2}}\right)\exp\frac{e^{-\pi\sqrt{3}}}{(1-e^{-\pi\sqrt{3}})^2} < 3.615.$$

Indeed the absolute value of the n-th sine term, including the Gaussian factor, is at most $2\exp(-\pi T(n+1/2-|\operatorname{Im} v|/T)^2)$, which is at most $2e^{-\pi T n^2}$; and $n^2 \ge 1 + 3(n-1)$ for $n \ge 1$.

For $T \ge 2$, a zero of $\wp$ in this period strip has $|\operatorname{Im} v| < 1$. To prove this, assume its absolute value is at least one and apply (8). With $r = e^{-4\pi}$, the absolute value of the terms other than $-1/3$ is at most

$$\frac{1}{\sinh^2\pi} + 8\left\{\frac{r}{(1-r)^3} + \frac{1}{2(1-r)}\left(\frac{e^{-2\pi}}{(1-e^{-2\pi})^2} + \frac{e^{-6\pi}}{(1-e^{-6\pi})^2}\right)\right\} < 0.016 < \frac{1}{3}.$$

Here $|\sin\pi v| \ge \sinh\pi$, and $|q|^n\cosh(2\pi n\operatorname{Im} v) \le (e^{-\pi T n} + e^{-3\pi T n})/2$. This contradicts $\wp(v) = 0$. The product expression for R now gives the uniform bound

$$\mathcal{N} \le 2\cosh\pi\, e^{-\pi/3}\exp\left(\frac{2\cosh(2\pi)r}{1-r} + \frac{r^2}{1-r^2}\right) < 8.151.$$

Insertion in (9) proves the stated upper bound. The script `cd_bounds.py` independently evaluates these monotone bounds by fifty-digit interval arithmetic: the larger resulting bound for D is contained, after outward rounding for display, in

$$[1.63421874605891012363,\ 1.63421874605891012365].$$

The output file retains the full fifty-digit endpoints. This encloses the coarse upper bound; the exact maximum is proved in C.2.3.

At the cusp $D_{\mathrm{part}} = -T + o(1)$, so it tends to minus infinity; continuity on the rest of the compact orbifold base gives attainment of its maximum. At z_2, C.1.1 gives $D_{\mathrm{part}} = -(3/\pi)\log 6$, proving the lower bound. Addition of c adds $\operatorname{Im} c$ to D; a zero is forced by continuity between a maximum and the cusp whenever $\operatorname{Im} c \ge -M$. For $\operatorname{Im} c < -M$, the real determinant TD is everywhere nonzero, proving the exact asserted form of the admissible disc. □

Proposition (C.2.2; Proved)**.** *The point $t = 1$ is a strict local maximum of D_{part}. A double-precision calculation on 5151 points of the half modular domain $0 \le \mathrm{Re}\,\tau \le 1/2$, $|\tau| \ge 1$, $\mathrm{Im}\,\tau \le 3$, has largest sampled value at $\tau = i$. This finite computation does not establish the global maximum.*

Proof. We give a quantitative neighbourhood, which will also specify the local region needed for a global interval cover. Put

$$h = v - (1-\tau)/2, \quad x = \tau - i, \quad G(h,\tau) = \wp_\tau((1-\tau)/2 + h), \quad H(h,\tau) = \log \frac{\vartheta_3(h,\tau)}{\eta_D(\tau)}.$$

The half-period theta identity gives exactly $\log \mathcal{N} = \mathrm{Re}\, H - \pi(\mathrm{Im}\, h)^2/T$. On the complex polydisc $|h| \le 1/5$, $|x| \le 1/10$, the theta series has $|\vartheta_3 - 1| < 1$, so the logarithm is holomorphic. There are no poles of G there: for every integer lattice point its imaginary difference from $(1-\tau)/2 + h$ has absolute value at least $0.9/2 - 0.2 > 0$. Both functions are even in h.

The following uniform bounds use only positive Gaussian sums:

$$|G| < 20, \qquad |H| < 0.6.$$

For explicit verification put $A_k = \sum_{n\ge 1} n^k e^{-0.9\pi n^2} \cosh(0.4\pi n)$, $a_0 = 2A_0$, $a_1 = 4\pi A_1$, $a_2 = 8\pi^2 A_2$, and $r = e^{-1.8\pi}$. The bounds used are

$$|G| \le \frac{a_2}{1-a_0} + \frac{a_1^2}{(1-a_0)^2} + \frac{\pi^2}{3}\left(1 + \frac{24r}{(1-r)^3}\right) < 18.373,$$
$$|H| \le -\log(1-a_0) + \frac{1.1\pi}{12} + \frac{r}{(1-r)^2} < 0.547.$$

The derivative identity behind the first bound is $G = -\partial_h^2 \log \vartheta_3 - \pi^2 E_2/3$. The script `cd_local_bound.py` evaluates these bounds by sixty-digit interval arithmetic. It truncates the positive sums at $n = 10$ and bounds the remaining terms by geometric sums using $n^2 \ge 121 + 23(n-11)$; the polynomial factors n^k are summed explicitly.

At $(0, i)$ the theta heat equation gives

$$G = 0, \quad G_{h^2} = -g_2(i)/4, \quad G_x = id, \qquad H = \log\sqrt{2}, \quad H_x = 0, \quad H_{h^2} = -\pi/2.$$

Subscripts h^2 denote power-series coefficients. The first two follow also from $\wp'' = 6\wp^2 - g_2/2$. Here d is the real positive derivative coefficient of $e_2(\tau) = (\pi^2/3)(\vartheta_2(0,\tau)^4 - \vartheta_4(0,\tau)^4)$ at i. Differentiating the convergent theta series, with explicit Gaussian tails, gives $10.03 < d < 10.04$. The value $H_x = 0$ follows by differentiating the modular S-invariance of $\vartheta_3(0,\tau)/\eta_D(\tau)$ at i. Differentiating $\eta_D(-1/\tau) = \sqrt{-i\tau}\,\eta_D(\tau)$ gives $E_2(i) = 3/\pi$, which with $\wp((1-i)/2, i) = 0$ gives $H_{h^2} = -\pi/2$. The positive Eisenstein series for g_2, with an explicit tail in the same script, gives $47.26 < g_2(i)/4 < 47.30$.

Set $\epsilon = 3/20000$, $r_h = 3/100$, $s = \epsilon/(1/10) = 0.0015$, and $r_2 = (r_h/(1/5))^2 = 0.0225$. Cauchy's estimates for the even power series of G give, on $|h| = r_h$, $x = 0$,

$$|G(h,i)| \ge \left(47.26 - \frac{500 r_2}{1-r_2}\right) r_h^2 > 0.03217.$$

For $|x| \le \epsilon$, the change of G on this circle is at most $20s/((1-s)(1-r_2)) < 0.03074$. Rouche's theorem therefore places both zeros of $G(\cdot,\tau)$ in $|h| < r_h$. They represent the two zeros of $\wp$ modulo its lattice, since an elliptic function with one double pole has two zeros with multiplicity.

Write $G = e_2(\tau) + h^2 Q(h,\tau)$. The same Cauchy bounds give

$$|Q + g_2(i)/4| \le \frac{500(r_2 + s/(1-s))}{1-r_2}, \quad |e_2(\tau) - idx| \le \frac{2000|x|^2}{1-s}.$$

At any zero they imply $|x| \le 7|h|^2$ and $|h|^2 < 1/20000$; the interval output records bounds 6.124 and 0.00004434, respectively. For H, its stated coefficients and Cauchy's estimates now bound the remainder, together with the change of $1/T$ in the Gaussian, by $5000|h|^4$. For example a direct upper bound for its coefficient is

$$\frac{0.6(70^2 + 25 \cdot 70 + 25^2)}{(1-s)(1-25/20000)} + \frac{7\pi}{1-\epsilon} < 5000.$$

Consequently, for $0 < |x| \le 3/20000$,

$$\log \mathcal{N} - \log\sqrt{2} \le -\frac{\pi}{2}|h|^2 + 5000|h|^4 < 0.$$

At $h = 0$ the equation $G = 0$ forces $x = 0$ in this disc by the lower bound for e_2. This proves strict local maximality, with an explicit radius, and covers the square $|\operatorname{Re}\tau| \le 10^{-4}$, $|\operatorname{Im}\tau - 1| \le 10^{-4}$.

The grid calculation uses thirty Fourier terms for $\wp$, complex Newton iteration, and the theta product. Five sample values were independently recomputed with eighty Fourier terms and the theta series at ninety-digit precision; discrepancies are below $7 \cdot 10^{-16}$. The source, precise grid rule, sampled maximum, and comparisons are in `cd_numerics.py` and its output. □

Theorem (C.2.3; Proved, with interval certificate)**.** *For the cusp normalization $\beta^{\mathrm{part}} + \tau \to 0$,*

$$M = -\frac{3}{\pi}\log 6 = -1.71100425815613411418\ldots, \qquad \arg\max_{B\setminus\{p_0\}} D_{\mathrm{part}} = \{1\}.$$

The exact parameter discs are

$$0 < |w| < 216, \qquad 0 < |u| < 46656$$

for the reference linearization $u = w^2$; an arbitrary reference constant λ_0 multiplies the latter radius by $|\lambda_0|$. The proof combines an analytic cusp bound, the quantitative local proof C.2.2, and a completed outward-rounded interval cover of the remaining compact domain.

Proof. By (9) it is enough to prove $\mathcal{N}(v,\tau) \le \sqrt{2}$ at every zero of $\wp_\tau$, with equality only in the modular orbit of i. Modular transformations, period translations and sign change preserve this norm. Complex conjugation also preserves it, through $(v,\tau) \mapsto (\bar v, -\bar\tau)$, as follows by conjugating the normally convergent theta and eta series. It therefore suffices to work on $0 \le \operatorname{Re}\tau \le 1/2$, $|\tau| \ge 1$.

For $T \ge 2$, the zero-strip argument in C.2.1 improves to $|\operatorname{Im} v| < b = 43/100$. Replace $1/\sinh^2\pi$ in its displayed error bound by $1/\sinh^2(\pi b)$; the resulting upper bound is less than $0.316 < 1/3$, as certified independently in `cd_special_intervals.py`. Retain the Gaussian in the product estimate. The function $\cosh(\pi y)\exp(-\pi y^2/T)$ increases on $0 \le y \le b$, $T \ge 2$: its logarithmic derivative is $\pi\tanh(\pi y) - 2\pi y/T \ge 0$, since $\tanh(\pi y) \ge y$ there. The latter follows from concavity and its endpoint values. With $r = e^{-4\pi}$ we get

$$\mathcal{N} \le 2\cosh(\pi b)e^{-\pi/3-\pi b^2/2}\exp\left(\frac{2\cosh(2\pi b)r}{1-r} + \frac{r^2}{1-r^2}\right) < 1.082 < \sqrt{2}.$$

The bound decreases for $T \ge 2$: the logarithmic derivative of the exponential $e^{-\pi T/6 - \pi b^2/T}$ is negative and the product majorant decreases with $|q|$. This excludes the whole cusp region, without a numerical truncation of the base.

We now specify the compact computation as a proof by enclosure. Its rational starting rectangle is $\mathcal{R} = [0, 1/2] + i[17/20, 2]$, which contains the compact half modular domain. For

every rational rectangle Q retained in this domain, the program selects a decimal complex centre v_0, a decimal nonzero preconditioner A, and a closed complex rectangle Z around v_0. Floating-point Newton iteration only selects these numbers. The proof uses the subsequent interval tests

$$v_0 - A\wp(v_0, Q) + (1 - A\partial_v\wp(Z, Q))(Z - v_0) \subset \operatorname{int} Z, \qquad \|1 - A\partial_v\wp(Z, Q)\|_\infty < 1.$$

Here the norm of multiplication by a complex number is bounded by the sum of the absolute real and imaginary parts. All displayed sets are outward-rounded intervals. The mean-value integral along a segment in Z proves that, for every $\tau \in Q$, the map $v \mapsto v - A\wp_\tau(v)$ maps Z into itself and is a contraction. Banach's theorem supplies a genuine zero in Z. Both zeros of $\wp$ are related by sign and lattice translations, so their norms agree; this one enclosure therefore controls the actual period section.

For reproducibility the Fourier remainders are explicit. With $q = e^{2\pi i\tau}$, $p = qe^{2\pi iv}$, $m = qe^{-2\pi iv}$, let $r \geq |q|$ and $s \geq \max(|p|, |m|)$ be the interval upper endpoints. The code requires $s < 1$. For $N = 24$ put

$$S_1(r) = r^{N+1}\left(\frac{N+1}{1-r} + \frac{r}{(1-r)^2}\right),$$

$$S_2(r) = r^{N+1}\left(\frac{(N+1)^2}{1-r} + \frac{2(N+1)r}{(1-r)^2} + \frac{r(1+r)}{(1-r)^3}\right).$$

The errors in $\wp$ and $\partial_v\wp$ are bounded respectively by $8\pi^2(S_1(r) + S_1(s))/(1 - r)$ and $16\pi^3 S_2(s)/(1 - r)$. These follow by absolute summation of (8) and its derivative. They are added as rectangles enclosing the corresponding complex discs. For the norm, use $h = v - (1 - \tau)/2$ and the even theta series $\vartheta_3 = 1 + 2\sum_{n\geq 1} e^{\pi in^2\tau}\cos(2\pi nh)$. With $T_* \leq \operatorname{Im}\tau$ and $H_* \geq |\operatorname{Im} h|$, its omitted terms after $n = 10$ have absolute value at most

$$\frac{2e^{-121\pi T_* + 22\pi H_*}}{1 - e^{-23\pi T_* + 2\pi H_*}},$$

and the denominator is tested positive. The logarithm of the eta product tail after $n = 17$ is bounded by $r^{18}/((1 - r)(1 - r^{18}))$. The resulting enclosure of $e^{-2\pi(\operatorname{Im} h)^2/T}|\vartheta_3/\eta_D|^2$ is required to have upper endpoint strictly below two.

The completed computation `f_interval_parallel.py`, using `f_interval_cert.py`, runs these tests at thirty decimal digits and records every rational box. It accepted 19701 norm boxes. A further 36 boxes lie inside the local square of C.2.2 and are covered by that analytic proof. Another 174 boxes satisfy $x_{\text{upper}}^2 + y_{\text{upper}}^2 < 1$ exactly and lie outside the modular domain. There are zero pending or undecided boxes. The independent program `f_interval_verify.py` reconstructs the entire binary partition from its leaves, checks all reported strict interval bounds, and verifies the exclusions by rational arithmetic. It consumes all 19911 leaves, with exact total area 23/40 and maximum depth 38. The largest reported squared-norm upper bound is the exact decimal rational

$$\frac{999999998278267313099467461630787 1}{500000000000000000000000000000000000} < 2.$$

The JSON certificate, complete outputs, and independently runnable source programs are included. Thus the compact domain is fully covered; the exploratory grid is not used as a substitute for any enclosure. The local proof gives equality only at $\tau = i$, and the cusp and other compact boxes give strict inequality. This proves the global bound and its equality set.

Finally C.1.1 evaluates D at i exactly. Since $j(\tau) = 1728t$, its modular orbit maps precisely to $t = 1$. Exponentiation gives $e^{-\pi M} = e^{3\log 6} = 216$ and $e^{-2\pi M} = 6^6 = 46656$. The two independent theta calculations and the sixty-digit logarithmic interval computation in C.1.2 verify the numerical value of M as well as the exact evaluation; the integer radii are checked against interval exponentiation in `cd_special_intervals.py`. □

3.3 C.3. The rank-three period quotient

Proposition (C.3.1; Proved)**.** *At every point with $D = 0$, $T > 0$, the real-linear map $\Pi : \mathbb{R}^4 \to \mathbb{C}^2$ has rank three. Write $\mu = a_0\tau + b_0$, $a_0, b_0 \in \mathbb{R}$, and put*

$$k = (1, -6a_0, -6b_0, 6a_0 \operatorname{Re}\mu - \operatorname{Re}\beta).$$

Its real kernel is $\mathbb{R}k$. The subgroup $\Pi\mathbb{Z}^4$ is discrete exactly when $k \in \mathbb{Q}^4$. In that case it is a rank-three lattice in a real hyperplane and its quotient in $\mathbb{C}^2$ is a noncompact complex Lie group. In every other case the topological quotient $\mathbb{C}^2/\Pi\mathbb{Z}^4$ is non-Hausdorff.

Proof. The real matrix, with output order $(\operatorname{Re}\zeta_1, \operatorname{Im}\zeta_1, \operatorname{Re}\zeta_2, \operatorname{Im}\zeta_2)$, is

$$\begin{pmatrix} 6\operatorname{Re}\mu & \operatorname{Re}\tau & 1 & 0 \\ 6\operatorname{Im}\mu & T & 0 & 0 \\ \operatorname{Re}\beta & \operatorname{Re}\mu & 0 & 1 \\ \operatorname{Im}\beta & \operatorname{Im}\mu & 0 & 0 \end{pmatrix}.$$

Its last three columns are real-linearly independent because $T > 0$. Its determinant is TD, so at zero determinant the rank is exactly three. Substitution gives the displayed kernel, including its signs. If the image subgroup is discrete, it is a lattice of rank at most three in its real span. The homomorphism $\mathbb{Z}^4 \to \Pi\mathbb{Z}^4$ therefore has a nonzero kernel, forcing the displayed kernel vector with first coordinate one to be rational. Conversely a rational line has a primitive integral generator extendable to a basis of $\mathbb{Z}^4$. The remaining three basis images are a real basis of the image hyperplane and hence generate a discrete lattice. Its quotient leaves one unbounded real direction. Finally a non-discrete finitely generated subgroup of this real hyperplane cannot be closed: a closed subgroup is a product of a real vector group and a discrete group, and a nonzero real vector group is uncountable. A quotient topological group is Hausdorff exactly when its identity subgroup is closed. This proves the last assertion. □

Proposition (C.3.2; Proved)**.** *The unique degenerate base point at the actual admissibility boundary is $t = 1$. The boundary constants $c = x + i(3/\pi)\log 6$ give*

$$\ker_{\mathbb{R}} \Pi = \mathbb{R}(1, 3, -3, -x - 3/2).$$

For rational $x + 3/2 = m/n$ in lowest terms, the quotient is a finite holomorphic quotient of $E_i \times \mathbb{C}^*$, with a covering of degree four. For irrational $x + 3/2$, the closure of the period subgroup is generated by the real line $\mathbb{R}(0, 1)$ and the two periods $(i, (1 - i)/2)$, $(1, 0)$. Its Hausdorff quotient is a real manifold homeomorphic to $\mathbb{R} \times (S^1)^2$.

Proof. C.1.1 gives $\operatorname{Im}\beta = 3/2$ at this value of c. The vector $\widehat{\gamma} + 3\widehat{u} - 3\widehat{w}$ maps to $(0, x + 3/2)$, while $\widehat{\delta}$ maps to $(0, 1)$. For $x + 3/2 = m/n$ in lowest terms, change coordinates by $(\zeta_1, \zeta_2) \mapsto (\zeta_1, \zeta_2 + \zeta_1/2)$. The vectors $2\widehat{u} - \widehat{\delta}$ and $2\widehat{w} - \widehat{\delta}$ then have periods $(2i, 0)$ and $(2, 0)$. Bezout's identity applied to $\widehat{\gamma} + 3\widehat{u} - 3\widehat{w}$ and $\widehat{\delta}$ gives the period $(0, 1/n)$. These three periods generate a sublattice of index four in the rank-three image: its two remaining generators are represented by $(i, 1/2)$ and $(1, 1/2)$, each of order two modulo this sublattice. The corresponding quotient is $E_i \times (\mathbb{C}/(n^{-1}\mathbb{Z}))$, and $[z] \mapsto e^{2\pi i n z}$ identifies its second factor with $\mathbb{C}^*$. Lattice inclusion gives the claimed degree-four map. For irrational $x + 3/2$, its additive subgroup with 1 is dense in the indicated real line. The other two displayed periods are independent modulo that line. This proves the closure and the quotient description. By C.2.3 this is the only degenerate torus cover at the admissibility boundary; every other finite-base period matrix retains real rank four. We have described the torus cover of the order-four filling, including the actual quotient and its topology. □

Remark (C.3.3; Open)**.** The remaining compactification problem at the actual boundary is now localized at the order-four fibre over $t = 1$. Its smallest undecided instance is a compact complex-space replacement of the rank-three quotient in C.3.2, over a neighbourhood of one boundary parameter, with a proof that it extends the original cyclic filling and its overlap with the ordinary torus family. Both rational and irrational values of $x + 3/2$ are included in the exact quotient description. The naive quotients cannot be central fibres of a proper complex-space family: for rational $x + 3/2$ they are noncompact, and for irrational values they are non-Hausdorff. C.3.4 supplies a further obstruction to preserving a holomorphic Weierstrass Albanese model in this parameter. These prove failure of the stated specific replacements. A compact modification that changes that model has not been constructed or ruled out. The requested general complex-space boundary filling remains unfinished, with its unique base-fibre location and exact local period obstruction determined.

Proposition (C.3.4; Proved)**.** *On its half-plane of definition, the j-invariant of* $\mathrm{Alb}(S_2)$ *has no meromorphic continuation across any point of the line* $\operatorname{Im} c = (3/\pi)\log 6$*. In particular the rank-three quotient at $t = 1$ cannot be replaced by a Weierstrass family of elliptic Albanese curves whose coefficients are holomorphic in the same parameter across that line.*

Proof. The explicit Albanese lattice in A.4 is $\mathbb{Z} + (\kappa_2/2)\mathbb{Z}$, with $\kappa_2 = c + 3/2 - i(3/\pi)\log 6$ by C.1.1. A positive-imaginary modulus is $z = -\kappa_2/2$, so the invariant is $j(z)$. At a reduced rational $r = a/b$ on the real boundary, choose an integral determinant-one modular matrix with bottom row $(b, -a)$. Its value at $r + iy$ has imaginary part $1/(b^2 y)$. The convergent Fourier expansion $j = q^{-1} + 744 + O(q)$ therefore gives growth of order $\exp(2\pi/(b^2 y))$ as $y \downarrow 0$. A meromorphic function at r is bounded by a fixed power of $1/y$, so continuation there is impossible. A meromorphic continuation near an irrational real point would also continue the same function near some rational real point in that neighbourhood, which has just been excluded. The affine change from c to z transfers this obstruction to the stated line. A Weierstrass family with holomorphic coefficients has a meromorphic j-invariant, expressed as the quotient of its cubic invariants, so it cannot supply the indicated continuation. This obstruction concerns the stated Albanese construction; it does not silently assume that every possible compact threefold filling preserves an Albanese model of its special surface. □

3.4 C.4. Classification of the non-normal cusp surface

Theorem (C.4.1; Proved)**.** *For the normalization of C.1.1, the three opposite-side identification constants of W_c, in the characters specified below, are*

$$(g_1, g_2, g_3) = (-a^{-1}, a^6, -a^{-7}e^{-2\pi i c}).$$

Two such compact complex surfaces are biholomorphic exactly when $c' - c \in \mathbb{Z}$. The vertical $\mathbb{C}^*$-action inherited from translation in the $\widehat{\delta}$-direction fixes exactly the double curve represented by the pair of edges of direction e_2.

Proof. Keep the lattice of toric cocharacters $\mathbb{Z}^2$, with $e_1 = \widehat{w}$, $e_2 = \widehat{\delta}$, and cusp logarithm $s = (2\pi i)^{-1}\log t_c$. The cusp period matrix is $[sB_0 + C(t_c) \mid I]$, where

$$B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C(0) = \begin{pmatrix} 6\mu_0 & h_0 \\ b_0 - h_0 & \mu_0 \end{pmatrix}, \quad b_0 = c.$$

For $v = e_1, e_2, e_1 - e_2$, the gluing map from the side of direction v to its opposite is the restriction of the toric action with lattice label $B_0 v$. A character row r with $rv = 0$ pulls back by this map to itself multiplied by $\exp(2\pi i\, rC(0)B_0 v)$: the remaining power of t_c is $rB_0^2 v = -rv = 0$. Taking respectively $r = (0,1), (1,0), (1,1)$ gives exponents $-\mu_0, 6\mu_0, -7\mu_0 - c$. This proves the

three constants, including the cancellation of h_0. These are morphisms of the whole projective boundary lines, since multiplication by a nonzero constant extends over both endpoints. The restrictions agree at both triple points, as the ambient toric action supplies the same vertex identifications.

A biholomorphism of the two surfaces lifts uniquely to a biholomorphism of their normalizations: integral closure of the reduced local rings is functorial under an isomorphism. It preserves the inverse image of the singular locus, the hexagon on dP_6. Its permutation of the six components preserves their cyclic incidence, hence lies in the order-twelve dihedral group. After composition with the corresponding toric lattice automorphism, it fixes each boundary component. Each torus character has its divisor supported on those components, so its pullback divided by itself has divisor zero on the compact normalization and is constant. Such an automorphism is therefore multiplication by an element of $(\mathbb{C}^*)^2$. These torus multipliers cancel in the opposite-side comparison because the same character is used on both sides.

The remaining six effective actions on the triple, with the central lattice automorphism $-I$ acting trivially, are described by the equations $g'_{k_j}{}^{\epsilon_j} = g_j$, where

$$((k_1,\epsilon_1),(k_2,\epsilon_2),(k_3,\epsilon_3)) \in \begin{cases} ((1,+),(2,+),(3,+)), & ((1,-),(3,+),(2,+)), \\ ((2,-),(3,-),(1,+)), & ((2,+),(1,+),(3,-)), \\ ((3,-),(2,-),(1,-)), & ((3,+),(1,-),(2,-)). \end{cases}$$

This finite calculation is reproduced by `cd_exact.py`, enumerating all integral automorphisms of the six rays and transporting the characters. Every nonidentity action requires one of $g_1^{-1} = g_1$, $g_2^{-1} = g_2$, or equality of a positive real number $g_2^{\pm 1}$ with a negative real number $g_1^{\pm 1}$. All fail for $a > 1$. Therefore $g'_3 = g_3$, precisely $e^{-2\pi i c'} = e^{-2\pi i c}$. Conversely equality of the three constants gives the identity gluing morphism on the normalization and hence a biholomorphism of the quotients. This proves the unmarked assertion.

For the action of the subtorus with cocharacter e_2, a one-dimensional toric orbit on the normalization is pointwise fixed exactly when its stabilizer line contains e_2. Among the six rays this occurs exactly for $\pm e_2$. Their images form one double curve; all zero-dimensional orbits map to its two endpoints. No dense two-dimensional orbit is fixed. The quotient action has the same fixed set: a continuous orbit cannot jump between the finitely many branches of a normalization fibre. □

4 D. Exceptional fibres, Neron–Severi classes, and compact subspaces

4.1 D.1. Elliptic quotients and the torsion locus

Proposition (D.1.1; Proved)**.** *The specified linear projection* $\mathrm{pr}_1 : \mathbb{C}^2 \to \mathbb{C}$ *induces a holomorphic group homomorphism* $F_t \to E_t = \mathbb{C}/(\mathbb{Z}\tau + \mathbb{Z})$ *exactly on* $6\mu = A\tau + B$*,* $A, B \in \mathbb{Z}$*. It is then surjective with connected kernel* $\mathbb{C}/(\mathbb{Z} + (A\mu - \beta)\mathbb{Z})$*. On the smooth base the two values of* t *are precisely*

$$t_\pm = \frac{64(47 \pm 27\sqrt{3})}{11}.$$

Proof. The images of the four lattice generators under the linear projection are $6\mu, \tau, 1, 0$. This proves descent in both directions. The integral basis change replacing $\widehat{\gamma}$ with $\widehat{\gamma} - A\widehat{u} - B\widehat{w}$ changes its period to $(0, \beta - A\mu)$; the remaining vertical generator is $(0, 1)$. The other two periods map onto the entire target lattice, so the kernel is connected. Its second modulus has imaginary part $A \operatorname{Im}\mu - \operatorname{Im}\beta = -D > 0$.

The elliptic model may be scaled by the explicit isomorphism $(X,Y)\mapsto(2X,2\sqrt{2}Y)$ to

$$Y^2=X^3-\frac{3}{4}t^3(t-1)X+\frac{1}{4}t^4(t-1)^2,\qquad P=(0,t^2(t-1)/2).$$

Its sixth division polynomial at P is

$$\psi_6(P)=\frac{9}{65536}t^{24}(t-1)^9(11t^2-6016t+8192).$$

For a smooth elliptic fibre, $\psi_n(P)=0$ is exactly the condition $nP=O$, since the multiplication rational map has these and only these poles in its affine coordinate. The two roots displayed above are simple and avoid $0,1$. We computed the same equation independently by the chord-and-tangent addition algorithm, obtaining

$$\begin{aligned}x(6P)&=\frac{(89t^2+4736t-4096)Q(t)}{1296(11t^2-6016t+8192)^2},\\ Q(t)&=881t^4+4745344t^3-12079104t^2\\&\quad+24641536t-16777216.\end{aligned}$$

The division-polynomial recurrence and the addition algorithm are in separate supplied programs; all polynomial operations use exact rational arithmetic. Thus the denominator has no unverified numerical root or cancellation. The decimal values 1.36510950191 and 545.543981407 are only convenient approximations to the exact roots. □

Proposition (D.1.2; Known, with derivation)**.** *The locus on B° where $P(t)$ is torsion is countable and dense. Every fibre over this locus is an abelian surface. The divisor on the smooth base cut out by $6kP=O$ has degree $3k^2-1$, including local intersection multiplicity, for every positive integer k.*

Proof. On a simply connected open subset write uniquely $\mu=a_0\tau+b_0$, a_0,b_0 real. The section is torsion precisely when $(a_0,b_0)\in\mathbb{Q}^2$. In a local complex parameter z, direct differentiation gives

$$\det\frac{\partial(a_0,b_0)}{\partial(\operatorname{Re}z,\operatorname{Im}z)}=-\frac{|\mu'(z)-a_0\tau'(z)|^2}{\operatorname{Im}\tau}.$$

This determinant vanishes on no open set. On an open set of vanishing, the two real differential equations imply $da_0=db_0=0$, hence $\mu=a_0\tau+b_0$ with constant real coefficients. Analytic continuation would give this identity on the cover. The cusp law $(\tau,\mu)\circ g_0=(\tau-1,\mu)$ then forces $a_0=0$, and the g_2-law makes a constant μ impossible. Thus every open subset contains a point where this real map is a local diffeomorphism, and its image contains rational pairs. This proves density. For each integer triple (n,A,B), $n\neq0$, the function $n\mu-A\tau-B$ is holomorphic and not identically zero by the same argument. Its zeros are discrete; a countable union proves countability.

For $N\mu=A\tau+B$, $A,B,N\in\mathbb{Z}$, $N>0$, use the linear coordinate change $\zeta_2'=\zeta_2-(A/N)\zeta_1$. The four integral vectors

$$N\widehat{u}-B\widehat{\delta},\quad N\widehat{w}+A\widehat{\delta},\quad N\widehat{\gamma}-6A\widehat{u}-6B\widehat{w},\quad \widehat{\delta}$$

span a sublattice of index N^3. Its first two periods lie on $\zeta_2'=0$, and its last two lie on $\zeta_1=0$. Both intersections are genuine rank-two lattices because $D<0$. Their product quotient is therefore a product of elliptic curves, and inclusion of this sublattice induces a finite surjective holomorphic homomorphism onto F_t. A positive integral form on the product can be summed over its finite translation kernel and multiplied by the index to descend an integral positive form to F_t. The Riemann polarization criterion proves projectivity.

For the degree statement we use the height computation actually read in [Eng26, Lemma 2.1 and proof]: $\langle P, P\rangle = 1/6$, $\chi(\mathcal{O}_S) = 1$, the component groups at the two reducible fibres have orders three and two, and the local corrections for P are $4/3$ and $1/2$. The application to $6kP$ is as follows. The corrected divisor class of a section is its class minus O and a fibre multiple, plus the unique rational combination of nonidentity fibre components that makes it perpendicular to those components. On restriction to the generic fibre this assignment is additive because the divisor class $[Q] - [O]$ is the group law in Pic^0. The perpendicular correction makes its self-intersection the negative of the height, so the height is quadratic. The element $6kP$ is in the identity component at both reducible fibres; its correction terms vanish. Hence

$$6k^2 = (6k)^2\langle P, P\rangle = 2 + 2((6kP) \cdot O),$$

which gives the formula. There are no missing intersection points at the three special fibres. The computed rational functions for $6P$ show that it has finite affine coordinates at $0, 1$, so it is a nonzero point of each additive identity component; multiplication by k remains nonzero in $\mathbb{C}$. At infinity its multiplicative coordinate is $(-a)^{6k} = a^{6k} \neq 1$. Thus all intersections counted by the height occur on B°. This derives the cited height input in precisely the form needed here; the first five degrees and the correction matrices are checked exactly in `cd_torsion_independent.py`. □

Theorem (D.1.3; Refuted)**.** *The assertion in Lead 6 that* existence of a surjection onto the specified E_t with elliptic kernel *is equivalent to* $6P(t) = O$ *is false. It remains true for the specified projection in D.1.1.*

Proof. Set $t = 32/5$. The exact fourth division polynomial is

$$\psi_4(P) = -\frac{1}{32}t^{10}(t-1)^4(5t-32),$$

whereas $\psi_2(P) = t^2(t-1) \neq 0$. Thus P has exact order four. On the rectangular branch write $\tau = iT$, $T > 1$, with $j(iT) = 1728 \cdot 32/5$. The chosen zero has real part $1/2$, negative imaginary part of absolute value less than $T/2$, and order four; therefore $\mu = (2-\tau)/4$. Choose the concrete parameter

$$c = \left(\frac{9}{8} - \frac{101}{4}\right)\tau - \frac{9}{4} - \beta^{\mathrm{part}}(t).$$

It is globally admissible. Indeed the positive-real Jacobi product at this (μ, τ) gives $\operatorname{Im}\beta^{\mathrm{part}} \geq -T - (3/\pi)\log 3$. Consequently $\operatorname{Im} c \leq -185T/8 + (3/\pi)\log 3 < -22 < -M$, by C.2.1. This inequality proves admissibility without using the numerical estimate $c = -2.25 - 34.02109724218209472308\,i$.

The holomorphic linear map $L : \mathbb{C}^2 \to \mathbb{C}$, $L(\zeta) = 3\zeta_1 + 4\zeta_2$, takes the four periods to

$$-101\tau, \qquad 2\tau + 2, \qquad 3, \qquad 4.$$

These generate exactly $\mathbb{Z}\tau + \mathbb{Z}$: the last two generate 1, and the first two then generate τ, since $\gcd(101, 2) = 1$. Thus L descends to a surjective holomorphic group homomorphism $F_{32/5} \to E_{32/5}$ with connected one-dimensional kernel. The kernel is compact as a closed subset of a compact torus, hence an elliptic curve. Nevertheless $6P = 2P \neq O$. Every equality, the division-polynomial noncancellation, and the four lattice images are checked by exact arithmetic in the supplied code. □

Proposition (D.1.4; Proved)**.** *For every smooth fibre and every admissible c, its Neron–Severi lattice is the following explicit subgroup of $\mathbb{Z}^6$:*

$$\left\{E : \; E_{12} - E_{13}\tau + (6E_{23} - E_{14})\mu + E_{24}\beta + E_{34}(6\mu^2 - \tau\beta) = 0\right\}. \tag{10}$$

The complex orientation has intersection form $E^2 = -2\,\mathrm{Pf}(E)$*, where* $\mathrm{Pf}(E) = E_{12}E_{34} - E_{13}E_{24} + E_{14}E_{23}$*. In particular the invariant class* $\eta = 6\gamma \wedge \delta + u \wedge w$ *has* $\int_{F_t} \eta^2 = -12$*. Its Hermitian form is indefinite. The fibres with lattice exactly* $\mathbb{Z}\eta$ *have no compact curves and algebraic dimension zero.*

Proof. On the universal cover the holomorphic two-form is the wedge of the two rows of Π. Its six coefficients in the integral coframe are $6\mu^2 - \tau\beta, -\beta, 6\mu, -\mu, \tau, 1$. An integral real two-form is of type $(1,1)$ exactly when its wedge with this form vanishes: its $(0,2)$ part is then zero, and reality also annihilates its $(2,0)$ part. Expanding the wedge gives (10). The exponential sequence identifies these integral $(1,1)$ classes with Chern classes of holomorphic line bundles, proving the lattice assertion.

The real period matrix in C.3.1 has determinant $TD < 0$. Thus the integral orientation $(\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta})$ is opposite to the complex orientation. Algebraically $E \wedge E = 2\,\mathrm{Pf}(E)\gamma \wedge u \wedge w \wedge \delta$, which proves the sign in the displayed intersection formula. In particular the expression "$\eta^2 = 12$" in AC26, Lemma 9.4 is correct as its displayed coefficient of $\gamma \wedge u \wedge w \wedge \delta$, and has the opposite sign as a complex-oriented numerical self-intersection. We retain the coefficient identity and record the orientation comparison explicitly.

For the precise Hermitian convention H_E defined in D.1.6, direct substitution of the four periods gives $H_\eta = -2G^{-1}$, where

$$G = \begin{pmatrix} 2T & 2U \\ 2U & \operatorname{Im}\beta/3 \end{pmatrix}, \qquad \det G = 2TD/3 < 0.$$

Thus H_η has one positive and one negative eigenvalue, and every nonzero multiple remains indefinite. The integration current of an effective divisor, averaged under translations of the compact torus, is a nonzero semipositive constant $(1,1)$-form in its Chern class. It cannot be an indefinite multiple of η. Thus no curve exists when $\mathrm{NS} = \mathbb{Z}\eta$. A nonconstant meromorphic function would have a nonempty polar divisor (after subtracting a constant if necessary), so the algebraic dimension is zero. □

Proposition (D.1.5; Proved)**.** *For a fixed* c*, all but countably many smooth fibres have* $\mathrm{NS} = \mathbb{Z}\eta$*. Conversely, for every fixed smooth* t*, there are admissible parameters tending to the cusp of the parameter disc for which* F_t *is an abelian surface, irrespective of the order of* $P(t)$*.*

Proof. We verify the required independence of five holomorphic functions. Suppose $A + B\tau + C\mu + D\beta + E(6\mu^2 - \tau\beta) = 0$. Apply g_0, using $(\tau, \mu, \beta) \circ g_0 = (\tau - 1, \mu, \beta + 1)$, and subtract. The result is $-B + D + E(\beta - \tau + 1) = 0$. A second cusp difference gives $2E = 0$, hence $E = 0, D = B$. Applying g_2 to the remaining relation and multiplying the difference by $-\tau$ gives

$$6B\mu^2 + C(\tau - 1)\mu + B(\tau^2 + 3\tau + 1) - C\tau = 0.$$

Its cusp difference gives $C\mu + 2B(\tau + 1) - C = 0$, whose next cusp difference forces $B = 0$. Then $C\mu = C$; the g_2-law forces $C = 0$, and finally $A = D = 0$. Thus an integral class outside $\mathbb{Z}\eta$ has a nonidentically-zero holomorphic left side in (10); its zeros are discrete. Countability of $\mathbb{Z}^6$ proves the first assertion, with explicit verification of its reduction.

For the second assertion put $\beta_n = -3n^2\tau + 12n\mu$, $c_n = \beta_n - \beta^{\mathrm{part}}(t)$. Its imaginary part tends to minus infinity, so the already proved finite bound on M makes every sufficiently large n admissible. The integral alternating matrix with nonzero upper entries

$$E_{13} = -3n^2, \quad E_{14} = 6n, \quad E_{23} = -n, \quad E_{24} = 1$$

satisfies (10) exactly at β_n, and $\mathrm{Pf}(E) = -3n^2 < 0$. Its complex-oriented square is positive. A nondegenerate Hermitian form in complex dimension two with positive determinant is definite. Choosing its sign gives a positive integral Riemann form, so F_t is projective. This supplies an explicit family of projective fibres even at any chosen non-torsion value of the section. □

Theorem (D.1.6; Known, with derivation)**.** *Fix any admissible c. For each integral alternating matrix E, let $Z_E(c)$ be the zero set on the marking cover of the explicit holomorphic function in (10), with $\beta = \beta^{\mathrm{part}} + c$ given by (7). Let π be the cover map to B°, and define*

$$\mathcal{E}_c = \pi\left(\bigcup_{E\neq 0,\ \mathrm{Pf}(E)=0} Z_E(c)\right), \qquad \mathcal{A}_c = \pi\left(\bigcup_{\mathrm{Pf}(E)<0} Z_E(c)\right).$$

These give the exact loci of all elliptic quotients and all abelian surfaces, respectively. The algebraic dimension of every smooth fibre is

$$a(F_t) = \begin{cases} 2, & t \in \mathcal{A}_c, \\ 1, & t \in \mathcal{E}_c \setminus \mathcal{A}_c, \\ 0, & t \notin \mathcal{E}_c \cup \mathcal{A}_c. \end{cases}$$

Together with (10), these are an exact classification for every t, by countable unions of explicitly given analytic zero sets; they do not assert that the loci have a finite algebraic description. They include all quotient maps, without requiring preservation of the first coordinate.

Proof. We apply the Appell–Humbert correspondence and the Riemann projectivity criterion in [Sch19, Theorems 4.1 and 4.4]; the reduction to the present explicit periods is as follows. The real vector-space isomorphism $\Pi : \mathbb{R}^4 \to \mathbb{C}^2$ transfers E to the actual alternating form $\widetilde{E}(v,w) = E(\Pi^{-1}v, \Pi^{-1}w)$ on the tangent space. Equation (10) gives J-invariance. The explicit comparison is

$$H_E(v,w) = \widetilde{E}(v, Jw) - i\widetilde{E}(v,w).$$

Direct substitution gives $H_E(Jv,w) = iH_E(v,w)$ and $H_E(w,v) = \overline{H_E(v,w)}$. Its associated real Chern form is $-\operatorname{Im} H_E = \widetilde{E}$, since $(i/2)(H_E(v,w) - H_E(w,v)) = \widetilde{E}(v,w)$. Thus the Appell–Humbert integrality condition is exactly integrality of the given E, with its sign fully specified. The complex-oriented square computed in D.1.4 shows that $\mathrm{Pf}(E) < 0$ is equivalent to definiteness of this Hermitian form. Its sign can be chosen positive. The cited projectivity criterion then gives precisely the asserted abelian locus, in both directions.

For nonzero E with Pfaffian zero, its real rank is two. Its nullspace $K_{\mathbb{R}} \subset \mathbb{R}^4$ is a rational plane because it is the kernel of an integral matrix. J-invariance makes $\Pi K_{\mathbb{R}}$ a complex line, and $K = K_{\mathbb{R}} \cap \mathbb{Z}^4$ is saturated. The maps

$$(\Pi K_{\mathbb{R}})/\Pi K \ \hookrightarrow F_t \ \longrightarrow\ \mathbb{C}^2/(\Pi K_{\mathbb{R}} + \Pi\mathbb{Z}^4)$$

are holomorphic group morphisms; the first is a closed embedded elliptic curve and the second is its quotient onto an elliptic curve, with connected kernel. A basis of the saturated K extends to an integral basis of $\mathbb{Z}^4$. The other two periods project to real-linearly independent vectors in the one-dimensional complex quotient, because Π is a real isomorphism. They therefore form a discrete lattice there. Compactness follows by taking the image of its compact real fundamental parallelogram. Conversely any elliptic quotient pulls an integral positive area form back to a nonzero rank-two integral $(1,1)$ form. This proves the characterization of $\mathcal{E}_c$. Equivalently all its kernels can be found by saturated rank-two lattices with a basis v, w satisfying $\det_{\mathbb{C}}(\Pi v, \Pi w) = 0$ and real independence. This is a second exact integer test, with explicit inclusion and quotient morphisms.

For the algebraic-dimension assertion, an effective divisor on a torus has a semipositive Hermitian form: averaging its integration current under translations gives that constant form. A nonzero semipositive form has rank one or two. Rank two gives the abelian case. Rank one gives the elliptic quotient just constructed. In the absence of a definite form, there cannot be two different such kernel subtori. Indeed addition from their product onto F_t would be

a surjective holomorphic homomorphism with finite kernel, and the product polarization would descend after multiplication by an integer, giving a definite form. Thus there is a single kernel K_t in this case. Every effective divisor is a sum of its translates: its intersection number with K_t is zero, whereas a curve mapping nonconstantly to the elliptic quotient has strictly positive intersection with a general quotient fibre. Every meromorphic function therefore factors through this quotient. One can check the last statement on a fibre avoiding its polar divisor, where it is a holomorphic function on a compact elliptic curve and hence constant; analytic continuation then gives the factorization. The quotient supplies nonconstant meromorphic functions, so the algebraic dimension is exactly one. If neither locus occurs, a nonconstant meromorphic function would have a polar divisor of one of the two forbidden ranks. This proves dimension zero. Finally monodromy acts by integral determinant-one changes of basis, preserving (10) and its Pfaffian. Thus the stated unions descend without any marking ambiguity. Each individual matrix and every locus is specified; no expected identification is left in these formulas. □

Proposition (D.1.7; Proved)**.** *There is an explicit nonempty locus of admissible members with ordinary fibres of algebraic dimension one. On a sufficiently small punctured cusp neighbourhood set*

$$c(z) = -6\tau(z) - 12\mu(z) - \beta^{\mathrm{part}}(z).$$

Outside a countable set of z, the fibre of $X_{c(z)}$ over $\pi(z)$ has Neron–Severi lattice $\mathbb{Z}\eta + \mathbb{Z}E$, where the nonzero upper entries of E are $E_{13} = -6$, $E_{14} = -6$, $E_{23} = 1$, $E_{24} = 1$. Its algebraic dimension is one. The section $P(\pi(z))$ can simultaneously be required to be non-torsion.

Proof. At the specified parameter $\beta = -6\tau - 12\mu$. Both E and η satisfy (10), and exact multiplication gives $\mathrm{Pf}(a\eta + bE) = 6a^2$. For any further integral class F, substitution in its period equation gives a linear relation among $1, \tau, \mu, (\mu+\tau)^2$. These four functions are linearly independent: applying g_0 to a relation $A + B\tau + C\mu + D(\mu+\tau)^2 = 0$ and subtracting yields $-B + D(1 - 2(\mu+\tau)) = 0$. A further cusp difference forces $D = 0$, and then $B = 0$; nonconstancy of μ forces $A = C = 0$. Comparison of coefficients now gives $F_{34} = F_{12} = 0$, $F_{13} = -6F_{24}$, and $6F_{23} - F_{14} = 12F_{24}$, exactly $F \in \mathbb{Z}\eta + \mathbb{Z}E$. Every other integral class therefore has a discrete zero set along this period locus. Excluding their countable union gives the asserted exact Neron–Severi lattice. It contains the nonzero isotropic E and no matrix of negative Pfaffian; D.1.6 gives algebraic dimension one. Finally $c = -5\tau - 12\mu - (\beta^{\mathrm{part}} + \tau)$ has imaginary part tending to minus infinity at the cusp. A sufficiently small cusp neighbourhood is therefore admissible by C.2.1. The torsion locus of P is countable by D.1.2, so its exclusion still leaves such points in every open subset of that neighbourhood. This supplies actual members and fibres, with all their parameter functions specified. □

4.2 D.2. Compact curves and surfaces

Proposition (D.2.1; Proved)**.** *For a standard-twist member X_c with underlying manifold S^6, every compact irreducible complex curve lies in one fibre of f, and every compact irreducible complex surface is one of the reduced fibres.*

Proof. The given diffeomorphism with S^6 implies $H^2(X_c; \mathbb{Z}) = 0$. For a compact irreducible curve C, let $\nu : \widetilde{C} \to C \hookrightarrow X_c$ be its normalization map. The holomorphic map $f\nu$ has $\deg(f\nu)^*\mathcal{O}_{\mathbb{P}^1}(1) = 0$, since its Chern class is pulled back from the zero class on X_c. A nonconstant holomorphic map of compact Riemann surfaces has positive degree: a regular value has a positive number of preimages, all with positive local multiplicity. Therefore $f\nu$, and hence $f|_C$, is constant.

An irreducible surface Z in the smooth threefold is an effective Cartier divisor. If its image were all of B, a general smooth fibre would meet it in a nonempty effective divisor D whose

line bundle is $\mathcal{O}_{X_c}(Z)|_{F_t}$. Its integral Chern class is zero. Choose a flat Kähler form ω on the torus F_t. The equality $\int_{F_t} c_1(D) \wedge \omega = \int_D \omega$ has left side zero and right side positive, a contradiction. Thus Z lies in a single fibre. Each reduced fibre is irreducible of dimension two, so containment forces equality. This uses the full map into X_c, not a transfer of a vanishing claim from its normalization. □

Proposition (D.2.2; Proved)**.** *There is no rational curve in $X_c \setminus W_c$. There are elliptic curves outside all three special fibres, in particular in each of the two smooth fibres $F_{t_\pm}$. All curves in W_c are precisely the images of irreducible algebraic curves in* dP_6 *under its finite normalization map. Distinct curves have the same image only for the three paired boundary lines. All curves in a bielliptic fibre S_j are images of irreducible curves in its finite product cover.*

Proof. By D.2.1 every curve is vertical. A holomorphic map from $\mathbb{P}^1$ into a torus lifts to its vector-space universal cover, since $\mathbb{P}^1$ is simply connected, and its coordinate functions are constant by compactness. Thus a smooth torus fibre contains no rational curve. A map from $\mathbb{P}^1$ into a bielliptic surface lifts through its finite unramified torus cover by the same covering criterion; the lifted map is constant. These arguments prove the first assertion, including possibly singular rational images. D.1.1 gives an embedded connected elliptic kernel in each $F_{t_\pm}$, proving the second.

For the third assertion let $\eta : \mathrm{dP}_6 \to W_c$ be the finite normalization morphism. Each component of the inverse image of a compact curve is compact and one-dimensional and maps onto that curve, because finite maps have zero-dimensional fibres. It is algebraic in the projective surface dP_6. Conversely its image is a compact analytic curve by properness. Away from the conductor, η is a biholomorphism, so two distinct curves can have the same image only when both lie in the conductor; its identifications are exactly the three opposite pairs computed in C.4.1. This is also an explicit algebraic parametrization: the curves of dP_6 are the three exceptional lines and strict transforms of irreducible plane curves under the blow-up of the three coordinate points. Their homogeneous equations and their three multiplicities specify the strict transforms. The finite-product cover statement for S_j follows from the identical proper-image and finite-preimage argument applied to its unramified morphism $E_j \times F_j \to S_j$; curves in the product are algebraic. □

Theorem (D.2.3; Known, with derivation)**.** *The following descriptions give all compact irreducible curves of X_c. In W_c and S_j they are exactly the images described in D.2.2. In a smooth fibre of algebraic dimension zero there are none. In a smooth fibre of algebraic dimension one they are exactly the translates of its unique elliptic kernel K_t. In an abelian fibre they are exactly the translates of elliptic subtori, together with the irreducible zero divisors of the positive Appell–Humbert theta sections described below. These descriptions, D.1.6, and D.2.1 constitute a classification including every smooth exceptional fibre and every compact irreducible surface.*

Proof. All curves are vertical by D.2.1, so the list of fibre types is exhaustive. The dimension-zero and dimension-one assertions, including uniqueness, were proved in D.1.6. Let now $F_t = \mathbb{C}^2/\Pi\Lambda$ be abelian. For every positive definite Hermitian form H whose imaginary part takes integer values on $\Pi\Lambda$, and every function $\alpha : \Pi\Lambda \to U(1)$ satisfying

$$\alpha(\lambda + \nu) = \alpha(\lambda)\alpha(\nu)e^{\pi i \operatorname{Im} H(\lambda,\nu)},$$

take the nonzero entire solutions $\theta : \mathbb{C}^2 \to \mathbb{C}$ of

$$\theta(\zeta + \lambda) = \alpha(\lambda)e^{\pi H(\zeta,\lambda)+\pi H(\lambda,\lambda)/2}\theta(\zeta). \tag{11}$$

The cocycle identity follows by multiplying the displayed factors; its integral imaginary cross-term has exponential one. It defines a holomorphic line bundle by quotienting $\mathbb{C}^2 \times \mathbb{C}$. The

invariant zero set of θ descends to a compact divisor on F_t. Retain precisely those zero divisors that are reduced and irreducible. This is an exact parametrization, with all periods and multiplier equations given; multiplying θ by a nonzero scalar does not change its divisor.

The correspondence [Sch19, Theorem 4.1 and equation (4.3)] proves that every holomorphic line bundle and every section on this actual quotient has this form. For the cohomology matrix the exact comparison is $H_E = \widetilde{E}(\,\cdot\,, J\,\cdot\,) - i\widetilde{E}$ from D.1.6; we choose the sign of E making H_E positive. Its imaginary part is $-\widetilde{E}$, which gives the integral semicharacter equation displayed above. A divisor class with positive Hermitian rank two gives exactly these positive theta sections. A divisor of Hermitian rank one is pulled back from its elliptic quotient and, when irreducible and reduced, is a single fibre. Thus it is a translate of an elliptic subtorus. The null lattice is computed by the explicit integer kernel in D.1.6. No other rank is possible for an effective nonempty divisor, by positivity of its averaged current. This proves exhaustiveness on the abelian fibre.

The same equations give the curves in the product covers of S_j: compute them on its two-elliptic-curve lattice and take images under the specified finite morphism. Conversely each curve downstairs is the image of a component of its finite inverse image, so that this procedure omits none. In W_c the analogous procedure uses the homogeneous polynomial parametrization of the normalization in D.2.2, with the three boundary pairings already identified. The maps used throughout are finite or closed embeddings, so they preserve compactness and give one-dimensional images; no assertion is transferred from a normalization in an invalid direction. □

5 E. The limit at the puncture

5.1 E.1. The untwisted pinching and its local smoothings

In this section S is the smooth minimal rational elliptic surface with Weierstrass equation

$$y^2 = x^3 - 3t^3(t-1)x + 2t^4(t-1)^2,$$

$\pi : S \to B$ is its elliptic fibration, P is the section used in the period dictionary, $T = t_{6P} : S \to S$, and $L = M_S = \mathcal{O}_S(P - O)$. We use the convention that $\mathbb{P}_S(\mathcal{O} \oplus L)$ contains the total space of L, with fibre coordinate a, zero section s_0, and infinity section s_∞. The two normal bundles are respectively L and L^{-1}. We write $R = \mathbb{P}_S(\mathcal{O} \oplus L)$.

Proposition (E.1.1; Proved)**.** *The quotient*

$$Q = R/\{s_0(x) \sim s_\infty(Tx) : x \in S\}$$

has a compact, reduced complex-space structure. Its normalization is the finite holomorphic map $\nu : R \to Q$. Its singular locus is a copy of S, and its local analytic equation along that locus is $ab = 0$, with two further coordinates on S. The induced map $Q \to B$ is proper. Its sheaf of first-order local smoothing parameters is

$$\mathcal{T}^1_Q = \mathcal{E}xt^1_{\mathcal{O}_Q}(\Omega_Q, \mathcal{O}_Q) \simeq L \otimes T^*L^{-1} \simeq \pi^*\mathcal{O}_B(1)$$

on the double locus.

Proof. The two sections of R are disjoint closed submanifolds. The equivalence relation consists of their two graphs and the diagonal; it is closed in $R \times R$. A quotient of a compact Hausdorff space by a closed equivalence relation is Hausdorff: for distinct equivalence classes, compactness gives disjoint saturated neighbourhoods by taking finite subcovers of the complements of the relation. Thus the displayed quotient is compact and Hausdorff.

For a coordinate neighbourhood $U \subset S$ and frames of $L|_U$ and $L^{-1}|_{T(U)}$, the two tubular coordinate charts are (x,a) at $s_0(U)$ and (Tx,b) at $s_\infty(T(U))$. We give their identification the ring

$$\{(f(x,a),g(x,b)) : f(x,0)=g(x,0)\} = \mathbb{C}\{x_1,x_2,a,b\}/(ab).$$

Changes of frame are holomorphic and invertible on each branch and have the same change of x on their intersection. They therefore glue these hypersurface charts to the inherited charts off the sections. This constructs the asserted complex space. The direct sum of the two regular branch rings is the integral closure: an integral element in the total ring of fractions is integral on each smooth branch, and conversely the two branch idempotents are integral. The resulting map ν is proper with at most two points in each fibre, and its local ring extension is finite. The map to B is well defined because $\pi T = \pi$; its properness follows from compactness of Q.

The conormal presentation for the hypersurface $ab=0$ gives $\mathcal{E}xt^1(\Omega,\mathcal{O}) = \mathcal{O}/(a,b)$. A change of normal frames $a' = g(x)a$, $b' = h(x)b$ changes the equation and its smoothing parameter by gh. Consequently these local modules glue as $N_{s_0/R} \otimes T^*N_{s_\infty/R} = L \otimes T^*L^{-1}$. For this particular surface, Engel's Lemma 2.1 states $\mathrm{Hom}(T^*L, L) = \pi^*\mathcal{O}_B(1)$ [Eng26, Lemma 2.1]. Its identification applies here with exactly $T = t_{6P}$ and $L = \mathcal{O}(P-O)$; there is no replacement of the translation by its inverse in this normal-bundle calculation. One can recover the degree in that lemma directly: the bundle has multidegree zero on every fibre, since T preserves every component; hence it is $\pi^*\mathcal{O}_B(n)$. Its degree on O is

$$n = (O-P)\cdot 6P + (P-O)\cdot O = (2-2) + (0-(-1)) = 1.$$

The section intersections used here are the explicit computations in the proof of that lemma. This derives the displayed sheaf in the normal-bundle convention of our pushout. □

Proposition (E.1.2; Proved)**.** *Over $S^\circ = S \setminus S_\infty$, the pushout in E.1.1 is the central fibre of a holomorphic plumbing with parameter u. On every relatively compact open disc $U \Subset B \setminus \{\infty\}$ this plumbing is proper and flat over $U \times \Delta_\epsilon$, for a sufficiently small positive ϵ. Its fibre at $u \neq 0$ is, over U, Engel's untwisted quotient by the contracting generator $u\phi$ covering T^{-1}. This assertion includes discs containing 0 or 1; it is made before the logarithmic transformations at those points.*

Proof. Let ϕ be the section of $L \otimes T^*L^{-1}$ with divisor S_∞, as in Corollary 2.2 of [Eng26]. On S° it is nowhere zero. At the pair $s_0(x), s_\infty(Tx)$ we use the node chart

$$a \otimes b = u\phi_x, \qquad a \in L_x, \quad b \in L_{Tx}^{-1}. \tag{E1}$$

This equation is invariant under changes of the two line-bundle frames, so it defines a hypersurface chart over $S^\circ \times \Delta$. We describe the gluing, including the required quotient comparison. Choose a Hermitian metric on L over $\pi^{-1}(\overline{U})$, and $0 < r < 1/4$. On this compact set the norm of ϕ has a finite upper bound. Shrink ϵ so that $|u|\|\phi_x\| < r^2$. Retain the core $r < \|a\| < r^{-1}$ of each projective fibre. Attach the node chart with $\|a\| < 2r$ and $\|b\| < 2r$, using the core coordinate a on its first collar and the coordinate $1/b$ over Tx on its second collar. The metric used for b is the dual metric over Tx. The two attachment collars in a node chart are disjoint because $\|a\|\|b\| < r^2$. The transition functions and their inverses are holomorphic on the overlaps.

These charts are Hausdorff after gluing. Indeed, the smaller closed core $r \le \|a\| \le r^{-1}$ and the closed subcharts $\|a\|, \|b\| \le 3r/2$ are compact over a compact subset of $U \times \Delta_\epsilon$ and have precisely the prescribed closed identifications on their collars. They cover after a harmless enlargement of the open cores. The finite closed-chart description separates distinct points, as in E.1.1, and proves properness over $U \times \Delta_\epsilon$.

For $u \neq 0$, solving (E1) gives

$$a_{Tx} = 1/b = a_x/(u\phi_x).$$

This is exactly the inverse of the contracting bundle map $L_{Tx} \to L_x$, $a_{Tx} \mapsto u\phi_x a_{Tx}$. Thus the map from the annular charts to $L^\times/\langle u\phi\rangle$ is holomorphic and identifies exactly the orbits of this generator. Every orbit meets the core or an attached collar by contraction, and two such representatives are related by the displayed chart identification. The comparison map is consequently a biholomorphism. At $u = 0$, (E1) is the pinching chart $ab = 0$.

The total node chart is smooth because the derivative of its defining equation with respect to u is $-\phi_x \neq 0$. The projection to u is flat: its local rings are torsion free over the convergent one-variable power-series ring. The joint map to (t, u) is also flat. On the core it is a product of π with the parameter disc; π is flat since a nonconstant map from a smooth connected surface to a smooth curve has torsion-free local rings over the curve. On a node chart, $t - t_0$ and $u - u_0$ form a regular sequence in the regular total local ring: after setting $t = t_0$, the equation (E1) has no component contained in a level set $u = u_0$. The two elements cut dimension by two. The local flatness criterion for a regular base (equivalently its elementary regular-sequence proof) gives flatness over (t, u). This argument uses smoothness of the surface S, not smoothness of the Kodaira fibre; it therefore applies at every point of the fibres IV^* and III. □

Proposition (E.1.3; Proved)**.** *The first simultaneous local plumbing at the nodal point of S_∞ has equation*

$$xy = uzv, \qquad q = zv, \tag{E2}$$

after a holomorphic change of one normal coordinate, where q is a local coordinate on B at infinity. This hypersurface is flat over the u-disc. For each $u \neq 0$, its three-dimensional fibre has an ordinary double point at $x = y = z = v = 0$. The projective modification defined by the blow-up of (x, z) resolves that point for $u \neq 0$. Its two strict-transform charts are

$$x = zr, \quad ry = uv; \qquad z = xa, \quad y = uav. \tag{E3}$$

The modified family remains flat over u. At $u = 0$, the normalization of its central fibre contains the blow-up of one branch of (E2) along the smooth curve $x = z = 0$; hence this particular modification changes the normalization of the unmodified pinching.

Proof. The surface S is smooth and its nodal elliptic fibre has local equation $q = zv$ at its node. The section ϕ has a simple zero along that fibre. Thus $\phi = zv$ times an invertible holomorphic function. Absorbing the function in one normal coordinate gives (E2). The convergent power-series ring $\mathbb{C}\{x, y, z, v, u\}$ is factorial, and u is coprime to $xy - uzv$. Hence $uh \in (xy - uzv)$ implies $h \in (xy - uzv)$. The quotient has no u-torsion. At a nonzero parameter the identical argument uses $u - u_0$. This proves flatness over the smooth parameter curve. For fixed $u \neq 0$ the four first derivatives vanish only at the origin, and the quadratic form $xy - uzv$ is nondegenerate. This proves the ordinary-double-point assertion by its defining equation.

The two standard affine charts of the blow-up give (E3), after removing the common factors z and x, respectively. For $u \neq 0$ the first chart solves for v and the second for y, so both are smooth. The map is an isomorphism away from the origin in the fixed u-fibre: where the ideal fails to be visibly principal, the hypersurface equation and one of y, v make it principal. The fibre over the origin is $\mathbb{P}^1$. The modification is projective over (E2), since it is a closed subspace of its product with $\mathbb{P}^1$. The first strict-transform equation is coprime to $u - u_0$ in its convergent power-series ring for every u_0; the second chart is smooth with free parameter u. Thus flatness over u persists.

At $u = 0$ the first chart has $ry = 0$. The branch $r = 0$ maps isomorphically to $x = 0$ in the original central fibre. The branch $y = 0$, together with the second chart, is the ordinary

blow-up of the branch $y = 0$ in coordinates (x, z, v) along (x, z). This description follows from the two affine substitutions defining that blow-up and is an identification of normalizations, not just a birational map. For $u \neq 0$, the base coordinate in the first chart is $q = zry/u$ and in the second it is $q = xav$. Thus the modification also exhibits the triple-coordinate strata that the cusp filling requires. The substitutions and Jacobian are independently checked in `code/e_degeneration.py`. □

Proposition (E.1.4; Proved)**.** *The compact candidate Q of E.1.1 has*

$$\pi_1(Q) = \mathbb{Z}, \qquad (H_i(Q;\mathbb{Z}))_{i=0}^{6} = (\mathbb{Z}, \mathbb{Z}, \mathbb{Z}^8, \mathbb{Z}^7, \mathbb{Z}^{10}, 0, \mathbb{Z}), \qquad \chi(Q) = 12.$$

These are invariants of the stated candidate; they do not identify it as a central fibre of the prescribed compact family.

Proof. The two section embeddings are cofibrations: a tubular neighbourhood retracts onto each section, and the two neighbourhoods are disjoint. The ordinary pushout is therefore homotopy equivalent to the space formed from R by attaching $S \times [0, 1]$ along s_0 and $s_\infty T$. This comparison follows by collapsing the intervals in the mapping cylinder; the tubular retractions construct its homotopy inverse. Both S and R are simply connected. Van Kampen applied to this mapping-cylinder model gives one free loop and no additional generator or relation, so $\pi_1(Q) = \mathbb{Z}$. The same model has the exact sequence

$$H_i(S) \xrightarrow{d_i = s_{0*} - s_{\infty *} T_*} H_i(R) \longrightarrow H_i(Q) \longrightarrow H_{i-1}(S). \tag{E4}$$

We compute its integral maps, including primitivity.

Put $F = [S_t]$, $m = P - O$, and $v_0 = [6P] - [O] - 3F \in H_2(S;\mathbb{Z})$. The intersection computations in [Eng26, Lemma 2.1] give

$$F^2 = Fv_0 = Fm = 0, \quad v_0^2 = -6, \quad mv_0 = -1, \quad m^2 = -2, \quad OF = 1, \quad Ov_0 = 0, \quad Om = 1. \tag{E5}$$

The translation preserves all components of the two reducible Kodaira fibres. Their root classes span a negative definite space of rank seven. Its orthogonal complement is generated over $\mathbb{Q}$ by F, O, v_0. Since $T_*O = O + v_0 + 3F$, preservation of the intersection form gives, on this complement and hence on the entire lattice,

$$T_*a = a + (aF)v_0 - (av_0 - 3aF)F. \tag{E6}$$

For clarity, its value on v_0 is $v_0 + 6F$: it has no O-component and has zero intersections with the rank-seven root space, so it is $\lambda v_0 + aF$. The already established identity $T^*m = m - F$ gives $mT_*v_0 = (m - F)v_0 = -1$, forcing $\lambda = 1$. Orthogonality to T_*O then forces $a = 6$. These values on F, O, v_0 and the root space verify (E6) on a rational basis, and both sides preserve the actual integral lattice. In particular $T^*m = m - F$, in agreement with E.1.1.

The rational elliptic surface is the blow-up of $\mathbb{P}^2$ at nine points, so its homology in even degrees is $\mathbb{Z}, \mathbb{Z}^{10}, \mathbb{Z}$ and vanishes in odd degrees; its intersection lattice is unimodular. The projective-bundle theorem follows here by local trivialization and the fibre hyperplane class: $H_2(R) = H_2(S) \oplus \mathbb{Z}f$ and $H_4(R) = \mathbb{Z} \oplus H_2(S)$, with f the projective-line fibre. The hyperplane class that vanishes on s_0 restricts to $-m$ on s_∞. Therefore

$$d_2(a) = (1 - T_*)a + (mT_*a)f = -(aF)v_0 + (av_0 - 3aF)F + (am - aF)f. \tag{E7}$$

The span of F, v_0, m is primitive. To check the only nontrivial part of this assertion, split off the unimodular plane spanned by F, O; $m - F$ and v_0 lie in its negative definite orthogonal complement. Their Gram determinant is $\det \left(\begin{smallmatrix} -2 & -1 \\ -1 & -6 \end{smallmatrix} \right) = 11$. The square of an index of a larger

integral lattice would divide eleven, so the index is one. The span of F, v_0 is also primitive: pairing with O, m gives determinant -1. Unimodularity consequently makes $a \mapsto (aF, av_0, am)$ surjective onto $\mathbb{Z}^3$. The coefficient matrix in (E7) has determinant -1. Thus d_2 has primitive image $\mathbb{Z}v_0 \oplus \mathbb{Z}F \oplus \mathbb{Z}f$, rank three, kernel $\mathbb{Z}^7$, and cokernel $\mathbb{Z}^8$.

The map d_4 sends $[S]$ to $s_0[S] - s_\infty[S]$, whose Poincare dual is $\pi_R^* m$. This is primitive and nonzero since $mO = 1$. Thus d_4 is injective with cokernel $\mathbb{Z}^{10}$. The map d_0 is zero. Substitution in (E4) proves every asserted homology group. The Euler characteristic is independently obtained by stratification: $\chi(R) - \chi(S) = 24 - 12 = 12$. The Smith computations and both Euler computations are recorded in `code/e_degeneration.out`. □

Problem (E.1.5; Open, with the precise remaining construction)**.** The existence of a flat proper filling of the prescribed compact untwisted family, with central fibre exactly Q, remains undecided here. E.1.1 constructs the compact candidate as an actual complex space; E.1.2 constructs its smoothing away from infinity, including the two other special fibres; E.1.3 computes the first unresolved two-parameter germ and a concrete flat modification of it.

The smallest missing instance is the following analytic gluing problem at one lift of the node of S_∞. The local model $xy = uzv$ must be replaced by charts whose restriction to every $u \neq 0$ is the chosen small resolution of the rank-two Mumford filling, and these charts must be equivariant under both period generators, with proper quotient and with central normalization R. The explicit blow-up in E.1.3 supplies the required smooth local general fibre but changes the normalization at $u = 0$. We have not proved that an alternative gluing preserves it, nor proved that every alternative must change it. Accordingly this computation establishes an obstruction to the unmodified plumbing argument, and proves neither falsity nor truth of Conjecture 8.2. A modified compact limit also requires an equivariant periodic fan and verification of its quotient near the crossing $(u, q) = (0, 0)$; those data have not been constructed in this work.

5.2 E.2. Finite bielliptic limits and the smooth monodromy

Proposition (E.2.1; Proved)**.** *After the auxiliary base change $w = s^6$, each of the two families of* reduced *special surfaces $S_j(w)$ has a proper flat semistable filling by explicit finite free quotients. The central surface for $j = 1$ is a cycle of four ruled surfaces over an elliptic curve; that for $j = 2$ is a cycle of three. They have*

	H_0	H_1	H_2	H_3	H_4	χ
$j = 1$	$\mathbb{Z}$	$\mathbb{Z} \oplus \mathbb{Z}/3$	$\mathbb{Z}^5$	$\mathbb{Z}^9$	$\mathbb{Z}^4$	0
$j = 2$	$\mathbb{Z}$	$\mathbb{Z} \oplus \mathbb{Z}/2$	$\mathbb{Z}^4$	$\mathbb{Z}^7$	$\mathbb{Z}^3$	0

with integral coefficients. Their fundamental groups are $\mathbb{Z}^2 \rtimes_{R_3} \mathbb{Z}$ and $\mathbb{Z}^2 \rtimes_{R_4} \mathbb{Z}$, respectively, where

$$R_3 = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \qquad R_4 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Proof. We retain the actual product presentations, which also specify the comparison with the surfaces in the question. Let $\rho = e^{\pi i/3}$,

$$E_j = \mathbb{C}/(\mathbb{Z} + \kappa_j \mathbb{Z}), \quad F_1 = \mathbb{C}/(\mathbb{Z}(\rho - 2) + \mathbb{Z}(-3)), \quad F_2 = \mathbb{C}/(\mathbb{Z}(i - 1) + \mathbb{Z}(-2)),$$

where $\kappa_1 = \beta_c(z_1) + 2\mu(z_1)$ and $\kappa_2 = \beta_c(z_2) + 3\mu(z_2)$ have negative imaginary parts. The holomorphic maps to the central tori are induced by $L_1(\xi, \eta) = (\eta, \xi - \eta/3)$ and $L_2(\xi, \eta) = (\eta, \xi - \eta/2)$. Their kernels are generated by

$$k_1 = (2/3, \rho), \qquad k_2 = (1/2, i).$$

The residual affine generators lift to

$$h_1(\xi,\eta) = (\xi + \kappa_1/3, (\rho-1)\eta), \qquad h_2(\xi,\eta) = (\xi - \kappa_2/4, -i\eta). \tag{E8}$$

These formulas are verified by substituting the four product periods in L_j; the lattice indices are three and two. Moreover $(\rho-1)\rho-\rho$ and $(-i)i-i$ are periods of the displayed F_j, so the generators commute. They give free groups $G_1 = C_3 \times C_3$ and $G_2 = C_2 \times C_4$. Indeed, equality of the first coordinates of $k_j^a h_j^b(\xi,\eta)$ and (ξ,η) forces $b = 0$ from the coefficient of the nonreal period κ_j, and then $a = 0$ from the real period. Thus the composite quotient maps are exactly the degree-nine and degree-eight covers of S_j, with the given signs, rather than abstract surface isogenies.

Use the multiplicative coordinate $v = e^{-2\pi i\xi}$ on E_j. Its Tate parameter is

$$q_j = e^{-2\pi i\kappa_j} = A_j w^2 = A_j s^{12}, \qquad A_j = e^{-2\pi i\kappa_j^{\mathrm{part}}} \neq 0.$$

Here A_j denotes a scalar only within this proof; it is not a monodromy matrix. Choose specifically $A_j^{1/12} = \exp(-2\pi i\kappa_j^{\mathrm{part}}/12)$ and put $r_j = A_j^{1/12} s$. This choice preserves the fractional period translations in (E8) exactly. The regular Tate model for $q_j = r_j^{12}$ has a central twelve-gon. We recall its charts to make the extension explicit. An infinite chain has node charts $x_i y_i = r_j$; on a branch overlap one uses $x_{i+1} = y_i^{-1}$ and $y_{i+1} = r_j y_i$. The coordinate $r_j^i x_i$ on a general fibre is unchanged on overlaps. The shift $i \mapsto i+12$ acts on that coordinate by r_j^{12}. Quotienting a tubular neighbourhood of the chain by this shift gives the proper twelve-gon family, and on $r_j \neq 0$ it is $\mathbb{C}^*/q_j^{\mathbb{Z}}$. Compactness follows from twelve projective components and a compact annular fundamental domain on a general fibre; the node charts give flatness and smoothness of its total space.

Multiplication by $q_1^{1/3} = r_1^4$ shifts the component index by four. Multiplication by $q_2^{-1/4} = r_2^{-3}$ shifts it by minus three. Hence the maps h_j in (E8) extend, together with their rotations on F_j. The maps k_j multiply v by roots of unity and translate F_j, and extend to every projective component. Their relations hold on the dense set $s \neq 0$ and consequently on all charts. The action is free on the central fibre as well. A nontrivial power of h_j moves every component and every node. A nontrivial power of k_j translates F_j by a nonzero torsion point, including above the nodes where the multiplicative action alone fixes a point. The finite free quotient is therefore a smooth total space, proper over s, with a reduced normal-crossing central surface. Flatness descends along the finite etale quotient. The component orbits have cardinality three and four, leaving four and three components. Quotienting an individual $\mathbb{P}^1 \times F_j$ by k_j is a ruled surface over $F_j/\langle\rho\rangle$ or $F_j/\langle i\rangle$, respectively. Its two distinguished sections have the same integral homology class: the corresponding line bundle has finite order and degree zero on the elliptic base.

For the topology, unwrap the component cycle into an infinite chain and then pass to the universal cover $\mathbb{C}$ of its elliptic base. The resulting union of simply connected ruled pieces, with simply connected intersections and a tree as incidence graph, is simply connected by van Kampen. Its deck group is generated by the enlarged lattice of translations on F_j and one step round the component cycle. The latter conjugates the lattice by the order-three or order-four rotation. This gives the asserted semidirect products; inverses of the displayed rotation matrices give isomorphic groups by reversing the cycle generator.

For homology, the normalization has $n = 4$ or 3 components, each with ranks $(1,2,2,2,1)$, and n elliptic double curves, each with ranks $(1,2,1)$. There are no triple points. The mapping-cylinder sequence for the normalization and the double curves applies integrally. In degree one the incidence matrix is a twisted cycle: its cokernel is $\operatorname{coker}(R_j - I)$ and its kernel is zero, because R_j has no eigenvalue one. The degree-zero incidence matrix contributes an additional $\mathbb{Z}$ to H_1. In degree two the two section classes on a component agree, so the incidence matrix

is the ordinary cycle matrix on their classes. It has kernel $\mathbb{Z}$ and primitive image of rank $n-1$. Thus $H_2 = \mathbb{Z}^{n+1}$, $H_3 = \mathbb{Z}^{2n+1}$, and $H_4 = \mathbb{Z}^n$. Finally $\operatorname{coker}(R_3 - I) = \mathbb{Z}/3$ and $\operatorname{coker}(R_4 - I) = \mathbb{Z}/2$. The exact Smith forms of both full cycle matrices and the independent 2×2 determinants are in `code/e_degeneration.out`. □

Proposition (E.2.2; Proved)**.** *For the retained S^6 family over the punctured w-disc, the monodromy acts as the identity on $H^*(S^6;\mathbb{Z})$. The remaining smooth monodromy is represented explicitly as follows. Set $c_0 = -i(M+1)$ and $c(r) = c_0 - 2r$, $0 \le r \le 1$. Let $H : X_{c_0} \to X_{c_0-2}$ be parallel transport in the proper smooth family over this interval. With the coherent isomorphism from A.3,*

$$\Phi = F_{2,c_0-2} \circ H : X_{c_0} \longrightarrow X_{c_0} \tag{E9}$$

represents the monodromy of the positively oriented w-circle. Its isotopy class is independent of the smooth connection. The associated oriented homotopy sphere $\Sigma_\Phi = D^7 \cup_\Phi (-D^7)$ reduces the mapping-class calculation to the explicit residue

$$[\Phi] = 28\mu(\Sigma_\Phi) = \frac{\langle p_1(W)^2, [W, \partial W]\rangle - 4\sigma(W)}{32} \pmod{28}, \tag{E10}$$

where W is a spin eight-manifold with boundary Σ_Φ and the Pontryagin class is lifted to relative rational cohomology. The identification of $\mathbb{Z}/28$ is oriented by a sphere with $\mu = 1/28$.

Proof. The proper submersion has a smooth horizontal distribution; integration along a compact path exists because the fibres are compact. The complex orientation varies continuously. Its monodromy therefore preserves the orientation class in H^6 and fixes the constant class in H^0. These are the only nonzero integral cohomology groups, proving the first assertion with integral coefficients.

Since $e^{-\pi i(c_0-2r)} = e^{-\pi i c_0} e^{2\pi i r}$, the path $c(r)$ is the lift of the positive circle. The descent gluing at its endpoint is exactly F_{2,c_0-2}, proving (E9). A path of horizontal distributions gives a smooth path of the resulting monodromy diffeomorphisms, so their isotopy classes agree. Changing the coherent descent maps by a member of the connected automorphism group also leaves this class unchanged. The diffeomorphism $X_{c_0} \simeq S^6$ is chosen with the complex orientation.

The glued manifold Σ_Φ is simply connected by van Kampen and has the integral homology of S^7 by Mayer–Vietoris; the Hurewicz and Whitehead arguments give a homotopy sphere. The gluing comparison $\pi_0 \operatorname{Diff}^+(S^6) = \Theta_7$ sends Φ to this construction. Here we use Cerf's Corollary 2 in dimension six [Cer71, p. 76]: isotopy classes are the group of sphere gluing maps modulo extension over a disc. The extension equivalence is exactly the equivalence of their two-disc gluings. In dimension seven, the simply connected h-cobordism comparison identifies these classes with oriented homotopy spheres; the comparison with oriented diffeomorphism classes is Smale's Theorem J [Sma61, pp. 393–394]. Thus the map being evaluated is the actual boundary gluing of (E9). Eells–Kuiper's seven-dimensional invariant is $(p_1^2 - 4\sigma)/896$ modulo one and distinguishes the twenty-eight oriented homotopy-seven-sphere classes [EK62, Section 7, pp. 100–103]. Multiplying that formula by twenty-eight gives (E10). Its hypotheses hold here: a homotopy seven-sphere is spin and has vanishing $H^3(-;\mathbb{Q})$ and $H^4(-;\mathbb{Q})$, so the relative lift is unique. The same theorem and its cobordism construction supply a spin coboundary. Thus (E10) is an invariant of the specific gluing (E9), rather than an inference from its action on cohomology. □

Problem (E.2.3; Open, with local fillings and an invariant reduction)**.** We have not assembled a flat proper filling of the compact S^6 family over $w = 0$. In particular no global central fibre is named whose Euler number, fundamental group, or integral homology could be asserted. E.2.1 constructs and computes the two reduced-surface degenerations after the stated additional

base change; it does not construct a neighbourhood of the nonreduced fibres $3S_1$ and $4S_2$ in a compact limit.

The smallest missing finite-place instance is the extension of the order-three affine deck action over a two-parameter plumbing in (t, s), with prescribed normal character and central multiplicity three, followed by descent from s to $w = s^6$. The order-four analogue must retain the normal character of order four. At infinity the unresolved periodic gluing is the one specified in E.1.5, now compatible with these finite-place quotients. The computed central surface homology groups cannot be substituted for the homology of that unconstructed threefold.

For the mapping-class part, E.2.2 gives the exact clutching map and its characteristic-number reduction. No Pontryagin-number or signature calculation for a coboundary of this particular Σ_Φ has been completed. The smallest undecided numeric instance is the residue (E10) for the single loop based at $c_0 = -i(M + 1)$; all other basepoints in the connected parameter disc give conjugate monodromy, and Θ_7 is abelian. The identity on cohomology proved above supplies no value for that residue.

6 F. The monodromy group and its period domain

We use the clockwise root deck generators and the lattice basis specified on the first page. In this section a group acts properly discontinuously when, for every compact subset K of its space, only finitely many group elements g satisfy $gK \cap K \neq \varnothing$. This definition permits finite stabilizers. We write $e = \widehat{\delta}$, $E = e \otimes \gamma$, and

$$J_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \Omega = \begin{pmatrix} 0 & 0 & 0 & 6 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -6 & 0 & 0 & 0 \end{pmatrix}.$$

Thus the alternating form is $\psi = u \wedge w + 6\gamma \wedge \delta$, an integral alternating form of type $(1, 6)$ on Λ. We use Ω for its matrix and retain J for a real complex structure. The domain $\mathcal{D}$ consists of the real endomorphisms J of $\Lambda_{\mathbb{R}}$ satisfying $J^2 = -I$, $J^{\mathrm{t}}\Omega J = \Omega$, and $\operatorname{sign}(\Omega J) = (2, 2)$. It is the stated $\mathrm{Sp}_4(\mathbb{R})/U(1, 1)$, with its complex structure as an open subset of the Lagrangian Grassmannian. An integral symplectic matrix acts by $J \mapsto gJg^{-1}$.

6.1 F.1. An exact arithmetic description

Theorem (F.1.1; Proved)**.** *Let $P_{e,\mathbb{Z}}$ be the integral symplectic group fixing e as a vector, and let $P_{\mathbb{Z}}$ be the integral symplectic group preserving $\mathbb{Z}e \subset \ker\gamma \subset \Lambda$. Every element of $P_{e,\mathbb{Z}}$ has a unique expression*

$$g(B, x, t) = \begin{pmatrix} 1 & 0 & 0 \\ 6x & B & 0 \\ t & -x^{\mathrm{t}} J_2 B & 1 \end{pmatrix}, \qquad B \in \mathrm{SL}_2(\mathbb{Z}), \quad x \in \mathbb{Z}^2, \quad t \in \mathbb{Z}, \tag{12}$$

where the block sizes are $1, 2, 1$. Put

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad a = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}.$$

There is a homomorphism $\alpha : \mathrm{SL}_2(\mathbb{Z}) \to \mathbb{Z}/12$ determined by $\alpha(S) = 9$ and $\alpha(T) = 1$; in particular $\alpha(a) = 4$. With $Q(x) = x_1 + x_2 + x_1x_2$, the actual monodromy group is

$$\Gamma = \{g(B, x, t) : t + 6Q(x) - \alpha(B) = 0 \text{ in } \mathbb{Z}/12\}. \tag{13}$$

Consequently $[P_{e,\mathbb{Z}} : \Gamma] = 12$ and $P_{\mathbb{Z}} = \{\pm I\} \times P_{e,\mathbb{Z}}$, so $[P_{\mathbb{Z}} : \Gamma] = 24$. Its image in the Levi quotient is exactly $\mathrm{SL}_2(\mathbb{Z})$, with scalar on e equal to 1.

Proof. The matrices in the question, in the fixed basis, are

$$A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 0 & 1 & 0 \\ -6 & -1 & -1 & 0 \\ -2 & 1 & 0 & 1 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -6 & 1 & 0 & 0 \\ 3 & 0 & 1 & 1 \end{pmatrix}.$$

Multiplication gives $A_1^3 = A_2^4 = I$ and $A_j^{\mathsf{t}} \Omega A_j = \Omega$. In particular $A_1 = g(a, (1,-1)^{\mathsf{t}}, -2)$ and $A_2 = g(S, (0,-1)^{\mathsf{t}}, 3)$. Their images a, S generate $\mathrm{SL}_2(\mathbb{Z})$ because $Sa = T$ and S, T perform the Euclidean algorithm on primitive integral columns.

A matrix fixing e and preserving Ω preserves $e^{\perp} = \ker \gamma$. Its first row is $(1,0,0,0)$, its last column is e, its middle block B is symplectic on $e^{\perp}/\mathbb{R}e$, and its remaining middle column, say v, satisfies $6r = -v^{\mathsf{t}} J_2 B$. Integrality of r and unimodularity of B give $v = 6x$ with $x \in \mathbb{Z}^2$. There is no further condition on the lower left entry t. This proves (12). Direct multiplication gives

$$g(B,x,t)g(C,y,s) = g(BC, x + By, t + s - 6x^{\mathsf{t}} J_2 B y). \tag{14}$$

For completeness, the presentation of the Levi group needed here follows from an elementary action on the real projective line. The images of a and S in $\mathrm{PSL}_2(\mathbb{Z})$ have orders 3 and 2. On positive real numbers, $a(z) = -1/(z+1)$ has image $(-1, 0)$ and $a^2(z) = -(z+1)/z$ has image $(-\infty, -1)$; S carries negative real numbers to positive real numbers. A cyclically reduced alternating word starting with a nonidentity power of a and ending with S therefore carries $(-\infty, 0)$ into one of these two proper subintervals. It cannot be the identity. Every reduced word is conjugate to such a word or to a nonidentity element of one factor. Together with the Euclidean algorithm, this proves the presentation $\mathrm{PSL}_2(\mathbb{Z}) = \langle \bar{a}, \bar{S} \mid \bar{a}^3 = \bar{S}^2 = 1 \rangle$. It follows that

$$\mathrm{SL}_2(\mathbb{Z}) = \langle a, S \mid a^3 = S^4 = 1,\ [a, S^2] = 1 \rangle. \tag{15}$$

Indeed the group on the right modulo its central element S^2 is the just-proved free product, and S^2 maps to the nonidentity matrix $-I$. Its kernel in the displayed matrix group is consequently trivial. The values $4, 9$ respect all these relations, proving the assertion about α; $T = Sa$ has value 1.

The quadratic function Q modulo 2 is zero at $(0,0)$ and one at all three other points of $\mathbb{F}_2^2$. Hence $Q(Bx) = Q(x) \pmod 2$ for every $B \in \mathrm{SL}_2(\mathbb{Z})$. Also

$$Q(x+y) - Q(x) - Q(y) \equiv x^{\mathsf{t}} J_2 y \pmod 2.$$

Equation (14) now shows that $h(g(B,x,t)) = t + 6Q(x) \pmod{12}$ is a homomorphism. The homomorphism $\chi = h - \alpha \circ B$ is surjective, since $\chi(g(I,0,1)) = 1$. Its values on A_1, A_2 are zero.

We verify equality with its kernel; containment alone would not determine the group. Let $N_{\mathbb{Z}}$ be the kernel of the map to the middle block. The kernel of $\langle g_1, g_2 \mid g_1^3 = g_2^4 = 1 \rangle \to \mathrm{SL}_2(\mathbb{Z})$ is normally generated by $[g_1, g_2^2]$, by (15). Its image in Γ is therefore normally generated by

$$K = [A_1, A_2^2] = g(I, (0,-1)^{\mathsf{t}}, 6).$$

Put

$$U = A_2 K A_2^{-1} = g(I, (1,0)^{\mathsf{t}}, -6), \quad V = K^{-1} = g(I, (0,1)^{\mathsf{t}}, -6), \quad Z = I + 12E.$$

Equation (14) gives $[U, V] = Z^{-1}$. Thus U, V, Z generate

$$H = \{ g(I,x,t) : t \in 6\mathbb{Z}, \quad t/6 \equiv Q(x) \pmod 2 \}. \tag{16}$$

More explicitly, $U^rV^sZ^k = g(I,(r,s)^{\mathsf{t}}, -6(r+s+rs)+12k)$, so these expressions give every member of the set. Conjugation by $g(B,x,t)$ sends $g(I,y,s)$ to $g(I,By,s-12x^{\mathsf{t}}J_2By)$; the invariance of Q modulo 2 proves that H is normal in $P_{e,\mathbb{Z}}$. The normal closure of K is contained in H and contains U,V,Z, so it is exactly H. We have proved $\Gamma\cap N_{\mathbb{Z}} = H$. Both Γ and $\ker\chi$ surject onto the same Levi group and have intersection H with $N_{\mathbb{Z}}$. Given $g\in\ker\chi$, choose $\gamma\in\Gamma$ with the same middle block. Then $g\gamma^{-1}\in H$, and $g\in\Gamma$. This proves (13) and the index 12.

Finally an integral automorphism preserving the primitive line $\mathbb{Z}e$ sends e to e or $-e$. Multiplication by the central matrix $-I$ converts the latter case to the former. The intersection of $\{\pm I\}$ with $P_{e,\mathbb{Z}}$ is trivial. This proves the direct product and the index 24. □

Proposition (F.1.2; Proved)**.** *The group in* (13) *has center* $\langle Z\rangle\cong\mathbb{Z}$ *and is the following explicit central extension of* $\mathbb{Z}^2\rtimes\mathrm{SL}_2(\mathbb{Z})$ *by* $\mathbb{Z}$*. Choose the representative* $a(B)\in\{0,\dots,11\}$ *of* $\alpha(B)$*. Its underlying set consists of triples* (B,x,k)*, with product*

$$(B,x,k)(C,y,l) = (BC, x+By, k+l+\kappa((B,x),(C,y))),$$

where

$$\begin{aligned}12\kappa((B,x),(C,y)) = a(B)+a(C)-a(BC)-6Q(x)-6Q(y)\\ -6x^{\mathsf{t}}J_2By+6Q(x+By).\end{aligned}$$

Its real and rational Zariski closures are respectively the full real and rational symplectic stabilizers of the vector e*. They have dimension* 6*, with three-dimensional Heisenberg unipotent radical and Levi quotient* SL_2*.*

Proof. The numerator defining κ is divisible by 12, by the two congruences proved above. The bijection

$$(B,x,k)\longmapsto g(B,x,a(B)-6Q(x)+12k)$$

intertwines the displayed multiplication with matrix multiplication by (14). This proves associativity as well as the claimed abstract description, without leaving an unspecified extension class. The kernel of the homomorphism to $\mathbb{Z}^2\rtimes\mathrm{SL}_2(\mathbb{Z})$ is $\langle Z\rangle$. That affine group has trivial center: commuting with all translations forces its linear part to be I, and commuting with S,T forces its translation part to vanish. Thus the center of Γ is exactly the stated kernel.

The logarithms $U-I, V-I, Z-I$ are linearly independent elements of the three-dimensional Heisenberg Lie algebra. The square of each is zero. A polynomial vanishing on all integral powers $I+n(U-I)$ vanishes on the whole one-parameter line; the same holds for V and Z. Hence the Zariski closure contains the full unipotent radical. Its Levi image contains S,T; the powers of T and of STS^{-1} similarly supply the two opposite unipotent subgroups of SL_2. Their Lie algebras and their bracket span $\mathfrak{sl}_2$. They generate the connected algebraic group SL_2. A subgroup containing the entire unipotent kernel and surjecting onto this Levi quotient is the entire vector stabilizer. Its dimension is $3+3=6$. This reasoning uses equations over $\mathbb{Q}$, so proves the rational assertion as well as the real assertion. □

6.2 F.2. Properness and the maximal open set

Define the real-analytic function

$$q_e:\mathcal{D}\longrightarrow\mathbb{R},\qquad q_e(J)=\psi(e,Je),\qquad \mathcal{O}=\{J: q_e(J)\neq 0\}.$$

The function q_e is invariant under $P_{\mathbb{Z}}$: replacing e by $-e$ changes both of its arguments. Write $\mathcal{O}_+$ and $\mathcal{O}_-$ for its two sign regions.

Proposition (F.2.1; Refuted)**.** *The assertion that* Γ *acts properly discontinuously on all of* $\mathcal{D}$ *is false. The following point of* $\mathcal{D}$ *has an explicit infinite stabilizer:*

$$J_0 = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -6 \\ 1 & 0 & 0 & 0 \\ 0 & 1/6 & 0 & 0 \end{pmatrix}, \qquad R = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 72 & 0 \\ 0 & 0 & 1 & 0 \\ -12 & 0 & 0 & 1 \end{pmatrix} \in \Gamma.$$

Every power of R *fixes* J_0*, and all these powers are distinct.*

Proof. Direct multiplication gives $J_0^2 = -I$ and $J_0^{\dagger}\Omega J_0 = \Omega$. Moreover

$$\Omega J_0 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 6 & 0 \end{pmatrix}$$

has eigenvalues $1, -1, 6, -6$, so $J_0 \in \mathcal{D}$ and $q_e(J_0) = 0$. The matrix R equals $g(T^{72}, 0, -12)$ and belongs to (13), because $\alpha(T^{72}) = 0 \pmod{12}$. Its commutator with J_0 is zero. Writing $R = I + N$ gives $N \neq 0$ and $N^2 = 0$, so $R^m = I + mN$ and the powers are distinct. The compact singleton $\{J_0\}$ therefore violates the stated definition of proper discontinuity. These identities are checked both with SymPy and with a separate rational matrix implementation in `code/f_exact.py`. □

Theorem (F.2.2; Proved)**.** *The set* $\mathcal{O}$ *is the largest* Γ*-invariant open subset of* $\mathcal{D}$ *on which the action is properly discontinuous. It is also the largest such open subset for* $P_{\mathbb{Z}}$*. Each sign region is a complex manifold in the analytic topology, and its quotient by either discrete group is a Hausdorff complex analytic orbifold.*

Proof. Let $P_{\mathbb{R}}$ be the real symplectic group preserving the real line $\mathbb{R}e$. We first prove properness on each sign region. For a fixed J, the form $h_J(v, w) = \psi(v, Jw) + i\psi(v, w)$ is a nondegenerate Hermitian form of signature $(1, 1)$ on the complex vector space $(\Lambda_{\mathbb{R}}, J)$, with the convention conjugate linear in the first entry. When $q_e(J) \neq 0$, the complex line $\mathbb{C}e$ is nondegenerate and its Hermitian orthogonal complement is a complex line of the opposite sign. Normalize e by a positive real scalar and choose a unit vector in this complement. Comparing these two Hermitian bases for two complex structures with the same sign gives a real symplectic map preserving $\mathbb{R}e$ and conjugating one complex structure to the other. Thus $P_{\mathbb{R}}$ is transitive on each sign region.

An element of the stabilizer of J inside $P_{\mathbb{R}}$ is complex linear and sends e to re for a real nonzero r. Preservation of $q_e(J)$ gives $r^2 = 1$. On the complementary complex line it acts by an element of $U(1)$. The stabilizer is consequently the compact group $\{\pm 1\} \times U(1)$. Each sign region is therefore a homogeneous space of $P_{\mathbb{R}}$ with compact stabilizer L. Local sections of $P_{\mathbb{R}} \to P_{\mathbb{R}}/L$, and a finite cover of a compact subset K, give a compact subset C of $P_{\mathbb{R}}$ mapping onto K. Every element carrying a point of K back into K belongs to CLC^{-1}, which is compact. This proves properness. Restricting a proper action to a closed discrete subgroup gives the required finite-intersection condition. Both Γ and $P_{\mathbb{Z}}$ are closed discrete matrix groups. The two sign regions are disjoint invariant open-and-closed subsets of $\mathcal{O}$, so the same finite-intersection condition holds on their union. Proper actions by discrete groups have finite stabilizers and Hausdorff quotients; their holomorphic local finite-group quotient charts supply the asserted analytic orbifolds.

It remains to prove maximality, which requires more than one bad point. The real vector stabilizer $P_{e,\mathbb{R}}$ acts transitively on $\mathcal{D}_0 = \{J : q_e(J) = 0\}$. Here is a direct comparison map. In the Hermitian space just described, e is a nonzero null vector. Choose v with $h_J(e, v) = 1$,

and replace v by $v - \frac{1}{2}h_J(v,v)e$. This preserves $h_J(e,v) = 1$ and makes v null. The ordered complex basis (e, v) has Gram matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. The complex-linear map between two such bases fixes e and preserves h, hence its imaginary part ψ. As a real map it belongs to $P_{e,\mathbb{R}}$ and conjugates the two complex structures.

The subgroup $P_{e,\mathbb{Q}}$ is dense in $P_{e,\mathbb{R}}$. For the unipotent coordinates in (12) this is density of $\mathbb{Q}^3$ in $\mathbb{R}^3$. For SL_2 one may approximate a, b, c by rational numbers in the chart $a \neq 0$ and set $d = (1 + bc)/a$; translates of that chart cover the group. Hence

$$\{rJ_0r^{-1} : r \in P_{e,\mathbb{Q}}\}$$

is dense in $\mathcal{D}_0$. Put $R_0 = g(T^6, 0, -1)$, so $R_0^{12} = R$ and $(R_0 - I)^2 = 0$. At rJ_0r^{-1} the nonzero rational nilpotent $L = r(R_0 - I)r^{-1}$ commutes with the complex structure and satisfies $L^2 = 0$. Choose a positive integer d clearing its denominators. The matrix $I + dL$ and its inverse $I - dL$ are integral, symplectic, and fix e; this follows either by conjugating the identity $(I + s(R_0 - I))^{\mathsf{t}}\Omega(I + s(R_0 - I)) = \Omega$ or by expanding it. The twelfth power of $I + dL$ lies in Γ, because $P_{e,\mathbb{Z}}/\Gamma \cong \mathbb{Z}/12$. It has infinite order and fixes rJ_0r^{-1}. Thus points with infinite Γ-stabilizer are dense in $\mathcal{D}_0$.

Every open neighborhood in $\mathcal{D}$ of any point of $\mathcal{D}_0$ therefore contains a point with infinite stabilizer. An open set on which the action is proper cannot meet $\mathcal{D}_0$. This proves the largest-open-set assertion for Γ. The same points have infinite $P_{\mathbb{Z}}$-stabilizer, so it proves that assertion for the full parabolic as well. □

Proposition (F.2.3; Proved)**.** *The positive sign region admits the global period chart*

$$\mathcal{O}_+ = \left\{(\tau, \mu, \beta) \in \mathbb{C}^3 : \operatorname{Im}\tau > 0, \quad \operatorname{Im}\beta - \frac{6(\operatorname{Im}\mu)^2}{\operatorname{Im}\tau} < 0\right\},$$

represented by the matrix Π *in the question. In this chart,*

$$q_e(J) = -\frac{6}{D}. \tag{17}$$

Every admissible period $\Pi_c(z)$ *belongs to* $\mathcal{O}_+$*, for every admissible* c*. Thus this single maximal proper-action open set contains all the period images in question.*

Proof. Let $J \in \mathcal{O}_+$. The two vectors $\widehat{w}, e$ are complex-linearly independent in $(\Lambda_{\mathbb{R}}, J)$. Otherwise $\widehat{w} = re + sJe$ for real r, s. Pairing with e gives $0 = sq_e(J)$; hence $s = 0$, contradicting the real linear independence of $\widehat{w}, e$. There is consequently a unique complex-linear isomorphism $(\Lambda_{\mathbb{R}}, J) \to \mathbb{C}^2$ sending these two vectors to $(1, 0)$ and $(0, 1)$. The corresponding period matrix has right block I. The equation $\Pi\Omega^{-1}\Pi^{\mathsf{t}} = 0$ says that its upper left entry is six times its lower middle entry, giving exactly the displayed matrix. This construction is the ordinary holomorphic Grassmannian chart, so its coordinates and inverse are holomorphic.

In the real symplectic basis $(\widehat{u}, \widehat{\gamma}/6, \widehat{w}, e)$ the normalized symmetric period matrix is $\left(\begin{smallmatrix} \tau & \mu \\ \mu & \beta/6 \end{smallmatrix}\right)$. Its imaginary part has signature $(1, 1)$, equivalently $6(\operatorname{Im}\mu)^2 - \operatorname{Im}\tau \operatorname{Im}\beta > 0$. Solving $\Pi(Je) = i\Pi e = (0, i)$ in real coefficients gives

$$\gamma(Je) = \frac{\operatorname{Im}\tau}{\operatorname{Im}\tau \operatorname{Im}\beta - 6(\operatorname{Im}\mu)^2}.$$

Since $\psi(e, v) = -6\gamma(v)$, positivity of q_e now gives $\operatorname{Im}\tau > 0$ and $D < 0$, and proves (17). Conversely these inequalities make the imaginary period matrix invertible with the stated signature and give $q_e > 0$. The admissibility inequality in the question is $D_c < 0$ everywhere, which proves the last assertion by this exact comparison map. □

6.3 F.3. The curve in the proper quotient

Theorem (F.3.1; Proved)**.** *For every admissible c, the marked period map on the covering of B° is a holomorphic immersion into $\mathcal{O}_+$. The induced map*

$$p_c : B^\circ \longrightarrow \Gamma\backslash\mathcal{O}_+$$

is injective and a holomorphic immersion. It extends over $t = 0, 1$ to a proper holomorphic embedding

$$\bar{p}_c : \mathbb{C} = B \setminus \{\infty\} \hookrightarrow \Gamma\backslash\mathcal{O}_+.$$

The closure of $p_c(B^\circ)$ in this quotient is exactly $\bar{p}_c(\mathbb{C})$.

Proof. For $B = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, changing the marking by $g(B, x, t)$ replaces Π by $\Pi g(B, x, t)^{-1}$ and then renormalizes its right block. On the quotient by the complex line of e, the two periods are the row $(\tau, 1)$. Right multiplication by B^{-1} sends it to $(d\tau - c, a - b\tau)$. Hence

$$\tau \longmapsto \frac{d\tau - c}{a - b\tau}. \tag{18}$$

The denominator never vanishes in the upper half plane. The function $j(\tau)$ is therefore a holomorphic invariant of this action and descends to a holomorphic function j_Γ on the proper quotient. Here j is the elliptic invariant, normalized by $j(E) = 1728\, 4A^3/(4A^3 + 27B^2)$ for $y^2 = x^3 + Ax + B$; changing a lattice basis preserves the isomorphism of elliptic curves and hence this function.

We derive the required value in the present family. The elliptic curve whose period is τ has the equation stipulated in the question,

$$y^2 = x^3 - 3t^3(t-1)x + 2t^4(t-1)^2.$$

Its discriminant is $1728t^8(t-1)^3$ and direct cancellation gives

$$j(\tau) = 1728t \qquad (t \in B^\circ). \tag{19}$$

Equivalently this is the defining lift of the elliptic modular map used for τ in the period construction. The computation is reproduced exactly in `f_exact.py`. In a local base coordinate t, the derivative of the composite $j \circ \tau$ equals 1728. Thus the derivative of the marked period map is nonzero. The same identity after passage to the quotient shows both injectivity and injectivity of the tangent map of p_c: $j_\Gamma/1728$ is a holomorphic left inverse.

At the two elliptic points, the period functions are holomorphic in the cyclic covering coordinates s_1, s_2, with $t = s_1^3$ and $t - 1 = s_2^4$ after choosing local coordinates as in the question. Their values are nondegenerate because $D_c < 0$ there. The maps from these discs to $\mathcal{O}_+$ are equivariant for A_1, A_2. Composing with the quotient makes them invariant, so they descend to holomorphic maps from the base discs. This is the finite-quotient comparison map: a holomorphic invariant germ on a disc is a convergent series in $s_j^{m_j}$, which proves descent on local function rings. The extensions agree with p_c off their centers and therefore with each other. Identity (19) extends by holomorphy, so the resulting $\bar{p}_c$ still has holomorphic left inverse $j_\Gamma/1728$.

The quotient is Hausdorff by F.2.2. For a compact subset K of it, $\bar{p}_c^{-1}(K)$ is closed in $\mathbb{C}$ and lies in the compact set $j_\Gamma(K)/1728$. Thus $\bar{p}_c$ is proper. A holomorphic section of a holomorphic map is locally a closed immersion: in local embeddings into polydiscs, its coordinate along the base is t and the remaining coordinates are its holomorphic component functions, whose graph defines the image. Applying this to the left inverse shows that $\bar{p}_c$ is a holomorphic embedding; properness makes its image closed. Since $\mathbb{C} \setminus \{0, 1\}$ is dense in $\mathbb{C}$, its image has the asserted closure. □

Proposition (F.3.2; Proved)**.** *At the two finite punctures the period map has the holomorphic finite-cover limits*

$$\Pi_1 = \begin{pmatrix} 2(2-\rho) & \rho & 1 & 0 \\ \beta_c(z_1) & (2-\rho)/3 & 0 & 1 \end{pmatrix}, \qquad \Pi_2 = \begin{pmatrix} 3(1-i) & i & 1 & 0 \\ \beta_c(z_2) & (1-i)/2 & 0 & 1 \end{pmatrix}, \quad \rho = e^{\pi i/3}.$$

Their monodromies are the given order-three and order-four matrices. At the cusp the curve leaves every compact subset of $\Gamma\backslash\mathcal{O}_+$. On the prescribed cusp branch its Hodge plane in the compact dual tends to $\mathbb{C}\gamma + \mathbb{C}u$. The latter plane has complex dimension two in common with its conjugate and therefore does not represent a point of $\mathcal{D}$.

Proof. The finite values of τ, μ follow by evaluating their transformation laws at the fixed points: $\tau(z_1) = \rho$, $\mu(z_1) = (2-\rho)/3$, $\tau(z_2) = i$, $\mu(z_2) = (1-i)/2$. Substitution in Π gives the two matrices. The finite covering comparison and its descent were proved in F.3.1. The unipotent cusp transformation is $\tau \mapsto \tau - 1$, $\mu \mapsto \mu$, $\beta \mapsto \beta + 1$. On the cusp branch, $\operatorname{Im}\tau \to +\infty$, $\mu \to \mu_0$, and $\beta_c + \tau \to b_0 + c$, from the holomorphic cusp expansions. Consequently

$$D_c = -\operatorname{Im}\tau + O(1), \qquad q_e(\Pi_c) \longrightarrow 0^+, \qquad j_\Gamma(p_c(t)) = 1728t \longrightarrow \infty.$$

The last equality alone excludes a convergent subnet in any compact subset of the proper quotient; the first two identities give the position relative to the maximal proper-action open set. The rows of Π_c, regarded as covectors, are $6\mu\gamma + \tau u + w$ and $\beta_c\gamma + \mu u + \delta$. Dividing each by τ gives limits u and $-\gamma$. Their exterior product has a nonzero limit, so their span converges in the Grassmannian to the stated plane. It is spanned by real covectors, hence equals its conjugate. A complex structure has $H^{1,0} \cap H^{0,1} = 0$; here the intersection is the entire two-dimensional plane. This explicit comparison proves the assertion about the compact-dual limit. □

Proposition (F.3.3; Proved)**.** *The determinant used in the workbench is generically nonzero, with the stated zero at z_1:*

$$\det \dot{Z} = 6\dot{\mu}^2 - \dot{\tau}\dot{\beta}, \qquad \det \dot{Z}(z_1) = 0.$$

The immersion established in F.3.1 holds at every point of B° without requiring this determinant to be everywhere nonzero.

Proof. In the cusp coordinate s, the expansion $Z = s\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) + C(q)$, $q = e^{2\pi i s}$, gives $\det(dZ/ds) = 1 + O(q)$, so the determinant is not the zero function. At z_1 the fixed-point transformation law of μ gives $\dot{\mu} = -\dot{\tau}/3$. One may obtain this by differentiating $\mu \circ g_1 = (1-\mu)/\tau$, using $g_1'(z_1) = -\rho$ and $\rho^2 - \rho + 1 = 0$. The law $\beta \circ g_1 = \beta + 2 - 6(1-\mu)^2/\tau$ then gives $\dot{\beta} = 2\dot{\tau}/3$. Substitution yields $6\dot{\tau}^2/9 - 2\dot{\tau}^2/3 = 0$. The derivative of a map from a curve to the three-dimensional period chart is a vector of three coordinate derivatives. Identity (19) proves that this vector is nonzero over B°; the determinant tests the rank of the associated two-by-two Kodaira–Spencer matrix. The maps compared here are explicitly the coordinate derivative and its determinant, and the displayed computation states exactly what each supplies. □

Sources, comparisons, and verification record

We have treated the attached original [CA26] as the source of its own formulas. We extracted its text, compared the relevant formulas with the supplied transcription, and inspected its original page 66 for the orientation computation used in D.1.4. The algebra, maps, finite quotient models and compatibility calculations used above are given in our proofs and in the

accompanying executed programs. The workbench, course and survey helped locate claims; none of their status labels enters a proof.

Our use of [Eng26] includes the statement and proof of Lemma 2.1, the reference linearization in Corollary 2.2, the geometric quotient in Section 2, the finite modifications in Section 3, and Sections 7–8. The passage from its linearization to our periods is B.1.1 and B.1.4. The height input is also computed in B.1.3 and applied explicitly in D.1.2. B.2.3 locates a global-parameter descent correction to Remark 8.1. E.1.5 identifies the remaining simultaneous degeneration needed for Conjecture 8.2; no statement about the topology of an unconstructed compact central fibre is made.

For the deformation comparison we read the fixed-parameter classification argument in [Yan26, Section 6.2] and the precise non-isotriviality statement in [Via26b, Proposition 4.7]. A.4.3 gives the maps between these methods and the present calculation. The additional starting surfaces and changed period families in [Via26c, Via26a, LR26, Wan26] were checked through their introductions and stated main theorems. The lattice in A.2 belongs to the displayed $(3, 4, \infty)$ period system. Substituting a different starting surface changes those matrices, and the proofs above do not perform that substitution. The coprime $(3k, 4l)$ substitution that we do perform is B.3.1. This states the exact scope of the classification through the actual period matrices and finite norm maps.

The zero of $\wp$ in C.1 is derived from the explicit section and the trigonometric cusp expansion. We read [DI08, Theorem 1] and its discussion of Eichler–Zagier, but did not use an unopened original paper as a premise. For the description of line bundles and their sections in D, we use the Appell–Humbert correspondence and the Riemann criterion in [Sch19, Theorems 4.1 and 4.4]; the substitution of the actual period lattice and the rational-kernel quotient are written in D.1 and D.2. E.2.2 states the exact gluing and characteristic-number reduction to which the theorems of [Cer71, Sma61, EK62] are applied.

We searched the public literature on 12 November 2026 using the paper titles and identifiers above and combinations of "complex six-sphere", "classification", "period parameter", "Klingler", "monodromy", "Engel Conjecture 8.2" and "degeneration". The report records the query groups and inspected sources. This search also located an important version change: [DGKX26a, Theorems 1.3–1.4] stated miniversality, the $2\mathbb{Z}$ identification and the full automorphism group; the current version [DGKX26b] omits those deformation and classification sections. We record the two texts separately and infer no reason for that revision. We then checked the actual root-curve, cusp cohomology and logarithmic-tangent arguments in the historical version. A.5.2 gives their explicit derivation for the present charts, including the coarse invariant exponents and the normal quotient at every cusp stratum. The resulting tangent dimensions and our Kodaira–Spencer calculation give the stated miniversal disc. The complete audit and independently executed weighted cohomology checks are recorded in the report. We claim no priority for any result.

All decimal tables identify working precision and independently computed expressions. Exact lattice, Smith-form and polynomial identities use integer or rational arithmetic. The scripts, their actual outputs, the item-by-item report and a statement ledger accompany this paper. Exploratory floating-point grids are identified as such; a certified enclosure is used only where its error bounds and completed computation are supplied.

References

[CA26] Claude and Levent Alpöge. The $(3, 4, \infty)$ modular family of 2-tori, completed at its three special points, is a complex structure on s^6. Attached original manuscript, 108 pages, 2026. verified: opened the attached original PDF and its transcription;

the locations used are recorded in the report. The transcription is a comparison aid.

[Cer71] Jean Cerf. The pseudo-isotopy theorem for simply connected differentiable manifolds. In *Manifolds—Amsterdam 1970*, volume 197 of *Lecture Notes in Mathematics*, pages 76–82. Springer, 1971. verified: opened the author's seven-page article and visually inspected Theorem 1 and Corollaries 1–2 on its first page; publication details checked against the publisher.

[DGKX26a] Yun-Heng Du, Bin Guo, Yuta Kusakabe, and Song-Yan Xie. Oka geometry and deformations of torus-fibred complex six-spheres. arXiv:2610.00250v1, 23 September 2026, 2026. verified: opened the full historical HTML version; Theorems 1.3 and 1.4 and Sections 6–7 inspected. The deformation sections are absent from version 2. Their actual root-curve and logarithmic-tangent computation is explicitly rederived and audited in A.5.2.

[DGKX26b] Yun-Heng Du, Bin Guo, Yuta Kusakabe, and Song-Yan Xie. Oka geometry of complex six-spheres. arXiv:2610.00250v2, 23 October 2026, 2026. verified: opened the current abstract and full HTML version and compared its contents with version 1. No reason for the version change is inferred.

[DI08] William Duke and Özlem Imamoğlu. The zeros of the Weierstrass $\wp$-function and hypergeometric series. *Mathematische Annalen*, 340(4):897–905, 2008. verified: opened the attached paper and read Theorem 1 and the discussion of Eichler–Zagier; publication data checked on Duke's author page. The original Eichler–Zagier paper was not opened or used as an authority.

[EK62] James Eells and Nicolaas H. Kuiper. An invariant for certain smooth manifolds. *Annali di Matematica Pura ed Applicata*, 60:93–110, 1962. verified: opened the original PDF and visually inspected the title page and Section 7, printed pages 100–103, including the characteristic-number formula and the classification theorem.

[Eng26] Philip Engel. Complex structures on s^6. arXiv:2609.38442v1, 29 September 2026, 2026. verified: opened the attached full text; Lemma 2.1 and proof, Corollary 2.2, Definition 2.3, Proposition 2.4, Sections 3 and 7, and Remarks 8.1–8.4 inspected.

[LR26] Wenfei Liu and Sönke Rollenske. Yet another family of complex structures on s^6. arXiv:2609.33826v2, 9 October 2026, 2026. verified: opened the attached full text; introduction and the stated Theorem 5.1 inspected. Used for scope comparison.

[Sch19] Christian Schnell. Lecture 4: The Appel–Humbert theorem. MAT 615, Stony Brook University, lecture of 6 February, 2019. verified: opened all five pages; Theorems 4.1 and 4.4, equation (4.3), and the proof on effective line bundles and quotient tori on pages 2–3 inspected.

[Sma61] Stephen Smale. Generalized Poincaré's conjecture in dimensions greater than four. *Annals of Mathematics*, 74(2):391–406, 1961. verified: opened the original article; Theorems A, I, J and the group comparisons on pages 391–394 inspected.

[Via26a] Jeff Viaclovsky. Combs and complex structures on the six-sphere. arXiv:2610.10451v1, 7 October 2026, 2026. verified: opened the attached full text; Theorem 1.1 and Section 10 inspected. Its modified period families are included in the scope discussion.

[Via26b] Jeff Viaclovsky. Logarithmic transformations on the six-sphere. arXiv:2610.39102v1, 29 October 2026, 2026. verified: opened the attached full text; Theorem 1.1, Questions 1.1, the translation data, and Proposition 4.7 inspected.

[Via26c] Jeff Viaclovsky. A two-parameter family of complex structures on s^6. arXiv:2609.33785v1, 27 September 2026, 2026. verified: opened the attached full text; introduction, initial elliptic surface, and description of the four critical fibres inspected. Used for scope comparison, not as a deformation theorem for the present family.

[Wan26] Zichang Wang. A complex structure on a rational homology six-sphere with two-torsion. arXiv:2609.11541v2, 8 November 2026, 2026. verified: opened the attached full text; introduction, Theorem 1.1 and initial elliptic surface inspected. Used for scope comparison.

[Yan26] Xiufan Yang. Compact complex threefolds fibred by two-dimensional complex tori. arXiv:2609.29666v1, 2026. verified: opened the attached full text; in particular Section 6.2, Lemmas 6.6–6.9, and Proposition 6.15 inspected for the comparison in A.4.